assignment_v2

April 28, 2026

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- **Project title** Sustainable Crop Yield prediction via Multi-modal AI

[1]:
```
import matplotlib
matplotlib.use('Agg')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import os, pathlib, random, warnings
import io
from IPython.display import display, Image

warnings.filterwarnings("ignore")
```

[2]:
```
# Global random seed for full reproducibility
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
try:
    import torch
    torch.manual_seed(SEED)
    if torch.cuda.is_available():
        torch.cuda.manual_seed_all(SEED)
    DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    print("PyTorch seed set. Device:", DEVICE)
except ImportError:
    DEVICE = None
    print("PyTorch not available, skipping torch seed.")

# Path constants
BASE = pathlib.Path(r"e:\learnings\University Of Hull␣
 ↪AI\assignment-september\MSc␣
 ↪project\multimodal_agriculture_prediction-notebook_only")
CROPNET       = BASE / "data" / "CropNet"
USDA_ROOT     = CROPNET / "USDA" / "data" / "Soybean"
```

```python
SENTINEL_ROOT = CROPNET / "Sentinel" / "data"
HRRR_ROOT     = CROPNET / "HRRR"

# Verify every root exists
paths = {"USDA_ROOT": USDA_ROOT, "SENTINEL_ROOT": SENTINEL_ROOT, "HRRR_ROOT":␣
↪HRRR_ROOT}
for name, p in paths.items():
    status = "OK" if p.exists() else "MISSING"
    print(f"{name}: {status}  ({p})")
```

```
PyTorch seed set. Device: cpu
USDA_ROOT: OK  (e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-
notebook_only\data\CropNet\USDA\data\Soybean)
SENTINEL_ROOT: OK  (e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-
notebook_only\data\CropNet\Sentinel\data)
HRRR_ROOT: OK  (e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-notebook_only\data\CropNet\HRRR)
```

[3]:

```python
# Build usda_index: full coverage across all available years and counties
#
# USDA data spans 2016-2022 (7 years) and covers all US soybean counties␣
↪(~500+).
# This is the primary label source for the weather-only Bi-LSTM pre-training␣
↪phase.
# The full 7-year, multi-county dataset (~3,030 county-year records) is loaded␣
↪here.

USDA_YEARS = ["2016", "2017", "2018", "2019", "2020", "2021", "2022"]
YIELD_COL  = "YIELD, MEASURED IN BU / ACRE"
PROD_COL   = "PRODUCTION, MEASURED IN BU"

frames = []
for yr in USDA_YEARS:
    csv_path = USDA_ROOT / yr / f"USDA_Soybean_County_{yr}.csv"
    if not csv_path.exists():
        print(f"WARNING: {csv_path} not found, skipping.")
        continue
    df = pd.read_csv(csv_path)
    if df.empty or YIELD_COL not in df.columns:
        print(f"WARNING: {yr} CSV is empty or missing yield column, skipping.")
        continue
    n_rows = len(df.dropna(subset=[YIELD_COL]))
    print(f"Loaded {yr}: {n_rows} valid county records")
    frames.append(df)
```

```python
usda_raw = pd.concat(frames, ignore_index=True)

# Drop rows with null FIPS components before constructing FIPS string.
# With large CSVs pandas may infer float64 for integer columns if any nulls␣
↪exist,
# causing astype(str) to produce "5.0" instead of "5". Explicit int conversion
# via fillna + astype(int) prevents malformed FIPS codes like "5.01.0".
usda_raw = usda_raw.dropna(subset=["state_ansi", "county_ansi"])
usda_raw["fips"] = (
    usda_raw["state_ansi"].astype(float).astype(int).astype(str).str.zfill(2) +
    usda_raw["county_ansi"].astype(float).astype(int).astype(str).str.zfill(3)
)

usda_index = (
    usda_raw[["fips", "year", "state_name", "county_name", YIELD_COL, PROD_COL]]
    .rename(columns={YIELD_COL: "yield_bu_acre", PROD_COL: "production_bu"})
    .dropna(subset=["yield_bu_acre"])
    .reset_index(drop=True)
)
usda_index["year"] = usda_index["year"].astype(str)

# Coverage summary
year_counts = usda_index.groupby("year").size()
fips_years  = usda_index.groupby("fips")["year"].nunique()

print()
print("USDA INDEX SUMMARY")
print(f"Total county-year records : {len(usda_index)}")
print(f"Unique FIPS codes         : {usda_index['fips'].nunique()}")
print(f"Years covered             : {sorted(usda_index['year'].unique())}")
print(f"States covered            : {usda_index['state_name'].nunique()}")
print()
print("Records per year:")
print(year_counts.to_string())
print()
print(f"Counties with all 7 years : {(fips_years == 7).sum()}")
print(f"Counties with >= 5 years  : {(fips_years >= 5).sum()}")
print(f"Counties with >= 3 years  : {(fips_years >= 3).sum()}")
print()
```

```
Loaded 2016: 541 valid county records
Loaded 2017: 543 valid county records
Loaded 2018: 473 valid county records
Loaded 2019: 429 valid county records
Loaded 2020: 546 valid county records
Loaded 2021: 483 valid county records
Loaded 2022: 520 valid county records
```

```
USDA INDEX SUMMARY
Total county-year records : 3535
Unique FIPS codes         : 649
Years covered             : ['2016', '2017', '2018', '2019', '2020', '2021',
'2022']
States covered            : 15

Records per year:
year
2016    541
2017    543
2018    473
2019    429
2020    546
2021    483
2022    520

Counties with all 7 years : 260
Counties with >= 5 years  : 493
Counties with >= 3 years  : 570
```

[4]:
```python
# Build sentinel_index by scanning all HDF5 files in the Sentinel tree
import h5py

sentinel_records = []
SENTINEL_AG_ROOT   = SENTINEL_ROOT / "AG"
SENTINEL_NDVI_ROOT = SENTINEL_ROOT / "NDVI"

for year_dir in sorted(SENTINEL_AG_ROOT.iterdir()):
    if not year_dir.is_dir():
        continue
    year = year_dir.name
    for state_dir in sorted(year_dir.iterdir()):
        if not state_dir.is_dir():
            continue
        state = state_dir.name
        for h5_file in sorted(state_dir.glob("*.h5")):
            # Parse quarter dates from filename:␣
↪Agriculture_NN_ST_YYYY-MM-DD_YYYY-MM-DD.h5
            parts = h5_file.stem.split("_")
            q_start = parts[3] if len(parts) > 4 else "unknown"
            q_end   = parts[4] if len(parts) > 4 else "unknown"

            # Open file to get FIPS codes, dates, and tile counts
            try:
                with h5py.File(h5_file, "r") as f:
```

```
                    for fips_key in f.keys():
                        date_keys = list(f[fips_key].keys())
                        n_dates   = len(date_keys)
                        # Sample tile count from first available date
                        n_tiles = 0
                        if n_dates > 0:
                            first_date = date_keys[0]
                            if "data" in f[fips_key][first_date]:
                                n_tiles = f[fips_key][first_date]["data"].
 ↪shape[0]
                        sentinel_records.append({
                            "fips"    : fips_key,
                            "year"    : year,
                            "state"   : state,
                            "q_start" : q_start,
                            "q_end"   : q_end,
                            "h5_path" : str(h5_file),
                            "n_dates" : n_dates,
                            "n_tiles" : n_tiles,
                        })
            except Exception as e:
                print(f"WARNING: Could not open {h5_file.name}: {e}")

sentinel_index = pd.DataFrame(sentinel_records)

print("sentinel_index shape:", sentinel_index.shape)
print("Years:", sorted(sentinel_index["year"].unique()))
print("States:", sorted(sentinel_index["state"].unique()))
print("Unique FIPS codes:", sentinel_index["fips"].nunique())
print("Total tiles across all files (approx):", sentinel_index["n_tiles"].sum())
print()
print(sentinel_index.groupby(["year","state"]).agg(
    files=("h5_path","count"),
    fips_codes=("fips","nunique"),
    total_dates=("n_dates","sum"),
    total_tiles=("n_tiles","sum")
).to_string())
```

```
sentinel_index shape: (116, 8)
Years: ['2021', '2022']
States: ['AL', 'AR', 'AZ', 'CA', 'CO', 'DE', 'FL', 'GA', 'IA', 'ID', 'IL', 'IN',
'KS', 'KY', 'LA']
Unique FIPS codes: 15
Total tiles across all files (approx): 3884

            files  fips_codes  total_dates  total_tiles
year state
2021 AL         4           1           24           60
```

	AR	4	1	24	148
	CA	4	1	24	120
	CO	4	1	24	120
	DE	4	1	24	80
	FL	4	1	24	116
	GA	4	1	24	76
	IA	4	1	24	48
	ID	4	1	24	132
	IL	4	1	24	96
	IN	4	1	24	24
	KS	4	1	24	32
	KY	4	1	24	52
	LA	4	1	24	76
2022	AL	4	1	24	60
	AR	4	1	24	148
	AZ	4	1	19	1524
	CA	4	1	24	120
	CO	4	1	24	120
	DE	4	1	24	80
	FL	4	1	24	116
	GA	4	1	24	76
	IA	4	1	24	48
	ID	4	1	24	132
	IL	4	1	24	96
	IN	4	1	24	24
	KS	4	1	24	32
	KY	4	1	24	52
	LA	4	1	24	76

[5]:
```
# Identify which Sentinel counties have matching USDA yield labels
#
# The Sentinel imagery was downloaded for one county per state (county ANSI␣
↪001).
# This cell programmatically looks up each Sentinel FIPS code in usda_index.
# No values are hardcoded. Whatever is found on disk is used; the rest is left␣
↪out.

sentinel_fips_list = sorted(sentinel_index["fips"].unique())
print("Sentinel FIPS codes (from imagery):", sentinel_fips_list)
print()

# Programmatic join: keep only Sentinel FIPS codes that exist in usda_index
sentinel_labels = (
    usda_index[usda_index["fips"].isin(sentinel_fips_list)]
    .copy()
    .reset_index(drop=True)
)
```

```python
print("=" * 60)
print("SENTINEL-USDA LABEL INTERSECTION")
print("=" * 60)
print(f"Sentinel FIPS codes queried : {len(sentinel_fips_list)}")
print(f"Matching USDA records found : {len(sentinel_labels)}")
print()

if len(sentinel_labels) == 0:
    print("No USDA yield records found for any Sentinel FIPS code.")
else:
    print("Matching records:")
    print(sentinel_labels[["fips", "year", "state_name", "county_name",
                           "yield_bu_acre"]].to_string(index=False))
```

```
Sentinel FIPS codes (from imagery): ['01001', '04001', '05001', '06001',
'08001', '10001', '12001', '13001', '16001', '17001', '18001', '19001', '20001',
'21001', '22001']

============================================================
SENTINEL-USDA LABEL INTERSECTION
============================================================
Sentinel FIPS codes queried : 15
Matching USDA records found : 105

Matching records:
 fips year state_name county_name  yield_bu_acre
13001 2016    GEORGIA     APPLING           31.4
08001 2016   COLORADO       ADAMS           32.8
21001 2016   KENTUCKY       ADAIR           43.9
18001 2016    INDIANA       ADAMS           59.3
19001 2016       IOWA       ADAIR           60.8
10001 2016   DELAWARE        KENT           44.5
04001 2016    ARIZONA      APACHE           30.1
05001 2016   ARKANSAS    ARKANSAS           48.4
06001 2016 CALIFORNIA     ALAMEDA           32.7
16001 2016      IDAHO         ADA           38.7
12001 2016    FLORIDA     ALACHUA           29.1
22001 2016  LOUISIANA      ACADIA           37.6
20001 2016     KANSAS    ANDERSON           45.4
17001 2016   ILLINOIS       ADAMS           55.5
01001 2016    ALABAMA      AUTAGA           61.8
13001 2017    GEORGIA     APPLING           32.2
08001 2017   COLORADO       ADAMS           35.1
21001 2017   KENTUCKY       ADAIR           44.5
18001 2017    INDIANA       ADAMS           54.1
19001 2017       IOWA       ADAIR           57.7
10001 2017   DELAWARE        KENT           43.2
```

04001	2017	ARIZONA	APACHE	29.8
05001	2017	ARKANSAS	ARKANSAS	45.8
06001	2017	CALIFORNIA	ALAMEDA	29.4
16001	2017	IDAHO	ADA	34.4
12001	2017	FLORIDA	ALACHUA	26.9
22001	2017	LOUISIANA	ACADIA	36.9
20001	2017	KANSAS	ANDERSON	45.0
17001	2017	ILLINOIS	ADAMS	56.1
01001	2017	ALABAMA	AUTAGA	48.0
13001	2018	GEORGIA	APPLING	27.3
08001	2018	COLORADO	ADAMS	29.9
21001	2018	KENTUCKY	ADAIR	38.2
18001	2018	INDIANA	ADAMS	52.9
19001	2018	IOWA	ADAIR	52.0
10001	2018	DELAWARE	KENT	41.4
04001	2018	ARIZONA	APACHE	25.3
05001	2018	ARKANSAS	ARKANSAS	43.5
06001	2018	CALIFORNIA	ALAMEDA	28.6
16001	2018	IDAHO	ADA	31.6
12001	2018	FLORIDA	ALACHUA	27.7
22001	2018	LOUISIANA	ACADIA	37.7
20001	2018	KANSAS	ANDERSON	43.3
17001	2018	ILLINOIS	ADAMS	51.0
01001	2018	ALABAMA	AUTAGA	65.4
13001	2019	GEORGIA	APPLING	31.7
08001	2019	COLORADO	ADAMS	27.0
21001	2019	KENTUCKY	ADAIR	36.0
18001	2019	INDIANA	ADAMS	47.6
19001	2019	IOWA	ADAIR	55.0
10001	2019	DELAWARE	KENT	45.0
04001	2019	ARIZONA	APACHE	23.9
05001	2019	ARKANSAS	ARKANSAS	38.1
06001	2019	CALIFORNIA	ALAMEDA	28.7
16001	2019	IDAHO	ADA	34.0
12001	2019	FLORIDA	ALACHUA	28.1
22001	2019	LOUISIANA	ACADIA	39.2
20001	2019	KANSAS	ANDERSON	43.1
17001	2019	ILLINOIS	ADAMS	48.6
01001	2019	ALABAMA	AUTAGA	39.3
13001	2020	GEORGIA	APPLING	34.2
08001	2020	COLORADO	ADAMS	33.5
21001	2020	KENTUCKY	ADAIR	42.5
18001	2020	INDIANA	ADAMS	55.3
19001	2020	IOWA	ADAIR	54.1
10001	2020	DELAWARE	KENT	44.2
04001	2020	ARIZONA	APACHE	28.8
05001	2020	ARKANSAS	ARKANSAS	45.8
06001	2020	CALIFORNIA	ALAMEDA	30.5

```
16001 2020       IDAHO        ADA            35.5
12001 2020     FLORIDA    ALACHUA            29.4
22001 2020   LOUISIANA     ACADIA            37.3
20001 2020      KANSAS   ANDERSON            46.5
17001 2020    ILLINOIS      ADAMS            52.8
01001 2020     ALABAMA     AUTAGA            47.2
13001 2021     GEORGIA    APPLING            33.6
08001 2021    COLORADO      ADAMS            31.0
21001 2021    KENTUCKY      ADAIR            44.2
18001 2021     INDIANA      ADAMS            58.3
19001 2021        IOWA      ADAIR            58.5
10001 2021    DELAWARE       KENT            47.6
04001 2021     ARIZONA     APACHE            28.4
05001 2021    ARKANSAS   ARKANSAS            47.2
06001 2021  CALIFORNIA    ALAMEDA            32.6
16001 2021       IDAHO        ADA            37.5
12001 2021     FLORIDA    ALACHUA            27.9
22001 2021   LOUISIANA     ACADIA            34.8
20001 2021      KANSAS   ANDERSON            46.2
17001 2021    ILLINOIS      ADAMS            57.5
01001 2021     ALABAMA     AUTAGA            54.3
13001 2022     GEORGIA    APPLING            31.6
08001 2022    COLORADO      ADAMS            32.6
21001 2022    KENTUCKY      ADAIR            40.6
18001 2022     INDIANA      ADAMS            54.5
19001 2022        IOWA      ADAIR            57.3
10001 2022    DELAWARE       KENT            41.6
04001 2022     ARIZONA     APACHE            26.6
05001 2022    ARKANSAS   ARKANSAS            45.2
06001 2022  CALIFORNIA    ALAMEDA            30.9
16001 2022       IDAHO        ADA            38.0
12001 2022     FLORIDA    ALACHUA            27.0
22001 2022   LOUISIANA     ACADIA            36.5
20001 2022      KANSAS   ANDERSON            43.4
17001 2022    ILLINOIS      ADAMS            53.2
01001 2022     ALABAMA     AUTAGA            47.1
```

[6]:
```python
# Define training universe splits for each model phase
#
# Phase 1 - Weather encoder (Bi-LSTM)
#   Source : usda_index - all USDA soybean records 2016-2022 (3,430 records,
↪634 counties)
#   Split  : 80/20 stratified by 5-bin yield quintile
#
# Phase 2 - Satellite encoder (ResNet18 + LSTM)
# Phase 3 - GMU fine-tuning (gated dual-stream)
#   Source : sentinel_labels - USDA records that match Sentinel FIPS codes
```

```
#   Split  : 80/20 stratified by 2-bin yield median (constrained by small n)
#   Note   : If sentinel_labels is empty (no USDA-Sentinel FIPS overlap on␣
 ↪disk),
#             Phase 2 and 3 are skipped and documented as a data gap.

from sklearn.model_selection import train_test_split

#   Phase 1: Weather branch
weather_df = usda_index.copy().reset_index(drop=True)
weather_df["yield_quintile"] = pd.qcut(
    weather_df["yield_bu_acre"], q=5, labels=False, duplicates="drop"
)
w_train_idx, w_test_idx = train_test_split(
    weather_df.index,
    test_size=0.20,
    random_state=SEED,
    stratify=weather_df["yield_quintile"]
)
weather_df["split"] = "test"
weather_df.loc[w_train_idx, "split"] = "train"
weather_train = weather_df[weather_df["split"] == "train"].
 ↪drop(columns=["yield_quintile", "split"]).reset_index(drop=True)
weather_test  = weather_df[weather_df["split"] == "test"].
 ↪drop(columns=["yield_quintile", "split"]).reset_index(drop=True)


print("PHASE 1 - WEATHER BRANCH (full USDA x HRRR, 2016-2022)")
print(f"Total records           : {len(weather_df)}")
print(f"Train (80%)             : {len(weather_train)}")
print(f"Test  (20%)             : {len(weather_test)}")
print(f"Unique counties - train : {weather_train['fips'].nunique()}")
print(f"Years in train          : {sorted(weather_train['year'].unique())}")
print(f"Yield range - train     : {weather_train['yield_bu_acre'].min():.1f} to␣
 ↪{weather_train['yield_bu_acre'].max():.1f} BU/ACRE")
print(f"Yield range - test      : {weather_test['yield_bu_acre'].min():.1f} to␣
 ↪{weather_test['yield_bu_acre'].max():.1f} BU/ACRE")
print()

#   Phase 2 & 3: Satellite + GMU
# Attach Sentinel tile counts to whatever sentinel_labels contains
sentinel_summary = (
    sentinel_index.groupby(["fips", "year"])
    .agg(total_dates=("n_dates", "sum"), total_tiles=("n_tiles", "sum"))
    .reset_index()
)
```

```python
print("PHASE 2 & 3 - SATELLITE / GMU")


if len(sentinel_labels) == 0:
    # No USDA yield labels match the Sentinel FIPS codes on disk.
    # Phase 2 and 3 cannot proceed. Create empty placeholders so downstream
    # cells that reference these variables do not raise NameError.
    multimodal_index = pd.DataFrame(columns=["fips", "year", "state_name",
                                              "county_name", "yield_bu_acre",
                                              "production_bu", "total_dates",
                                              "total_tiles", "split"])
    train_set = multimodal_index.copy()
    test_set  = multimodal_index.copy()
    print("Sentinel-USDA label intersection is empty.")
    print()
    print("codes that appear in the Sentinel imagery:")
    print(sorted(sentinel_index["fips"].unique()))
else:
    mm_df = sentinel_labels.copy().reset_index(drop=True)
    # Normalise types so fips/year match usda_index (sentinel_index keys are␣
 ↪strings)
    sentinel_summary["fips"] = sentinel_summary["fips"].astype(str).str.zfill(5)
    sentinel_summary["year"] = sentinel_summary["year"].astype(int)
    mm_df["fips"] = mm_df["fips"].astype(str).str.zfill(5)
    mm_df["year"] = mm_df["year"].astype(int)
    mm_df = mm_df.merge(sentinel_summary, on=["fips", "year"], how="left")

    n_mm = len(mm_df)
    # Use 2-bin stratification when test set is too small for more bins
    mm_df["yield_bin"] = (mm_df["yield_bu_acre"] >= mm_df["yield_bu_acre"].
 ↪median()).astype(int)

    if n_mm < 5:
        # Too few records to split - use all as train, none as test
        mm_df["split"] = "train"
        print(f"WARNING: Only {n_mm} records in sentinel_labels. Too few for an␣
 ↪80/20 split.")
        print("All records assigned to train. Test set is empty.")
    else:
        mm_train_idx, mm_test_idx = train_test_split(
            mm_df.index,
            test_size=0.20,
            random_state=SEED,
            stratify=mm_df["yield_bin"]
        )
        mm_df["split"] = "test"
        mm_df.loc[mm_train_idx, "split"] = "train"
```

```
    multimodal_index = mm_df.drop(columns=["yield_bin"]).reset_index(drop=True)
    train_set = multimodal_index[multimodal_index["split"] == "train"].
↪reset_index(drop=True)
    test_set  = multimodal_index[multimodal_index["split"] == "test"].
↪reset_index(drop=True)

    print(f"Total records          : {len(multimodal_index)}")
    print(f"Train (80%)            : {len(train_set)}")
    print(f"Test  (20%)            : {len(test_set)}")
    print(f"Approx train tiles     : {int(train_set['total_tiles'].sum())}")
    print(f"Approx test tiles      : {int(test_set['total_tiles'].sum())}")
    print()
    print("Train set:")
    print(train_set[["fips", "state_name", "county_name", "year",
                     "yield_bu_acre", "total_tiles"]].to_string(index=False))
    print()
    print("Test set:")
    print(test_set[["fips", "state_name", "county_name", "year",
                    "yield_bu_acre", "total_tiles"]].to_string(index=False))

print()
print("COVERAGE SUMMARY")
print(f"Weather branch  : {len(weather_train)} train records, "
      f"{weather_train['fips'].nunique()} counties, "
      f"{weather_train['state_name'].nunique()} states, "
      f"years 2016-2022")
if len(train_set) > 0:
    print(f"Satellite branch: {len(train_set)} train county-years, "
          f"~{int(train_set['total_tiles'].sum())} tiles, "
          f"years {sorted(train_set['year'].unique())}")

else:
    print("Satellite branch: 0 records - data gap (no USDA-Sentinel FIPS␣
↪overlap on disk)")
print(f"Split seed: {SEED}, method: 80/20 stratified random")
```

```
PHASE 1 - WEATHER BRANCH (full USDA x HRRR, 2016-2022)
Total records           : 3535
Train (80%)             : 2828
Test  (20%)             : 707
Unique counties - train : 638
Years in train          : ['2016', '2017', '2018', '2019', '2020', '2021',
'2022']
Yield range - train     : 9.3 to 80.4 BU/ACRE
Yield range - test      : 9.4 to 75.2 BU/ACRE
```

```
PHASE 2 & 3 - SATELLITE / GMU
Total records         : 105
Train (80%)           : 84
Test  (20%)           : 21
Approx train tiles    : 3420
Approx test tiles     : 464

Train set:
 fips state_name county_name  year  yield_bu_acre  total_tiles
13001    GEORGIA     APPLING  2016           31.4          NaN
08001   COLORADO       ADAMS  2016           32.8          NaN
21001   KENTUCKY       ADAIR  2016           43.9          NaN
19001       IOWA       ADAIR  2016           60.8          NaN
10001   DELAWARE        KENT  2016           44.5          NaN
04001    ARIZONA      APACHE  2016           30.1          NaN
05001   ARKANSAS    ARKANSAS  2016           48.4          NaN
06001 CALIFORNIA     ALAMEDA  2016           32.7          NaN
16001      IDAHO         ADA  2016           38.7          NaN
12001    FLORIDA     ALACHUA  2016           29.1          NaN
22001  LOUISIANA      ACADIA  2016           37.6          NaN
20001     KANSAS    ANDERSON  2016           45.4          NaN
08001   COLORADO       ADAMS  2017           35.1          NaN
21001   KENTUCKY       ADAIR  2017           44.5          NaN
18001    INDIANA       ADAMS  2017           54.1          NaN
19001       IOWA       ADAIR  2017           57.7          NaN
10001   DELAWARE        KENT  2017           43.2          NaN
04001    ARIZONA      APACHE  2017           29.8          NaN
05001   ARKANSAS    ARKANSAS  2017           45.8          NaN
16001      IDAHO         ADA  2017           34.4          NaN
12001    FLORIDA     ALACHUA  2017           26.9          NaN
22001  LOUISIANA      ACADIA  2017           36.9          NaN
17001   ILLINOIS       ADAMS  2017           56.1          NaN
01001    ALABAMA      AUTAGA  2017           48.0          NaN
08001   COLORADO       ADAMS  2018           29.9          NaN
21001   KENTUCKY       ADAIR  2018           38.2          NaN
19001       IOWA       ADAIR  2018           52.0          NaN
10001   DELAWARE        KENT  2018           41.4          NaN
04001    ARIZONA      APACHE  2018           25.3          NaN
16001      IDAHO         ADA  2018           31.6          NaN
22001  LOUISIANA      ACADIA  2018           37.7          NaN
20001     KANSAS    ANDERSON  2018           43.3          NaN
17001   ILLINOIS       ADAMS  2018           51.0          NaN
01001    ALABAMA      AUTAGA  2018           65.4          NaN
08001   COLORADO       ADAMS  2019           27.0          NaN
21001   KENTUCKY       ADAIR  2019           36.0          NaN
18001    INDIANA       ADAMS  2019           47.6          NaN
19001       IOWA       ADAIR  2019           55.0          NaN
10001   DELAWARE        KENT  2019           45.0          NaN
```

```
04001    ARIZONA      APACHE  2019          23.9         NaN
05001   ARKANSAS    ARKANSAS  2019          38.1         NaN
06001 CALIFORNIA     ALAMEDA  2019          28.7         NaN
12001    FLORIDA     ALACHUA  2019          28.1         NaN
22001  LOUISIANA      ACADIA  2019          39.2         NaN
20001     KANSAS    ANDERSON  2019          43.1         NaN
01001    ALABAMA      AUTAGA  2019          39.3         NaN
13001    GEORGIA     APPLING  2020          34.2         NaN
08001   COLORADO       ADAMS  2020          33.5         NaN
21001   KENTUCKY       ADAIR  2020          42.5         NaN
18001    INDIANA       ADAMS  2020          55.3         NaN
19001       IOWA       ADAIR  2020          54.1         NaN
10001   DELAWARE        KENT  2020          44.2         NaN
04001    ARIZONA      APACHE  2020          28.8         NaN
05001   ARKANSAS    ARKANSAS  2020          45.8         NaN
06001 CALIFORNIA     ALAMEDA  2020          30.5         NaN
16001      IDAHO         ADA  2020          35.5         NaN
12001    FLORIDA     ALACHUA  2020          29.4         NaN
22001  LOUISIANA      ACADIA  2020          37.3         NaN
17001   ILLINOIS       ADAMS  2020          52.8         NaN
01001    ALABAMA      AUTAGA  2020          47.2         NaN
08001   COLORADO       ADAMS  2021          31.0       120.0
21001   KENTUCKY       ADAIR  2021          44.2        52.0
18001    INDIANA       ADAMS  2021          58.3        24.0
19001       IOWA       ADAIR  2021          58.5        48.0
10001   DELAWARE        KENT  2021          47.6        80.0
04001    ARIZONA      APACHE  2021          28.4         NaN
05001   ARKANSAS    ARKANSAS  2021          47.2       148.0
06001 CALIFORNIA     ALAMEDA  2021          32.6       120.0
12001    FLORIDA     ALACHUA  2021          27.9       116.0
22001  LOUISIANA      ACADIA  2021          34.8        76.0
20001     KANSAS    ANDERSON  2021          46.2        32.0
01001    ALABAMA      AUTAGA  2021          54.3        60.0
13001    GEORGIA     APPLING  2022          31.6        76.0
08001   COLORADO       ADAMS  2022          32.6       120.0
21001   KENTUCKY       ADAIR  2022          40.6        52.0
19001       IOWA       ADAIR  2022          57.3        48.0
10001   DELAWARE        KENT  2022          41.6        80.0
04001    ARIZONA      APACHE  2022          26.6      1524.0
05001   ARKANSAS    ARKANSAS  2022          45.2       148.0
06001 CALIFORNIA     ALAMEDA  2022          30.9       120.0
16001      IDAHO         ADA  2022          38.0       132.0
12001    FLORIDA     ALACHUA  2022          27.0       116.0
20001     KANSAS    ANDERSON  2022          43.4        32.0
17001   ILLINOIS       ADAMS  2022          53.2        96.0

Test set:
 fips state_name county_name  year  yield_bu_acre  total_tiles
```

```
18001    INDIANA       ADAMS  2016           59.3          NaN
17001   ILLINOIS       ADAMS  2016           55.5          NaN
01001    ALABAMA      AUTAGA  2016           61.8          NaN
13001    GEORGIA     APPLING  2017           32.2          NaN
06001 CALIFORNIA     ALAMEDA  2017           29.4          NaN
20001     KANSAS    ANDERSON  2017           45.0          NaN
13001    GEORGIA     APPLING  2018           27.3          NaN
18001    INDIANA       ADAMS  2018           52.9          NaN
05001   ARKANSAS    ARKANSAS  2018           43.5          NaN
06001 CALIFORNIA     ALAMEDA  2018           28.6          NaN
12001    FLORIDA     ALACHUA  2018           27.7          NaN
13001    GEORGIA     APPLING  2019           31.7          NaN
16001      IDAHO         ADA  2019           34.0          NaN
17001   ILLINOIS       ADAMS  2019           48.6          NaN
20001     KANSAS    ANDERSON  2020           46.5          NaN
13001    GEORGIA     APPLING  2021           33.6         76.0
16001      IDAHO         ADA  2021           37.5        132.0
17001   ILLINOIS       ADAMS  2021           57.5         96.0
18001    INDIANA       ADAMS  2022           54.5         24.0
22001  LOUISIANA      ACADIA  2022           36.5         76.0
01001    ALABAMA      AUTAGA  2022           47.1         60.0

COVERAGE SUMMARY
Weather branch  : 2828 train records, 638 counties, 15 states, years 2016-2022
Satellite branch: 84 train county-years, ~3420 tiles, years [np.int64(2016),
np.int64(2017), np.int64(2018), np.int64(2019), np.int64(2020), np.int64(2021),
np.int64(2022)]
Split seed: 42, method: 80/20 stratified random
 fips state_name county_name  year  yield_bu_acre  total_tiles
13001    GEORGIA     APPLING  2016           31.4          NaN
08001   COLORADO       ADAMS  2016           32.8          NaN
21001   KENTUCKY       ADAIR  2016           43.9          NaN
19001       IOWA       ADAIR  2016           60.8          NaN
10001   DELAWARE        KENT  2016           44.5          NaN
04001    ARIZONA      APACHE  2016           30.1          NaN
05001   ARKANSAS    ARKANSAS  2016           48.4          NaN
06001 CALIFORNIA     ALAMEDA  2016           32.7          NaN
16001      IDAHO         ADA  2016           38.7          NaN
12001    FLORIDA     ALACHUA  2016           29.1          NaN
22001  LOUISIANA      ACADIA  2016           37.6          NaN
20001     KANSAS    ANDERSON  2016           45.4          NaN
08001   COLORADO       ADAMS  2017           35.1          NaN
21001   KENTUCKY       ADAIR  2017           44.5          NaN
18001    INDIANA       ADAMS  2017           54.1          NaN
19001       IOWA       ADAIR  2017           57.7          NaN
10001   DELAWARE        KENT  2017           43.2          NaN
04001    ARIZONA      APACHE  2017           29.8          NaN
05001   ARKANSAS    ARKANSAS  2017           45.8          NaN
```

16001	IDAHO	ADA	2017	34.4	NaN
12001	FLORIDA	ALACHUA	2017	26.9	NaN
22001	LOUISIANA	ACADIA	2017	36.9	NaN
17001	ILLINOIS	ADAMS	2017	56.1	NaN
01001	ALABAMA	AUTAGA	2017	48.0	NaN
08001	COLORADO	ADAMS	2018	29.9	NaN
21001	KENTUCKY	ADAIR	2018	38.2	NaN
19001	IOWA	ADAIR	2018	52.0	NaN
10001	DELAWARE	KENT	2018	41.4	NaN
04001	ARIZONA	APACHE	2018	25.3	NaN
16001	IDAHO	ADA	2018	31.6	NaN
22001	LOUISIANA	ACADIA	2018	37.7	NaN
20001	KANSAS	ANDERSON	2018	43.3	NaN
17001	ILLINOIS	ADAMS	2018	51.0	NaN
01001	ALABAMA	AUTAGA	2018	65.4	NaN
08001	COLORADO	ADAMS	2019	27.0	NaN
21001	KENTUCKY	ADAIR	2019	36.0	NaN
18001	INDIANA	ADAMS	2019	47.6	NaN
19001	IOWA	ADAIR	2019	55.0	NaN
10001	DELAWARE	KENT	2019	45.0	NaN
04001	ARIZONA	APACHE	2019	23.9	NaN
05001	ARKANSAS	ARKANSAS	2019	38.1	NaN
06001	CALIFORNIA	ALAMEDA	2019	28.7	NaN
12001	FLORIDA	ALACHUA	2019	28.1	NaN
22001	LOUISIANA	ACADIA	2019	39.2	NaN
20001	KANSAS	ANDERSON	2019	43.1	NaN
01001	ALABAMA	AUTAGA	2019	39.3	NaN
13001	GEORGIA	APPLING	2020	34.2	NaN
08001	COLORADO	ADAMS	2020	33.5	NaN
21001	KENTUCKY	ADAIR	2020	42.5	NaN
18001	INDIANA	ADAMS	2020	55.3	NaN
19001	IOWA	ADAIR	2020	54.1	NaN
10001	DELAWARE	KENT	2020	44.2	NaN
04001	ARIZONA	APACHE	2020	28.8	NaN
05001	ARKANSAS	ARKANSAS	2020	45.8	NaN
06001	CALIFORNIA	ALAMEDA	2020	30.5	NaN
16001	IDAHO	ADA	2020	35.5	NaN
12001	FLORIDA	ALACHUA	2020	29.4	NaN
22001	LOUISIANA	ACADIA	2020	37.3	NaN
17001	ILLINOIS	ADAMS	2020	52.8	NaN
01001	ALABAMA	AUTAGA	2020	47.2	NaN
08001	COLORADO	ADAMS	2021	31.0	120.0
21001	KENTUCKY	ADAIR	2021	44.2	52.0
18001	INDIANA	ADAMS	2021	58.3	24.0
19001	IOWA	ADAIR	2021	58.5	48.0
10001	DELAWARE	KENT	2021	47.6	80.0
04001	ARIZONA	APACHE	2021	28.4	NaN
05001	ARKANSAS	ARKANSAS	2021	47.2	148.0

```
06001 CALIFORNIA     ALAMEDA  2021           32.6        120.0
12001    FLORIDA     ALACHUA  2021           27.9        116.0
22001  LOUISIANA      ACADIA  2021           34.8         76.0
20001     KANSAS    ANDERSON  2021           46.2         32.0
01001    ALABAMA      AUTAGA  2021           54.3         60.0
13001    GEORGIA     APPLING  2022           31.6         76.0
08001   COLORADO       ADAMS  2022           32.6        120.0
21001   KENTUCKY       ADAIR  2022           40.6         52.0
19001       IOWA       ADAIR  2022           57.3         48.0
10001   DELAWARE        KENT  2022           41.6         80.0
04001    ARIZONA      APACHE  2022           26.6       1524.0
05001   ARKANSAS    ARKANSAS  2022           45.2        148.0
06001 CALIFORNIA     ALAMEDA  2022           30.9        120.0
16001      IDAHO         ADA  2022           38.0        132.0
12001    FLORIDA     ALACHUA  2022           27.0        116.0
20001     KANSAS    ANDERSON  2022           43.4         32.0
17001   ILLINOIS       ADAMS  2022           53.2         96.0

Test set:
 fips state_name county_name  year  yield_bu_acre  total_tiles
18001    INDIANA       ADAMS  2016           59.3          NaN
17001   ILLINOIS       ADAMS  2016           55.5          NaN
01001    ALABAMA      AUTAGA  2016           61.8          NaN
13001    GEORGIA     APPLING  2017           32.2          NaN
06001 CALIFORNIA     ALAMEDA  2017           29.4          NaN
20001     KANSAS    ANDERSON  2017           45.0          NaN
13001    GEORGIA     APPLING  2018           27.3          NaN
18001    INDIANA       ADAMS  2018           52.9          NaN
05001   ARKANSAS    ARKANSAS  2018           43.5          NaN
06001 CALIFORNIA     ALAMEDA  2018           28.6          NaN
12001    FLORIDA     ALACHUA  2018           27.7          NaN
13001    GEORGIA     APPLING  2019           31.7          NaN
16001      IDAHO         ADA  2019           34.0          NaN
17001   ILLINOIS       ADAMS  2019           48.6          NaN
20001     KANSAS    ANDERSON  2020           46.5          NaN
13001    GEORGIA     APPLING  2021           33.6         76.0
16001      IDAHO         ADA  2021           37.5        132.0
17001   ILLINOIS       ADAMS  2021           57.5         96.0
18001    INDIANA       ADAMS  2022           54.5         24.0
22001  LOUISIANA      ACADIA  2022           36.5         76.0
01001    ALABAMA      AUTAGA  2022           47.1         60.0

COVERAGE SUMMARY
Weather branch  : 2828 train records, 638 counties, 15 states, years 2016-2022
Satellite branch: 84 train county-years, ~3420 tiles, years [np.int64(2016),
np.int64(2017), np.int64(2018), np.int64(2019), np.int64(2020), np.int64(2021),
np.int64(2022)]
Split seed: 42, method: 80/20 stratified random
```

1 USDA Soybean EDA (Target Variable)

This part fully characterises the regression target - county-level soybean yield in BU/ACRE. All analysis uses the usda_index and multimodal_index tables built earlier.

- Full county inventory, data quality check, FIPS resolution
- Target distribution: histogram, boxplot, normality test
- Year-over-year yield change per county
- Geographic choropleth map with Sentinel coverage overlay
- Freeze and export final label tables

[7]:
```
#Full county inventory, quality check, FIPS resolution



print("USDA INDEX - FULL COVERAGE SUMMARY")

print(f"Total records (all years, all counties): {len(usda_index)}")
print(f"Unique FIPS codes:                        {usda_index['fips'].
 ↪nunique()}")
print(f"Years with data:                          {sorted(usda_index['year'].
 ↪astype(str).unique())}")
print(f"States with data:                        ␣
 ↪{sorted(usda_index['state_name'].unique())}")
print()

# Completeness check: every FIPS must have both years
pivot = usda_index.pivot_table(index="fips", columns="year",␣
 ↪values="yield_bu_acre", aggfunc="first")
print("County x Year yield matrix (BU/ACRE):")
print(pivot.to_string())
print()

missing_either_year = pivot[pivot.isnull().any(axis=1)]
if len(missing_either_year) > 0:
    print("WARNING: Counties missing yield for at least one year:")
    print(missing_either_year.to_string())
else:
    print("All counties with USDA data have at least one year of yield records.
 ↪")
print()

# FIPS resolution and intersection report
usda_fips     = set(usda_index["fips"].unique())
sentinel_fips = set(sentinel_index["fips"].unique())
overlap_fips  = usda_fips & sentinel_fips
```

```
print("INTERSECTION REPORT")
print(f"USDA counties:                    {len(usda_fips)}")
print(f"Sentinel counties:                {len(sentinel_fips)}")
print(f"Multimodal intersection:          {len(overlap_fips)}")
print(f"Sentinel-only (no USDA):          {len(sentinel_fips - usda_fips)}")
print(f"  -> {sorted(sentinel_fips - usda_fips)}")
print()

# Show the full multimodal_index with all metadata
print("Full multimodal_index:")
print(multimodal_index[["fips","state_name","county_name","year","yield_bu_acre",
                          "production_bu","total_tiles","split"]].
 ↪to_string(index=False))
```

```
USDA INDEX - FULL COVERAGE SUMMARY
Total records (all years, all counties): 3535
Unique FIPS codes:                        649
Years with data:                          ['2016', '2017', '2018', '2019',
'2020', '2021', '2022']
States with data:                         ['ALABAMA', 'ARIZONA', 'ARKANSAS',
'CALIFORNIA', 'COLORADO', 'DELAWARE', 'FLORIDA', 'GEORGIA', 'IDAHO', 'ILLINOIS',
'INDIANA', 'IOWA', 'KANSAS', 'KENTUCKY', 'LOUISIANA']

County x Year yield matrix (BU/ACRE):
year   2016  2017  2018  2019  2020  2021  2022
fips
01001  61.8  48.0  65.4  39.3  47.2  54.3  47.1
01003  41.8  42.0  45.4   NaN  37.2  44.3  37.1
01009  25.0  48.6  39.3  35.3  47.0  42.5  37.4
01015   NaN   NaN   NaN   NaN  35.1  44.7  45.5
01019  20.8  42.1  45.3  30.1  36.3  45.1  39.2
01031  43.6   NaN   NaN   NaN   NaN  31.0   NaN
01033  31.2  48.2  36.1  38.7  42.2   NaN  35.6
01035  33.8   NaN   NaN   NaN   NaN   NaN   NaN
01039  32.6   NaN   NaN   NaN  35.1   NaN   NaN
01043  30.3  55.3  44.6  42.8  51.1  50.1  47.4
01047  32.4  43.2  42.0  26.1  42.7  58.6  41.3
01049  29.2  46.1  49.9  30.6  39.9  44.9  48.8
01051  20.3  39.9  42.8   NaN   NaN  50.4   NaN
01053  42.4  46.2   NaN  22.3   NaN  50.7   NaN
01055  25.6  52.7   NaN  30.4  41.9  34.7  46.4
01057  28.7  25.2  26.4  24.9  38.8  40.6  42.0
01059  39.3  41.9  37.5  37.7  38.8   NaN   NaN
01061  33.8  48.4   NaN   NaN   NaN   NaN   NaN
01065  23.0   NaN   NaN   NaN   NaN   NaN   NaN
01069  23.3  47.2  28.4  46.2  48.7   NaN  42.6
01071  21.5  40.3  39.0  29.1  40.1  42.2  45.5
01075  26.2  27.9   NaN   NaN   NaN   NaN   NaN
```

01077	29.9	46.3	35.4	38.6	39.6	47.2	42.5
01079	33.0	54.3	39.4	39.3	43.4	51.5	42.9
01083	31.8	50.3	43.3	40.5	36.9	45.4	37.1
01085	29.3	NaN	NaN	NaN	NaN	NaN	NaN
01087	NaN	32.1	NaN	NaN	44.3	30.8	42.5
01089	37.6	42.4	38.2	39.5	42.1	49.6	47.1
01091	30.2	31.4	NaN	NaN	30.0	NaN	NaN
01093	NaN	NaN	NaN	NaN	48.0	NaN	46.2
01095	NaN	43.2	36.1	31.0	42.4	43.8	43.1
01099	38.3	NaN	NaN	NaN	NaN	49.1	NaN
01103	35.3	47.0	NaN	39.5	47.4	44.2	33.4
01105	36.5	39.9	37.6	NaN	37.4	39.1	39.3
01109	40.1	NaN	NaN	51.6	NaN	NaN	33.8
01117	39.7	NaN	NaN	NaN	NaN	NaN	NaN
01119	30.7	40.7	43.1	38.0	NaN	45.3	NaN
01121	NaN	41.9	38.2	NaN	46.7	NaN	46.3
01125	38.5	51.3	46.3	31.8	41.2	53.0	39.2
01127	NaN	NaN	38.5	32.4	NaN	NaN	NaN
04001	30.1	29.8	25.3	23.9	28.8	28.4	26.6
05001	48.4	45.8	43.5	38.1	45.8	47.2	45.2
05003	58.2	56.8	54.9	50.6	55.4	63.7	59.7
05007	NaN	NaN	NaN	NaN	NaN	41.2	NaN
05017	52.7	51.2	52.0	54.1	53.9	55.2	54.1
05021	46.8	54.8	50.8	48.0	53.3	44.0	54.7
05029	NaN	39.4	37.0	NaN	33.2	32.9	32.0
05031	48.8	53.1	50.0	46.3	48.7	NaN	46.5
05035	43.8	52.7	NaN	46.1	50.4	52.3	53.2
05037	47.7	51.1	53.3	52.0	50.4	53.3	51.9
05041	55.7	56.7	59.5	59.8	57.3	52.1	58.8
05043	53.5	54.5	56.9	57.5	56.7	58.6	61.9
05045	37.2	42.0	43.2	NaN	33.6	33.2	39.9
05047	NaN	35.5	NaN	NaN	37.3	NaN	NaN
05055	43.7	51.0	41.5	45.8	46.9	NaN	52.0
05063	38.5	43.4	NaN	40.9	43.1	NaN	42.9
05067	39.2	42.2	40.1	40.3	40.6	46.8	38.4
05069	52.2	53.1	56.5	54.3	55.8	57.0	57.6
05071	NaN	NaN	NaN	NaN	34.7	NaN	NaN
05073	NaN	NaN	NaN	NaN	53.1	NaN	NaN
05075	35.7	42.0	40.4	39.9	42.9	NaN	48.7
05077	43.5	48.5	44.4	NaN	55.6	53.5	57.7
05079	56.2	57.8	55.4	57.2	54.6	53.2	60.8
05081	NaN	NaN	NaN	30.2	NaN	NaN	31.3
05083	NaN	NaN	NaN	NaN	37.5	35.3	NaN
05085	46.3	42.5	NaN	48.2	47.6	47.5	49.6
05093	48.9	56.7	50.5	47.6	53.4	NaN	59.3
05095	43.3	47.2	47.3	47.7	50.9	45.1	44.2
05107	48.9	52.9	50.5	NaN	55.5	58.8	55.4
05111	51.0	55.5	56.4	51.0	51.9	54.2	49.2

05115	NaN	NaN	NaN	NaN	NaN	NaN	43.8
05117	50.0	50.9	53.6	44.0	50.0	55.5	50.6
05119	NaN	46.7	44.3	NaN	35.9	38.8	35.6
05121	37.9	48.2	NaN	42.5	45.3	NaN	NaN
05123	44.5	49.7	46.2	50.3	50.2	51.9	52.0
05131	NaN	NaN	NaN	NaN	33.3	NaN	NaN
05145	37.9	42.6	NaN	43.0	47.9	45.8	36.4
05147	35.7	44.6	45.8	43.8	50.8	48.4	40.1
05149	NaN	42.6	NaN	NaN	42.5	38.2	34.6
06001	32.7	29.4	28.6	28.7	30.5	32.6	30.9
08001	32.8	35.1	29.9	27.0	33.5	31.0	32.6
10001	44.5	43.2	41.4	45.0	44.2	47.6	41.6
10003	47.0	NaN	45.8	47.2	48.4	46.9	36.9
10005	38.3	49.5	41.3	44.8	47.8	49.7	42.5
12001	29.1	26.9	27.7	28.1	29.4	27.9	27.0
13001	31.4	32.2	27.3	31.7	34.2	33.6	31.6
13005	NaN	NaN	NaN	NaN	37.7	36.8	NaN
13015	17.0	52.3	NaN	12.6	NaN	NaN	NaN
13027	49.4	45.7	NaN	36.2	NaN	48.7	47.8
13031	20.7	30.8	NaN	NaN	NaN	NaN	NaN
13033	32.4	34.0	NaN	NaN	51.3	41.1	38.3
13043	21.6	NaN	NaN	NaN	NaN	NaN	NaN
13055	21.4	41.5	37.2	40.4	42.0	47.1	NaN
13069	36.4	36.9	NaN	25.2	NaN	NaN	46.0
13071	38.8	44.8	43.5	NaN	NaN	NaN	NaN
13075	NaN	NaN	NaN	NaN	NaN	48.8	48.4
13087	43.3	NaN	NaN	NaN	NaN	NaN	NaN
13091	29.3	NaN	NaN	NaN	NaN	NaN	NaN
13093	NaN	42.3	NaN	NaN	42.0	NaN	NaN
13099	22.7	NaN	NaN	NaN	NaN	NaN	NaN
13105	NaN	39.4	45.5	24.7	48.7	47.9	31.8
13107	34.4	51.0	29.9	23.8	39.5	36.3	31.4
13115	15.3	53.0	NaN	NaN	NaN	42.7	NaN
13119	9.8	37.3	NaN	NaN	NaN	NaN	NaN
13129	NaN	NaN	NaN	NaN	NaN	NaN	44.1
13131	36.6	40.8	NaN	NaN	51.3	NaN	52.3
13147	16.0	36.4	41.2	23.2	52.3	NaN	NaN
13151	NaN	42.1	NaN	NaN	NaN	NaN	NaN
13153	40.7	33.9	NaN	NaN	NaN	NaN	49.9
13161	NaN	NaN	NaN	NaN	NaN	NaN	47.4
13163	28.7	NaN	38.1	34.0	46.2	36.1	NaN
13165	29.6	53.1	32.9	NaN	NaN	NaN	NaN
13167	21.3	NaN	NaN	NaN	NaN	58.2	NaN
13175	NaN	NaN	NaN	NaN	37.1	40.9	35.8
13177	NaN	NaN	35.6	NaN	NaN	40.2	NaN
13185	37.5	39.8	NaN	NaN	36.7	36.9	NaN
13193	37.1	36.7	NaN	28.4	59.7	47.1	NaN
13195	16.1	NaN	NaN	NaN	30.8	NaN	NaN

13197	16.3	35.3	NaN	NaN	NaN	40.1	41.3
13205	41.2	47.9	41.6	NaN	NaN	NaN	NaN
13209	35.3	45.4	NaN	NaN	NaN	NaN	NaN
13211	36.3	NaN	NaN	NaN	NaN	NaN	NaN
13213	23.2	NaN	NaN	NaN	NaN	NaN	NaN
13221	NaN	NaN	35.3	NaN	NaN	NaN	NaN
13225	36.8	NaN	26.9	NaN	34.8	NaN	38.9
13229	23.9	NaN	NaN	NaN	NaN	NaN	NaN
13233	21.8	NaN	NaN	NaN	NaN	NaN	NaN
13243	34.1	53.5	49.1	NaN	57.1	NaN	41.9
13251	29.7	42.7	NaN	NaN	36.1	NaN	NaN
13253	49.1	50.8	38.7	NaN	NaN	NaN	NaN
13267	24.1	NaN	NaN	NaN	NaN	NaN	NaN
13269	NaN	40.8	NaN	NaN	NaN	NaN	37.4
13273	12.3	24.0	NaN	NaN	NaN	45.2	49.1
13275	NaN	44.7	46.1	NaN	NaN	53.8	52.1
13279	34.0	41.8	NaN	NaN	NaN	42.7	NaN
13295	NaN	47.2	NaN	NaN	NaN	42.3	NaN
13299	20.3	NaN	NaN	NaN	44.4	NaN	NaN
13303	NaN	NaN	NaN	NaN	48.8	NaN	NaN
16001	38.7	34.4	31.6	34.0	35.5	37.5	38.0
17001	55.5	56.1	51.0	48.6	52.8	57.5	53.2
17003	38.4	42.3	NaN	NaN	48.7	58.5	51.3
17005	NaN	54.5	67.0	51.7	59.0	65.0	58.6
17007	62.3	47.4	60.3	50.0	53.0	69.2	62.2
17009	52.0	57.1	60.3	56.2	61.4	59.4	64.4
17011	64.2	59.1	68.5	57.4	64.8	66.4	67.5
17013	NaN	53.6	58.6	46.9	54.1	NaN	57.8
17015	69.0	64.3	68.9	64.7	68.1	68.6	68.9
17017	62.4	62.9	71.9	NaN	63.0	68.4	64.6
17019	67.7	64.3	67.9	52.8	66.1	75.3	69.4
17021	62.9	65.6	73.6	NaN	65.9	71.2	69.5
17023	NaN	56.5	64.1	NaN	58.4	65.5	61.3
17025	52.5	44.3	58.5	40.5	52.4	59.2	57.3
17027	54.1	50.8	60.1	51.9	52.5	63.6	54.6
17029	62.6	64.3	74.9	63.7	66.6	NaN	NaN
17031	NaN	NaN	NaN	NaN	NaN	45.8	NaN
17033	56.4	52.7	63.0	49.5	56.3	65.0	61.8
17035	59.0	54.1	67.2	50.6	56.5	NaN	67.2
17037	67.5	53.6	61.1	55.6	59.0	63.9	64.9
17039	62.1	65.9	75.0	NaN	66.4	70.7	68.0
17041	67.4	63.7	78.1	62.4	67.1	NaN	69.0
17045	65.0	64.7	69.5	59.5	61.0	70.8	67.9
17047	47.7	55.6	44.1	NaN	54.7	63.1	53.5
17049	58.2	55.3	72.1	51.5	57.9	64.7	59.9
17051	55.6	53.5	59.5	48.9	54.4	63.5	NaN
17053	60.5	62.4	60.7	52.6	56.3	66.7	66.7
17055	45.2	46.4	47.5	38.9	51.2	53.3	47.0

17057	56.4	60.1	62.4	53.9	58.4	65.5	62.5
17059	48.4	54.5	50.0	47.5	57.6	62.5	55.3
17061	62.5	63.9	70.1	60.7	65.2	NaN	66.6
17063	58.0	57.3	58.5	49.0	56.2	59.5	63.1
17065	46.1	51.3	51.2	NaN	51.8	NaN	50.2
17067	63.7	66.1	69.8	NaN	66.6	67.3	67.6
17069	39.8	NaN	43.3	44.8	50.8	50.6	NaN
17071	57.8	61.6	60.2	NaN	62.1	66.6	67.2
17073	61.4	58.9	63.2	52.6	62.3	68.1	66.8
17075	59.1	57.1	55.3	45.3	60.0	57.9	63.8
17077	NaN	50.1	48.9	39.6	52.2	49.9	51.1
17079	58.1	54.6	60.7	53.9	54.5	NaN	55.8
17081	48.4	40.9	51.4	37.4	47.6	47.7	48.1
17083	55.3	56.9	NaN	60.1	65.3	NaN	63.2
17085	60.6	53.4	NaN	55.6	59.1	58.7	62.6
17087	NaN	48.0	NaN	NaN	45.8	43.8	45.1
17089	64.5	53.6	58.9	50.3	56.0	67.7	64.4
17091	55.8	52.2	54.2	52.3	54.8	61.7	59.7
17093	NaN	58.9	60.8	57.4	57.5	66.3	NaN
17095	63.6	63.7	64.1	55.9	63.9	68.4	67.0
17097	52.5	NaN	41.3	48.2	51.8	50.3	57.3
17099	61.2	60.8	67.1	55.4	59.3	NaN	65.7
17101	NaN	51.0	54.1	47.8	57.9	60.4	58.0
17103	60.8	59.4	63.7	59.7	64.7	67.6	66.8
17105	60.6	58.8	61.0	56.9	56.0	66.0	63.5
17107	66.6	64.7	70.1	NaN	68.7	NaN	71.8
17109	62.5	64.7	69.3	60.2	65.1	NaN	66.3
17111	59.1	47.2	58.9	NaN	52.5	NaN	NaN
17113	63.8	66.8	67.5	61.0	63.9	67.4	68.8
17115	68.2	66.2	75.2	63.8	66.2	74.4	73.8
17117	63.4	61.3	69.7	58.3	67.4	70.8	65.1
17119	54.7	56.7	62.5	56.6	58.5	61.0	59.6
17121	53.3	45.9	60.5	46.5	54.9	60.1	54.3
17123	63.3	60.1	65.5	53.8	63.1	NaN	64.1
17125	59.2	62.1	NaN	55.4	61.2	64.7	62.6
17127	47.5	NaN	NaN	43.5	51.6	52.5	50.0
17129	63.6	63.9	68.6	62.0	67.3	66.4	67.9
17131	59.9	60.0	63.2	51.9	59.4	NaN	65.0
17133	48.4	43.4	54.1	47.2	55.7	56.3	53.1
17135	61.5	61.9	73.0	61.5	63.2	NaN	68.6
17137	60.4	60.2	79.8	64.1	62.9	NaN	70.0
17139	68.5	66.4	75.9	59.3	65.0	72.4	65.7
17141	65.6	55.0	64.3	59.9	61.9	61.8	67.6
17143	60.0	63.9	66.6	55.4	65.2	69.7	64.0
17145	48.0	42.7	52.5	35.6	48.2	53.2	43.4
17147	71.3	69.4	77.4	NaN	71.7	77.3	74.2
17149	58.1	57.7	61.2	NaN	62.1	NaN	NaN
17151	35.1	44.6	NaN	NaN	53.1	NaN	NaN

17153	44.4	44.5	48.2	43.2	54.8	56.8	53.5
17155	63.3	57.2	NaN	NaN	61.8	62.2	64.6
17157	50.0	44.4	52.1	41.6	53.4	55.3	50.3
17159	51.8	48.8	56.3	NaN	54.0	NaN	56.1
17161	58.0	58.6	60.5	NaN	61.3	64.6	66.3
17163	52.4	50.9	56.0	53.2	52.7	59.6	53.6
17165	46.8	49.9	49.4	NaN	59.0	59.2	52.8
17167	66.2	70.3	80.4	65.5	68.8	72.1	72.7
17169	56.9	58.6	59.9	NaN	58.1	63.2	63.1
17171	57.9	62.9	NaN	NaN	65.1	64.9	72.1
17173	57.5	61.0	70.4	56.2	60.4	67.3	67.1
17175	65.2	64.6	NaN	58.8	63.0	70.1	70.8
17177	66.3	59.2	63.3	56.4	64.1	64.5	62.3
17179	65.0	66.8	72.1	63.8	63.8	70.8	71.0
17181	43.8	51.6	NaN	34.0	50.2	44.5	48.0
17183	64.6	64.9	71.7	NaN	61.8	67.4	67.6
17185	55.2	54.4	NaN	NaN	54.5	60.8	58.1
17187	62.6	63.3	67.5	58.5	66.6	NaN	64.5
17189	48.8	48.3	57.4	43.9	51.8	58.7	50.5
17191	48.3	47.5	52.9	43.4	52.0	NaN	56.4
17193	47.0	52.8	53.0	46.6	55.0	63.4	54.5
17195	60.8	58.7	62.7	NaN	NaN	67.0	66.4
17197	55.2	50.9	57.8	47.8	54.5	58.4	57.2
17199	42.5	47.2	45.6	NaN	52.8	NaN	45.9
17201	58.7	47.2	56.4	55.8	56.7	56.9	62.8
17203	67.0	60.4	64.7	61.4	63.3	68.1	69.1
18001	59.3	54.1	52.9	47.6	55.3	58.3	54.5
18003	54.8	49.9	58.2	52.3	53.8	59.5	54.7
18005	54.8	54.2	55.5	NaN	59.9	NaN	57.3
18007	61.9	59.7	66.5	54.5	64.1	60.5	61.4
18009	63.5	54.7	57.4	53.8	NaN	62.4	56.1
18011	62.2	54.5	61.2	52.4	65.0	62.7	62.4
18013	51.7	49.8	NaN	NaN	NaN	54.4	53.8
18015	65.1	58.5	63.9	58.7	67.9	NaN	NaN
18017	60.0	53.5	61.8	NaN	59.4	NaN	57.1
18019	45.0	47.8	49.8	40.6	54.4	50.6	52.6
18021	NaN	55.6	52.8	48.4	56.4	58.7	58.1
18023	65.6	58.0	65.2	NaN	67.4	61.5	61.0
18025	NaN	NaN	47.9	46.3	53.4	NaN	NaN
18027	55.4	58.3	54.3	53.5	60.9	68.2	58.6
18029	NaN	47.9	54.1	30.9	51.4	47.5	47.4
18031	61.1	59.5	62.9	53.9	68.4	61.6	68.3
18033	49.2	48.2	49.3	52.4	45.6	55.6	54.2
18035	59.4	50.8	57.0	48.8	59.3	60.8	57.9
18037	53.2	63.6	59.0	51.4	61.7	63.6	58.4
18039	54.2	54.9	57.6	48.3	55.7	58.0	58.6
18041	57.8	52.1	57.0	51.5	60.6	58.0	57.3
18043	NaN	50.3	NaN	37.1	54.8	NaN	NaN

18045	59.4	56.5	NaN	NaN	64.4	NaN	NaN
18047	56.7	53.3	59.0	41.6	56.5	53.5	57.0
18049	56.3	55.8	57.5	51.4	54.3	56.3	NaN
18051	55.3	58.5	56.3	55.0	62.1	64.7	62.0
18053	59.3	52.9	59.4	NaN	59.9	62.1	56.3
18055	50.8	46.3	53.2	NaN	55.2	61.4	52.3
18057	60.0	58.5	62.8	48.8	65.2	61.6	59.7
18059	56.4	48.9	57.1	42.3	59.3	57.5	56.2
18061	48.1	50.5	51.6	43.6	59.3	55.0	56.1
18063	61.7	53.7	60.2	53.1	64.4	62.2	59.8
18065	60.8	49.2	56.9	45.8	61.5	61.4	55.8
18067	62.8	54.8	62.5	55.4	66.0	66.4	64.2
18069	57.6	51.4	58.5	54.3	57.9	62.6	57.5
18071	55.0	50.5	51.2	NaN	58.9	NaN	59.3
18073	54.7	52.6	56.9	NaN	55.1	58.2	54.6
18075	58.9	50.0	59.5	49.3	55.7	NaN	55.3
18077	43.8	45.0	48.3	32.1	50.3	49.8	49.4
18079	51.9	52.3	55.0	45.7	57.7	52.9	52.8
18081	NaN	59.1	NaN	53.6	60.9	58.0	62.2
18083	56.6	56.6	52.5	NaN	62.5	67.2	57.4
18085	53.3	51.1	60.6	49.0	52.1	56.8	54.9
18087	54.9	50.2	52.9	47.5	52.7	58.5	56.0
18089	54.8	50.5	55.7	48.3	52.0	NaN	51.8
18091	56.0	50.8	58.3	50.1	49.7	55.0	52.0
18093	47.5	46.9	52.5	NaN	54.7	52.0	50.0
18095	62.0	50.2	60.5	47.6	64.5	65.4	56.7
18097	NaN	44.0	NaN	47.6	NaN	47.0	57.8
18099	54.4	52.0	53.3	NaN	50.0	57.8	NaN
18101	55.3	NaN	NaN	43.8	60.6	65.0	57.6
18103	55.8	48.9	55.0	49.1	54.5	NaN	57.1
18105	52.2	45.8	44.7	NaN	51.3	48.9	55.1
18107	64.0	62.2	63.8	58.5	63.0	65.0	64.5
18109	54.4	49.0	56.9	NaN	61.3	60.0	59.3
18111	61.5	54.4	59.0	52.0	60.8	55.3	59.3
18113	54.0	49.7	51.6	54.3	52.2	57.0	59.1
18115	56.1	52.0	50.0	30.8	52.7	42.2	52.6
18117	55.0	63.2	55.8	56.6	64.9	65.1	63.0
18119	48.9	44.8	NaN	NaN	54.0	55.4	50.9
18121	60.6	55.8	56.5	52.3	61.7	58.9	57.4
18123	NaN	NaN	45.0	47.2	57.0	NaN	51.6
18125	51.0	NaN	NaN	51.8	56.7	NaN	50.4
18127	57.1	53.0	59.7	52.1	50.5	60.5	54.8
18129	53.9	56.6	54.4	56.3	66.5	NaN	60.7
18131	50.1	51.0	54.5	44.3	49.9	NaN	51.0
18133	58.8	55.5	57.4	58.8	61.0	58.8	60.3
18135	58.5	51.6	58.2	50.9	62.5	62.2	55.8
18137	52.1	57.4	51.1	42.2	53.9	52.3	51.2
18139	62.2	56.5	57.9	56.9	63.2	60.4	61.8

18141	53.4	49.4	54.4	47.3	50.4	55.5	53.4
18143	NaN	50.6	49.6	NaN	NaN	49.1	NaN
18145	61.9	57.6	56.0	51.8	61.8	56.8	63.6
18147	52.4	59.0	51.1	53.3	58.5	60.0	NaN
18149	50.4	48.9	53.8	NaN	46.9	NaN	50.6
18151	50.4	48.2	46.9	NaN	44.7	55.9	54.2
18153	52.9	53.7	55.5	50.4	55.6	61.9	54.9
18155	52.1	48.5	47.6	NaN	57.4	49.0	55.2
18157	NaN	56.1	63.2	54.7	66.6	58.8	58.4
18159	66.8	61.3	NaN	56.4	69.8	69.2	62.2
18161	60.0	59.5	59.8	51.0	65.5	60.6	59.5
18163	57.9	62.9	55.5	60.9	69.4	73.3	59.1
18165	62.8	54.2	60.3	51.4	57.1	64.4	NaN
18167	NaN	55.2	51.5	49.3	50.6	58.4	55.7
18169	57.4	49.5	59.2	53.1	51.5	55.7	58.6
18171	64.5	64.9	66.5	60.7	67.5	63.4	64.7
18173	50.2	60.9	49.8	53.6	59.4	NaN	57.2
18175	52.5	51.1	52.3	44.1	56.7	NaN	54.7
18177	58.7	55.4	59.2	50.1	61.7	60.4	55.9
18179	61.8	50.9	61.0	NaN	58.7	66.0	57.4
18181	61.1	58.8	60.4	52.8	59.3	NaN	56.2
18183	58.0	51.7	54.7	53.2	51.3	61.7	54.6
19001	60.8	57.7	52.0	55.0	54.1	58.5	57.3
19003	56.5	52.6	53.2	NaN	NaN	NaN	NaN
19005	56.8	52.6	52.8	NaN	58.8	61.9	61.2
19007	55.9	50.7	45.1	NaN	39.1	57.5	40.4
19009	62.0	57.2	52.3	58.9	48.3	67.3	57.9
19011	61.1	57.6	61.0	57.9	56.7	67.1	66.1
19013	61.8	59.5	58.4	51.9	54.2	60.1	64.5
19015	60.2	56.7	54.6	53.2	50.9	64.4	60.4
19017	62.1	56.8	57.3	52.5	NaN	62.0	66.7
19019	60.2	55.9	57.2	55.2	56.7	63.4	65.0
19021	60.2	56.5	55.5	53.5	NaN	63.8	57.2
19023	59.9	55.9	56.1	55.0	54.2	60.8	61.6
19025	62.3	57.5	53.1	55.1	47.7	63.5	61.2
19027	58.7	60.4	59.0	58.8	54.8	65.3	58.9
19029	59.6	56.9	52.6	56.9	49.3	63.3	51.2
19031	60.5	59.6	63.3	NaN	56.4	65.3	67.0
19033	58.6	55.2	51.7	52.7	59.8	61.0	60.8
19035	69.6	62.4	64.3	58.8	57.2	NaN	59.6
19037	56.0	50.4	53.2	54.5	55.9	NaN	63.6
19039	49.9	NaN	47.9	NaN	45.9	57.7	40.2
19041	61.9	57.6	52.6	50.9	50.3	62.4	56.9
19043	63.2	59.0	58.8	56.4	60.1	NaN	64.5
19045	59.8	59.7	58.3	59.5	57.0	63.9	66.0
19047	64.3	61.3	61.7	61.9	54.2	63.5	62.8
19049	58.7	54.6	52.8	54.4	47.1	63.6	59.8
19051	57.4	37.6	50.2	54.6	42.0	56.7	44.6

19053	57.2	NaN	48.0	NaN	47.4	57.0	37.9
19055	63.8	63.0	63.9	NaN	61.1	66.4	67.4
19057	60.4	59.6	63.8	55.1	60.7	NaN	66.9
19059	56.1	57.7	49.8	51.3	55.5	56.0	58.8
19061	57.8	59.6	62.0	60.2	NaN	70.5	68.6
19063	58.5	58.9	46.7	NaN	56.0	59.6	62.8
19065	62.3	56.3	56.5	55.5	55.9	64.4	63.4
19067	57.5	54.1	49.8	54.3	54.0	NaN	61.9
19069	60.5	57.0	58.5	54.1	55.4	59.7	62.8
19071	55.5	60.1	50.6	NaN	54.8	63.1	53.4
19073	63.5	56.7	55.5	54.2	49.0	64.1	59.0
19075	64.8	63.6	61.9	56.3	60.9	66.5	64.9
19077	58.4	59.2	52.0	54.2	45.5	NaN	58.6
19079	61.0	57.2	49.9	51.4	51.5	62.2	61.3
19081	62.4	57.4	53.2	49.6	60.0	63.5	63.0
19083	59.9	59.7	59.3	53.9	52.8	65.0	63.4
19085	58.2	53.7	52.6	NaN	52.1	NaN	52.4
19087	57.2	60.5	62.3	56.9	54.9	58.3	60.3
19089	56.4	51.6	53.7	52.1	54.5	60.3	60.9
19091	61.0	58.0	50.4	54.9	56.9	62.4	59.0
19093	67.6	62.1	63.8	NaN	55.5	65.6	63.1
19095	57.4	56.7	56.0	52.3	49.8	NaN	61.6
19097	61.1	59.2	58.1	NaN	54.7	NaN	66.3
19099	60.3	60.5	60.7	56.5	NaN	67.3	60.1
19101	55.1	46.1	62.0	53.3	53.8	53.9	50.7
19103	56.1	56.4	59.9	55.5	51.8	NaN	62.1
19105	60.4	54.9	60.1	57.2	54.0	64.5	61.7
19107	57.1	51.6	57.7	55.8	51.6	NaN	53.1
19109	63.4	60.4	54.2	57.4	60.2	60.4	56.2
19111	61.4	53.5	64.7	54.8	52.1	59.7	54.6
19113	60.0	56.0	57.7	56.5	NaN	NaN	64.6
19115	56.4	59.0	58.1	54.3	53.3	NaN	61.4
19117	54.4	41.4	41.5	NaN	43.8	54.1	43.1
19119	66.1	63.8	60.6	54.8	60.8	63.6	62.2
19121	57.1	52.5	50.9	52.3	51.4	58.8	55.4
19123	59.1	54.4	59.7	53.7	NaN	62.0	59.3
19125	56.6	52.9	58.3	54.0	NaN	60.6	57.8
19127	62.6	63.5	61.9	60.6	NaN	67.1	65.0
19129	56.3	55.7	NaN	NaN	49.3	NaN	52.1
19131	61.0	54.4	55.6	49.8	59.4	62.5	64.6
19133	56.1	57.1	55.3	55.8	56.0	61.8	48.6
19135	57.3	51.0	50.4	56.1	45.4	59.3	47.4
19137	57.8	56.2	51.8	58.3	49.2	63.0	54.8
19139	60.8	60.3	61.7	55.2	55.6	62.6	65.5
19141	63.9	62.3	59.9	56.0	62.0	66.4	59.4
19143	56.1	59.7	57.2	50.5	NaN	62.9	56.3
19145	57.5	55.4	50.5	55.9	52.4	62.6	54.0
19147	59.7	57.8	48.8	55.6	NaN	61.6	57.9

19149	66.1	59.6	61.3	NaN	NaN	NaN	47.8
19151	60.6	58.2	49.7	NaN	54.6	65.0	56.5
19153	55.2	57.1	53.4	51.0	50.2	64.6	54.1
19155	60.6	57.0	NaN	58.5	48.0	64.4	51.0
19157	57.5	56.6	58.2	53.4	54.8	NaN	60.4
19159	51.7	48.8	47.2	NaN	50.6	54.0	41.8
19161	63.3	57.8	60.7	59.3	47.1	NaN	61.3
19163	64.4	63.4	63.7	60.5	61.0	68.1	69.1
19165	60.5	58.2	57.8	58.7	52.8	NaN	58.0
19167	65.8	65.2	65.8	60.8	NaN	70.5	62.9
19169	58.4	57.9	52.8	50.0	52.2	63.6	60.0
19171	63.5	59.5	60.1	53.4	56.2	66.3	61.4
19173	51.6	52.8	NaN	49.9	47.0	61.3	52.9
19175	56.0	44.5	48.6	53.7	52.1	61.9	41.4
19177	56.6	36.6	54.2	53.9	44.2	53.0	43.6
19179	58.1	45.0	60.2	55.5	52.0	59.0	53.7
19181	54.3	49.9	52.0	52.3	49.5	NaN	50.2
19183	55.8	61.7	61.5	54.7	53.2	NaN	57.2
19185	54.2	NaN	49.0	52.5	44.6	57.1	NaN
19187	58.4	57.9	51.1	49.1	50.4	64.6	57.5
19189	62.5	59.9	51.8	54.2	60.8	61.6	64.4
19191	58.4	54.6	51.7	55.6	53.6	61.8	66.3
19193	61.2	58.1	59.5	56.8	56.6	NaN	56.4
19195	58.2	54.8	53.7	51.0	58.2	60.1	62.0
19197	58.8	57.9	52.6	54.1	55.9	61.0	60.6
20001	45.4	45.0	43.3	43.1	46.5	46.2	43.4
20003	46.9	47.1	48.0	34.2	37.4	41.8	23.2
20005	55.2	51.9	47.4	51.9	49.4	53.7	44.3
20007	NaN	NaN	NaN	26.7	NaN	28.6	20.2
20009	NaN	31.9	NaN	29.9	43.2	37.7	19.1
20011	NaN	40.8	38.2	35.9	35.0	33.1	17.0
20013	62.5	49.4	48.6	56.5	57.2	64.8	53.3
20015	46.1	35.4	47.6	37.6	34.3	35.9	18.3
20017	47.9	43.7	NaN	NaN	39.7	41.8	33.6
20019	31.4	33.5	33.8	40.3	NaN	18.8	12.2
20021	NaN	NaN	NaN	38.9	39.6	32.2	16.0
20023	NaN	56.2	NaN	NaN	62.1	NaN	62.4
20027	48.9	39.7	46.2	45.3	51.9	NaN	28.5
20029	52.5	37.1	NaN	NaN	44.7	39.8	26.3
20031	46.7	39.2	38.7	33.9	38.2	36.2	27.2
20035	42.4	30.9	NaN	37.5	37.4	26.6	9.5
20037	39.2	NaN	38.7	NaN	32.4	34.2	NaN
20041	47.7	28.8	43.6	41.8	40.9	40.1	28.0
20043	62.3	58.5	53.3	NaN	58.3	64.3	NaN
20045	46.1	46.3	33.4	41.0	43.4	36.1	29.8
20047	NaN	53.4	NaN	NaN	63.9	55.9	36.0
20049	41.9	34.3	43.6	39.9	NaN	33.7	21.0
20053	NaN	NaN	48.1	NaN	41.3	34.3	14.5

20055	68.2	NaN	NaN	56.2	NaN	NaN	NaN
20057	NaN	NaN	NaN	NaN	61.6	NaN	NaN
20059	49.9	41.7	43.4	NaN	40.1	36.9	23.5
20061	55.5	41.8	44.6	46.3	50.2	40.6	41.1
20063	NaN	NaN	50.9	NaN	NaN	NaN	NaN
20065	NaN	50.2	54.5	NaN	NaN	NaN	NaN
20067	NaN	NaN	NaN	NaN	NaN	53.1	NaN
20069	68.1	62.8	NaN	53.4	NaN	59.7	NaN
20073	NaN	36.4	34.8	NaN	32.0	30.8	19.6
20077	NaN	22.0	29.4	31.0	32.4	24.0	9.4
20079	48.7	30.8	48.8	37.3	35.5	42.4	25.9
20081	64.5	59.7	64.5	65.0	63.8	62.6	30.5
20083	NaN	55.7	57.1	NaN	NaN	NaN	49.2
20085	47.7	42.9	39.7	46.2	47.1	45.0	NaN
20087	49.2	52.8	NaN	49.2	46.2	52.0	36.7
20089	55.5	39.3	42.9	49.4	47.0	41.2	34.2
20091	NaN	NaN	34.1	NaN	NaN	38.8	32.8
20095	45.9	26.7	46.0	26.7	34.5	35.4	22.5
20097	57.8	NaN	63.3	NaN	53.3	50.0	41.0
20099	NaN	39.4	33.7	NaN	36.7	24.4	9.3
20103	NaN	NaN	NaN	46.0	47.3	42.0	38.2
20105	NaN	NaN	NaN	NaN	42.2	NaN	12.3
20107	41.8	NaN	NaN	NaN	40.0	33.5	20.9
20111	41.7	38.6	NaN	34.0	35.9	36.1	23.2
20113	45.1	29.3	45.5	35.1	30.2	34.4	16.9
20115	52.0	25.7	42.4	35.1	29.7	38.3	17.9
20117	45.3	28.3	38.6	48.0	42.7	47.9	38.8
20119	NaN	70.9	NaN	NaN	73.1	69.1	67.6
20121	47.8	49.7	42.2	43.3	48.9	46.1	29.5
20123	53.4	NaN	43.0	43.3	44.0	29.1	24.7
20125	35.9	37.0	39.6	35.6	40.7	27.9	10.7
20127	42.8	38.8	32.8	44.9	40.2	36.7	32.7
20131	56.1	33.3	41.5	NaN	45.7	57.6	39.8
20133	43.6	35.7	38.9	NaN	31.4	29.5	11.0
20137	NaN	NaN	47.8	NaN	32.5	33.4	NaN
20139	45.1	42.8	35.5	41.0	37.5	39.5	25.1
20141	50.3	27.5	NaN	NaN	38.5	NaN	13.6
20143	45.0	29.9	NaN	NaN	43.9	NaN	20.5
20145	49.5	43.1	NaN	NaN	56.7	NaN	44.3
20147	54.0	NaN	NaN	43.8	27.4	38.7	25.7
20149	54.5	48.1	40.0	49.5	45.8	40.7	NaN
20151	57.4	51.8	54.0	NaN	54.7	44.9	37.8
20153	NaN	NaN	NaN	NaN	NaN	36.5	NaN
20155	45.9	23.1	NaN	32.8	38.0	NaN	21.4
20157	55.7	44.8	51.0	45.1	50.5	54.6	43.3
20159	43.3	29.3	NaN	NaN	35.7	37.6	16.4
20161	50.9	37.0	38.7	46.1	48.4	47.8	35.0
20163	NaN	25.5	NaN	NaN	29.7	30.3	13.9

20165	44.1	NaN	37.3	NaN	NaN	NaN	NaN
20167	41.7	25.2	NaN	30.3	41.9	33.5	NaN
20169	42.4	22.6	NaN	43.0	39.4	34.4	19.7
20171	NaN	NaN	51.5	NaN	NaN	NaN	NaN
20173	45.9	32.0	43.5	36.3	35.3	32.2	25.2
20175	59.3	72.5	65.8	64.4	60.6	58.9	57.6
20177	48.9	39.0	37.7	NaN	41.5	42.1	35.2
20179	NaN	NaN	NaN	NaN	59.5	59.1	NaN
20181	NaN	44.4	NaN	NaN	NaN	NaN	NaN
20183	NaN	39.6	45.8	NaN	37.5	43.8	24.7
20185	48.7	NaN	NaN	NaN	43.8	NaN	NaN
20189	61.3	58.1	NaN	NaN	62.0	57.9	64.1
20191	39.6	26.5	35.7	31.0	30.4	21.4	12.5
20193	NaN	NaN	NaN	NaN	45.6	NaN	40.2
20195	NaN	NaN	NaN	34.8	NaN	NaN	NaN
20197	NaN	36.8	31.0	NaN	41.9	39.8	34.2
20201	43.4	33.2	NaN	NaN	45.2	48.8	42.0
20203	NaN	NaN	52.5	NaN	NaN	NaN	NaN
20205	44.1	37.7	NaN	39.4	29.2	32.3	15.8
20207	NaN	36.0	NaN	NaN	33.1	31.7	NaN
21001	43.9	44.5	38.2	36.0	42.5	44.2	40.6
21003	NaN	NaN	NaN	NaN	54.2	41.3	NaN
21005	NaN	NaN	NaN	NaN	NaN	53.4	NaN
21007	50.1	49.4	NaN	51.5	56.9	NaN	49.2
21009	53.3	56.7	NaN	44.0	60.9	54.5	52.8
21011	46.4	NaN	NaN	48.1	43.4	47.3	52.1
21015	52.3	60.3	55.9	53.4	53.1	51.3	54.8
21017	47.5	50.9	47.4	NaN	50.4	61.6	NaN
21021	54.3	59.7	56.0	40.0	56.1	58.8	57.9
21023	45.4	51.3	51.9	NaN	NaN	51.3	NaN
21027	53.2	53.9	49.0	36.0	55.7	50.8	53.6
21029	47.8	51.8	52.4	30.1	52.6	47.5	NaN
21031	40.8	50.5	50.8	51.0	54.0	57.8	55.2
21033	50.6	50.3	48.1	46.5	58.3	56.6	41.2
21035	41.0	43.7	43.0	NaN	52.4	57.7	42.0
21039	48.1	NaN	51.2	NaN	56.0	54.2	45.3
21041	39.6	NaN	NaN	NaN	50.2	47.6	53.2
21045	NaN	54.0	NaN	47.0	55.1	56.8	51.0
21047	48.1	54.4	51.6	44.5	54.9	55.3	44.0
21049	47.9	58.1	49.3	58.0	55.0	59.6	54.4
21051	NaN	NaN	NaN	NaN	49.2	46.1	44.8
21053	49.4	NaN	NaN	NaN	54.8	NaN	53.8
21055	50.0	49.7	51.5	NaN	54.1	55.1	43.4
21057	NaN	NaN	46.3	53.9	59.9	NaN	50.5
21059	54.2	60.3	56.2	54.5	60.0	62.7	59.2
21061	46.9	49.9	52.2	48.0	54.9	50.1	53.8
21065	42.9	42.5	35.2	35.0	50.3	NaN	NaN
21067	50.6	54.1	52.9	44.5	51.4	53.4	50.0

21069	45.7	44.9	NaN	40.5	48.5	49.8	50.5
21073	NaN	NaN	NaN	NaN	48.2	51.9	NaN
21075	49.4	52.1	NaN	48.0	54.7	57.1	45.5
21077	42.0	52.0	NaN	NaN	NaN	NaN	NaN
21079	NaN	49.0	NaN	NaN	NaN	56.7	NaN
21081	NaN	NaN	NaN	49.5	51.7	48.2	50.0
21083	50.2	51.4	45.4	NaN	52.0	53.1	42.8
21085	51.8	51.7	50.4	NaN	50.5	53.7	51.3
21087	51.5	55.5	55.0	NaN	55.8	56.2	59.7
21089	46.5	NaN	46.7	NaN	42.3	41.9	48.0
21091	50.6	59.0	NaN	54.0	58.6	66.5	NaN
21093	53.4	52.4	52.9	36.5	54.3	54.0	52.7
21097	46.9	NaN	NaN	NaN	50.7	NaN	NaN
21099	54.8	51.3	55.7	46.0	54.5	55.9	53.2
21101	NaN	58.8	52.0	53.0	59.0	NaN	58.8
21103	40.0	45.0	48.5	42.5	51.3	55.6	NaN
21105	47.4	53.6	50.8	52.0	53.8	56.7	46.6
21107	NaN	NaN	NaN	NaN	53.2	NaN	NaN
21111	NaN	NaN	NaN	25.0	50.7	56.6	NaN
21113	42.5	47.6	51.8	31.0	45.9	48.7	47.1
21117	NaN	NaN	NaN	NaN	NaN	51.3	NaN
21123	NaN	58.0	NaN	47.0	58.5	57.6	56.9
21135	50.2	NaN	NaN	44.0	46.7	45.8	50.7
21137	53.4	NaN	49.9	37.0	54.1	56.9	51.6
21139	48.4	46.0	NaN	47.0	54.0	49.3	41.3
21141	48.7	49.1	47.8	43.0	52.5	50.8	45.3
21143	NaN	NaN	46.0	43.0	51.6	NaN	45.0
21145	NaN	NaN	NaN	NaN	51.4	49.9	NaN
21149	NaN	55.2	58.3	55.0	58.1	60.4	54.7
21151	47.5	53.9	48.6	30.0	54.2	57.9	NaN
21155	52.9	59.2	57.2	38.5	59.6	58.5	52.7
21157	NaN	46.7	43.0	34.0	52.2	52.8	NaN
21161	53.8	55.0	56.4	44.0	54.4	55.6	57.6
21163	56.5	58.8	55.0	43.0	59.9	58.4	54.8
21167	54.8	54.0	NaN	42.0	NaN	NaN	59.7
21169	53.0	54.0	NaN	NaN	53.2	58.9	58.3
21171	NaN	NaN	49.9	NaN	NaN	NaN	57.8
21173	NaN	NaN	NaN	NaN	44.9	NaN	NaN
21177	NaN	NaN	47.3	NaN	51.0	54.5	50.4
21179	52.1	50.3	NaN	33.5	50.3	53.4	NaN
21181	44.7	49.3	NaN	36.0	NaN	NaN	49.0
21183	NaN	56.4	58.5	53.0	53.6	57.6	60.7
21185	45.5	46.4	46.1	39.5	51.7	50.0	NaN
21187	NaN	NaN	34.7	NaN	53.1	53.4	NaN
21191	46.0	60.9	46.8	NaN	46.1	NaN	50.5
21197	NaN	NaN	NaN	NaN	NaN	NaN	46.7
21199	54.9	58.9	45.1	43.0	49.1	53.0	51.9
21203	45.4	NaN	NaN	44.0	46.2	43.4	NaN

21205	36.7	46.8	NaN	NaN	NaN	NaN	NaN
21207	50.8	51.3	50.4	47.5	52.0	51.4	58.6
21209	50.8	57.5	NaN	41.0	50.0	NaN	56.1
21211	48.2	53.3	56.8	40.5	56.9	55.2	56.7
21213	51.5	53.3	51.2	45.5	54.5	51.4	46.8
21215	53.7	50.3	57.9	38.5	53.5	56.8	52.6
21217	48.7	54.8	45.8	48.0	54.2	55.8	55.9
21219	52.1	50.7	49.5	43.5	51.9	54.8	45.5
21221	46.2	51.5	46.7	41.0	51.1	55.8	40.6
21223	43.8	46.4	45.1	35.0	52.6	50.9	46.6
21225	53.9	60.4	57.4	56.5	63.9	67.0	60.6
21227	51.5	53.3	53.0	47.5	56.8	53.0	56.9
21229	54.0	52.7	NaN	30.5	52.7	56.5	53.2
21231	52.2	60.4	48.1	48.0	52.9	54.2	55.7
21233	50.5	53.4	54.1	51.5	60.1	60.4	57.4
21239	50.9	54.6	56.9	50.5	56.4	57.2	55.7
22001	37.6	36.9	37.7	39.2	37.3	34.8	36.5
22003	25.9	NaN	NaN	NaN	NaN	NaN	NaN
22007	46.2	39.7	NaN	38.1	NaN	42.6	50.0
22009	46.0	49.0	44.2	40.8	46.4	43.4	48.3
22011	NaN	37.1	NaN	NaN	NaN	NaN	NaN
22015	29.7	56.0	NaN	NaN	NaN	37.6	NaN
22017	37.9	48.9	40.4	42.3	46.9	33.8	33.5
22019	NaN	34.1	NaN	NaN	NaN	NaN	NaN
22025	46.8	51.4	44.3	45.2	52.6	49.1	44.2
22029	55.3	55.3	54.7	55.1	61.8	51.9	48.3
22035	59.8	65.6	60.6	60.5	65.0	61.9	57.4
22039	29.5	38.2	29.3	31.9	34.9	39.0	32.4
22041	57.9	59.0	61.7	54.6	50.3	56.1	54.7
22043	45.4	NaN	39.8	38.4	38.7	46.4	49.3
22045	34.4	42.3	32.0	38.2	36.5	42.4	40.4
22047	51.9	58.0	57.6	47.3	51.5	NaN	NaN
22053	24.1	31.3	36.9	NaN	NaN	NaN	27.9
22055	33.6	37.6	NaN	NaN	NaN	NaN	NaN
22065	51.8	56.8	59.4	50.1	55.0	57.4	48.5
22067	54.0	61.5	54.1	NaN	53.6	60.0	50.7
22069	37.6	47.0	37.7	42.0	37.7	40.4	32.0
22073	38.8	56.9	NaN	NaN	44.5	NaN	35.1
22077	48.6	54.0	52.1	51.3	54.1	47.3	43.1
22079	49.7	49.4	47.8	41.7	40.4	44.2	39.9
22081	NaN	NaN	36.2	36.6	39.6	NaN	NaN
22083	48.7	57.0	56.5	NaN	57.6	61.5	50.6
22093	NaN	NaN	NaN	NaN	NaN	38.3	NaN
22097	38.3	43.2	38.3	40.1	42.2	41.1	36.7
22099	36.3	41.1	30.1	37.4	40.0	33.6	36.5
22101	42.9	45.8	NaN	NaN	38.7	38.4	42.7
22107	54.6	63.8	61.8	54.2	58.0	58.8	46.1
22113	27.1	35.6	26.3	41.8	46.6	40.0	43.8

```
22117  55.8   NaN   NaN  35.4  50.8  38.0  45.0
22121   NaN  57.4   NaN   NaN  50.5  58.7  52.1
22123  51.4  61.5  60.5  54.3  59.5  57.0  56.0

WARNING: Counties missing yield for at least one year:
year   2016  2017  2018  2019  2020  2021  2022
fips
01003  41.8  42.0  45.4   NaN  37.2  44.3  37.1
01015   NaN   NaN   NaN   NaN  35.1  44.7  45.5
01031  43.6   NaN   NaN   NaN   NaN  31.0   NaN
01033  31.2  48.2  36.1  38.7  42.2   NaN  35.6
01035  33.8   NaN   NaN   NaN   NaN   NaN   NaN
01039  32.6   NaN   NaN   NaN  35.1   NaN   NaN
01051  20.3  39.9  42.8   NaN   NaN  50.4   NaN
01053  42.4  46.2   NaN  22.3   NaN  50.7   NaN
01055  25.6  52.7   NaN  30.4  41.9  34.7  46.4
01059  39.3  41.9  37.5  37.7  38.8   NaN   NaN
01061  33.8  48.4   NaN   NaN   NaN   NaN   NaN
01065  23.0   NaN   NaN   NaN   NaN   NaN   NaN
01069  23.3  47.2  28.4  46.2  48.7   NaN  42.6
01075  26.2  27.9   NaN   NaN   NaN   NaN   NaN
01085  29.3   NaN   NaN   NaN   NaN   NaN   NaN
01087   NaN  32.1   NaN   NaN  44.3  30.8  42.5
01091  30.2  31.4   NaN   NaN  30.0   NaN   NaN
01093   NaN   NaN   NaN   NaN  48.0   NaN  46.2
01095   NaN  43.2  36.1  31.0  42.4  43.8  43.1
01099  38.3   NaN   NaN   NaN   NaN  49.1   NaN
01103  35.3  47.0   NaN  39.5  47.4  44.2  33.4
01105  36.5  39.9  37.6   NaN  37.4  39.1  39.3
01109  40.1   NaN   NaN  51.6   NaN   NaN  33.8
01117  39.7   NaN   NaN   NaN   NaN   NaN   NaN
01119  30.7  40.7  43.1  38.0   NaN  45.3   NaN
01121   NaN  41.9  38.2   NaN  46.7   NaN  46.3
01127   NaN   NaN  38.5  32.4   NaN   NaN   NaN
05007   NaN   NaN   NaN   NaN   NaN  41.2   NaN
05029   NaN  39.4  37.0   NaN  33.2  32.9  32.0
05031  48.8  53.1  50.0  46.3  48.7   NaN  46.5
05035  43.8  52.7   NaN  46.1  50.4  52.3  53.2
05045  37.2  42.0  43.2   NaN  33.6  33.2  39.9
05047   NaN  35.5   NaN   NaN  37.3   NaN   NaN
05055  43.7  51.0  41.5  45.8  46.9   NaN  52.0
05063  38.5  43.4   NaN  40.9  43.1   NaN  42.9
05071   NaN   NaN   NaN   NaN  34.7   NaN   NaN
05073   NaN   NaN   NaN   NaN  53.1   NaN   NaN
05075  35.7  42.0  40.4  39.9  42.9   NaN  48.7
05077  43.5  48.5  44.4   NaN  55.6  53.5  57.7
05081   NaN   NaN   NaN  30.2   NaN   NaN  31.3
05083   NaN   NaN   NaN   NaN  37.5  35.3   NaN
```

05085	46.3	42.5	NaN	48.2	47.6	47.5	49.6
05093	48.9	56.7	50.5	47.6	53.4	NaN	59.3
05107	48.9	52.9	50.5	NaN	55.5	58.8	55.4
05115	NaN	NaN	NaN	NaN	NaN	NaN	43.8
05119	NaN	46.7	44.3	NaN	35.9	38.8	35.6
05121	37.9	48.2	NaN	42.5	45.3	NaN	NaN
05131	NaN	NaN	NaN	NaN	33.3	NaN	NaN
05145	37.9	42.6	NaN	43.0	47.9	45.8	36.4
05149	NaN	42.6	NaN	NaN	42.5	38.2	34.6
10003	47.0	NaN	45.8	47.2	48.4	46.9	36.9
13005	NaN	NaN	NaN	NaN	37.7	36.8	NaN
13015	17.0	52.3	NaN	12.6	NaN	NaN	NaN
13027	49.4	45.7	NaN	36.2	NaN	48.7	47.8
13031	20.7	30.8	NaN	NaN	NaN	NaN	NaN
13033	32.4	34.0	NaN	NaN	51.3	41.1	38.3
13043	21.6	NaN	NaN	NaN	NaN	NaN	NaN
13055	21.4	41.5	37.2	40.4	42.0	47.1	NaN
13069	36.4	36.9	NaN	25.2	NaN	NaN	46.0
13071	38.8	44.8	43.5	NaN	NaN	NaN	NaN
13075	NaN	NaN	NaN	NaN	NaN	48.8	48.4
13087	43.3	NaN	NaN	NaN	NaN	NaN	NaN
13091	29.3	NaN	NaN	NaN	NaN	NaN	NaN
13093	NaN	42.3	NaN	NaN	42.0	NaN	NaN
13099	22.7	NaN	NaN	NaN	NaN	NaN	NaN
13105	NaN	39.4	45.5	24.7	48.7	47.9	31.8
13115	15.3	53.0	NaN	NaN	NaN	42.7	NaN
13119	9.8	37.3	NaN	NaN	NaN	NaN	NaN
13129	NaN	NaN	NaN	NaN	NaN	NaN	44.1
13131	36.6	40.8	NaN	NaN	51.3	NaN	52.3
13147	16.0	36.4	41.2	23.2	52.3	NaN	NaN
13151	NaN	42.1	NaN	NaN	NaN	NaN	NaN
13153	40.7	33.9	NaN	NaN	NaN	NaN	49.9
13161	NaN	NaN	NaN	NaN	NaN	NaN	47.4
13163	28.7	NaN	38.1	34.0	46.2	36.1	NaN
13165	29.6	53.1	32.9	NaN	NaN	NaN	NaN
13167	21.3	NaN	NaN	NaN	NaN	58.2	NaN
13175	NaN	NaN	NaN	NaN	37.1	40.9	35.8
13177	NaN	NaN	35.6	NaN	NaN	40.2	NaN
13185	37.5	39.8	NaN	NaN	36.7	36.9	NaN
13193	37.1	36.7	NaN	28.4	59.7	47.1	NaN
13195	16.1	NaN	NaN	NaN	30.8	NaN	NaN
13197	16.3	35.3	NaN	NaN	NaN	40.1	41.3
13205	41.2	47.9	41.6	NaN	NaN	NaN	NaN
13209	35.3	45.4	NaN	NaN	NaN	NaN	NaN
13211	36.3	NaN	NaN	NaN	NaN	NaN	NaN
13213	23.2	NaN	NaN	NaN	NaN	NaN	NaN
13221	NaN	NaN	35.3	NaN	NaN	NaN	NaN
13225	36.8	NaN	26.9	NaN	34.8	NaN	38.9

13229	23.9	NaN	NaN	NaN	NaN	NaN	NaN
13233	21.8	NaN	NaN	NaN	NaN	NaN	NaN
13243	34.1	53.5	49.1	NaN	57.1	NaN	41.9
13251	29.7	42.7	NaN	NaN	36.1	NaN	NaN
13253	49.1	50.8	38.7	NaN	NaN	NaN	NaN
13267	24.1	NaN	NaN	NaN	NaN	NaN	NaN
13269	NaN	40.8	NaN	NaN	NaN	NaN	37.4
13273	12.3	24.0	NaN	NaN	NaN	45.2	49.1
13275	NaN	44.7	46.1	NaN	NaN	53.8	52.1
13279	34.0	41.8	NaN	NaN	NaN	42.7	NaN
13295	NaN	47.2	NaN	NaN	NaN	42.3	NaN
13299	20.3	NaN	NaN	NaN	44.4	NaN	NaN
13303	NaN	NaN	NaN	NaN	48.8	NaN	NaN
17003	38.4	42.3	NaN	NaN	48.7	58.5	51.3
17005	NaN	54.5	67.0	51.7	59.0	65.0	58.6
17013	NaN	53.6	58.6	46.9	54.1	NaN	57.8
17017	62.4	62.9	71.9	NaN	63.0	68.4	64.6
17021	62.9	65.6	73.6	NaN	65.9	71.2	69.5
17023	NaN	56.5	64.1	NaN	58.4	65.5	61.3
17029	62.6	64.3	74.9	63.7	66.6	NaN	NaN
17031	NaN	NaN	NaN	NaN	NaN	45.8	NaN
17035	59.0	54.1	67.2	50.6	56.5	NaN	67.2
17039	62.1	65.9	75.0	NaN	66.4	70.7	68.0
17041	67.4	63.7	78.1	62.4	67.1	NaN	69.0
17047	47.7	55.6	44.1	NaN	54.7	63.1	53.5
17051	55.6	53.5	59.5	48.9	54.4	63.5	NaN
17061	62.5	63.9	70.1	60.7	65.2	NaN	66.6
17065	46.1	51.3	51.2	NaN	51.8	NaN	50.2
17067	63.7	66.1	69.8	NaN	66.6	67.3	67.6
17069	39.8	NaN	43.3	44.8	50.8	50.6	NaN
17071	57.8	61.6	60.2	NaN	62.1	66.6	67.2
17077	NaN	50.1	48.9	39.6	52.2	49.9	51.1
17079	58.1	54.6	60.7	53.9	54.5	NaN	55.8
17083	55.3	56.9	NaN	60.1	65.3	NaN	63.2
17085	60.6	53.4	NaN	55.6	59.1	58.7	62.6
17087	NaN	48.0	NaN	NaN	45.8	43.8	45.1
17093	NaN	58.9	60.8	57.4	57.5	66.3	NaN
17097	52.5	NaN	41.3	48.2	51.8	50.3	57.3
17099	61.2	60.8	67.1	55.4	59.3	NaN	65.7
17101	NaN	51.0	54.1	47.8	57.9	60.4	58.0
17107	66.6	64.7	70.1	NaN	68.7	NaN	71.8
17109	62.5	64.7	69.3	60.2	65.1	NaN	66.3
17111	59.1	47.2	58.9	NaN	52.5	NaN	NaN
17123	63.3	60.1	65.5	53.8	63.1	NaN	64.1
17125	59.2	62.1	NaN	55.4	61.2	64.7	62.6
17127	47.5	NaN	NaN	43.5	51.6	52.5	50.0
17131	59.9	60.0	63.2	51.9	59.4	NaN	65.0
17135	61.5	61.9	73.0	61.5	63.2	NaN	68.6

17137	60.4	60.2	79.8	64.1	62.9	NaN	70.0
17147	71.3	69.4	77.4	NaN	71.7	77.3	74.2
17149	58.1	57.7	61.2	NaN	62.1	NaN	NaN
17151	35.1	44.6	NaN	NaN	53.1	NaN	NaN
17155	63.3	57.2	NaN	NaN	61.8	62.2	64.6
17159	51.8	48.8	56.3	NaN	54.0	NaN	56.1
17161	58.0	58.6	60.5	NaN	61.3	64.6	66.3
17165	46.8	49.9	49.4	NaN	59.0	59.2	52.8
17169	56.9	58.6	59.9	NaN	58.1	63.2	63.1
17171	57.9	62.9	NaN	NaN	65.1	64.9	72.1
17175	65.2	64.6	NaN	58.8	63.0	70.1	70.8
17181	43.8	51.6	NaN	34.0	50.2	44.5	48.0
17183	64.6	64.9	71.7	NaN	61.8	67.4	67.6
17185	55.2	54.4	NaN	NaN	54.5	60.8	58.1
17187	62.6	63.3	67.5	58.5	66.6	NaN	64.5
17191	48.3	47.5	52.9	43.4	52.0	NaN	56.4
17195	60.8	58.7	62.7	NaN	NaN	67.0	66.4
17199	42.5	47.2	45.6	NaN	52.8	NaN	45.9
18005	54.8	54.2	55.5	NaN	59.9	NaN	57.3
18009	63.5	54.7	57.4	53.8	NaN	62.4	56.1
18013	51.7	49.8	NaN	NaN	NaN	54.4	53.8
18015	65.1	58.5	63.9	58.7	67.9	NaN	NaN
18017	60.0	53.5	61.8	NaN	59.4	NaN	57.1
18021	NaN	55.6	52.8	48.4	56.4	58.7	58.1
18023	65.6	58.0	65.2	NaN	67.4	61.5	61.0
18025	NaN	NaN	47.9	46.3	53.4	NaN	NaN
18029	NaN	47.9	54.1	30.9	51.4	47.5	47.4
18043	NaN	50.3	NaN	37.1	54.8	NaN	NaN
18045	59.4	56.5	NaN	NaN	64.4	NaN	NaN
18049	56.3	55.8	57.5	51.4	54.3	56.3	NaN
18053	59.3	52.9	59.4	NaN	59.9	62.1	56.3
18055	50.8	46.3	53.2	NaN	55.2	61.4	52.3
18071	55.0	50.5	51.2	NaN	58.9	NaN	59.3
18073	54.7	52.6	56.9	NaN	55.1	58.2	54.6
18075	58.9	50.0	59.5	49.3	55.7	NaN	55.3
18081	NaN	59.1	NaN	53.6	60.9	58.0	62.2
18083	56.6	56.6	52.5	NaN	62.5	67.2	57.4
18089	54.8	50.5	55.7	48.3	52.0	NaN	51.8
18093	47.5	46.9	52.5	NaN	54.7	52.0	50.0
18097	NaN	44.0	NaN	47.6	NaN	47.0	57.8
18099	54.4	52.0	53.3	NaN	50.0	57.8	NaN
18101	55.3	NaN	NaN	43.8	60.6	65.0	57.6
18103	55.8	48.9	55.0	49.1	54.5	NaN	57.1
18105	52.2	45.8	44.7	NaN	51.3	48.9	55.1
18109	54.4	49.0	56.9	NaN	61.3	60.0	59.3
18119	48.9	44.8	NaN	NaN	54.0	55.4	50.9
18123	NaN	NaN	45.0	47.2	57.0	NaN	51.6
18125	51.0	NaN	NaN	51.8	56.7	NaN	50.4

18129	53.9	56.6	54.4	56.3	66.5	NaN	60.7
18131	50.1	51.0	54.5	44.3	49.9	NaN	51.0
18143	NaN	50.6	49.6	NaN	NaN	49.1	NaN
18147	52.4	59.0	51.1	53.3	58.5	60.0	NaN
18149	50.4	48.9	53.8	NaN	46.9	NaN	50.6
18151	50.4	48.2	46.9	NaN	44.7	55.9	54.2
18155	52.1	48.5	47.6	NaN	57.4	49.0	55.2
18157	NaN	56.1	63.2	54.7	66.6	58.8	58.4
18159	66.8	61.3	NaN	56.4	69.8	69.2	62.2
18165	62.8	54.2	60.3	51.4	57.1	64.4	NaN
18167	NaN	55.2	51.5	49.3	50.6	58.4	55.7
18173	50.2	60.9	49.8	53.6	59.4	NaN	57.2
18175	52.5	51.1	52.3	44.1	56.7	NaN	54.7
18179	61.8	50.9	61.0	NaN	58.7	66.0	57.4
18181	61.1	58.8	60.4	52.8	59.3	NaN	56.2
19003	56.5	52.6	53.2	NaN	NaN	NaN	NaN
19005	56.8	52.6	52.8	NaN	58.8	61.9	61.2
19007	55.9	50.7	45.1	NaN	39.1	57.5	40.4
19017	62.1	56.8	57.3	52.5	NaN	62.0	66.7
19021	60.2	56.5	55.5	53.5	NaN	63.8	57.2
19031	60.5	59.6	63.3	NaN	56.4	65.3	67.0
19035	69.6	62.4	64.3	58.8	57.2	NaN	59.6
19037	56.0	50.4	53.2	54.5	55.9	NaN	63.6
19039	49.9	NaN	47.9	NaN	45.9	57.7	40.2
19043	63.2	59.0	58.8	56.4	60.1	NaN	64.5
19053	57.2	NaN	48.0	NaN	47.4	57.0	37.9
19055	63.8	63.0	63.9	NaN	61.1	66.4	67.4
19057	60.4	59.6	63.8	55.1	60.7	NaN	66.9
19061	57.8	59.6	62.0	60.2	NaN	70.5	68.6
19063	58.5	58.9	46.7	NaN	56.0	59.6	62.8
19067	57.5	54.1	49.8	54.3	54.0	NaN	61.9
19071	55.5	60.1	50.6	NaN	54.8	63.1	53.4
19077	58.4	59.2	52.0	54.2	45.5	NaN	58.6
19085	58.2	53.7	52.6	NaN	52.1	NaN	52.4
19093	67.6	62.1	63.8	NaN	55.5	65.6	63.1
19095	57.4	56.7	56.0	52.3	49.8	NaN	61.6
19097	61.1	59.2	58.1	NaN	54.7	NaN	66.3
19099	60.3	60.5	60.7	56.5	NaN	67.3	60.1
19103	56.1	56.4	59.9	55.5	51.8	NaN	62.1
19107	57.1	51.6	57.7	55.8	51.6	NaN	53.1
19113	60.0	56.0	57.7	56.5	NaN	NaN	64.6
19115	56.4	59.0	58.1	54.3	53.3	NaN	61.4
19117	54.4	41.4	41.5	NaN	43.8	54.1	43.1
19123	59.1	54.4	59.7	53.7	NaN	62.0	59.3
19125	56.6	52.9	58.3	54.0	NaN	60.6	57.8
19127	62.6	63.5	61.9	60.6	NaN	67.1	65.0
19129	56.3	55.7	NaN	NaN	49.3	NaN	52.1
19143	56.1	59.7	57.2	50.5	NaN	62.9	56.3

19147	59.7	57.8	48.8	55.6	NaN	61.6	57.9
19149	66.1	59.6	61.3	NaN	NaN	NaN	47.8
19151	60.6	58.2	49.7	NaN	54.6	65.0	56.5
19155	60.6	57.0	NaN	58.5	48.0	64.4	51.0
19157	57.5	56.6	58.2	53.4	54.8	NaN	60.4
19159	51.7	48.8	47.2	NaN	50.6	54.0	41.8
19161	63.3	57.8	60.7	59.3	47.1	NaN	61.3
19165	60.5	58.2	57.8	58.7	52.8	NaN	58.0
19167	65.8	65.2	65.8	60.8	NaN	70.5	62.9
19173	51.6	52.8	NaN	49.9	47.0	61.3	52.9
19181	54.3	49.9	52.0	52.3	49.5	NaN	50.2
19183	55.8	61.7	61.5	54.7	53.2	NaN	57.2
19185	54.2	NaN	49.0	52.5	44.6	57.1	NaN
19193	61.2	58.1	59.5	56.8	56.6	NaN	56.4
20007	NaN	NaN	NaN	26.7	NaN	28.6	20.2
20009	NaN	31.9	NaN	29.9	43.2	37.7	19.1
20011	NaN	40.8	38.2	35.9	35.0	33.1	17.0
20017	47.9	43.7	NaN	NaN	39.7	41.8	33.6
20019	31.4	33.5	33.8	40.3	NaN	18.8	12.2
20021	NaN	NaN	NaN	38.9	39.6	32.2	16.0
20023	NaN	56.2	NaN	NaN	62.1	NaN	62.4
20027	48.9	39.7	46.2	45.3	51.9	NaN	28.5
20029	52.5	37.1	NaN	NaN	44.7	39.8	26.3
20035	42.4	30.9	NaN	37.5	37.4	26.6	9.5
20037	39.2	NaN	38.7	NaN	32.4	34.2	NaN
20043	62.3	58.5	53.3	NaN	58.3	64.3	NaN
20047	NaN	53.4	NaN	NaN	63.9	55.9	36.0
20049	41.9	34.3	43.6	39.9	NaN	33.7	21.0
20053	NaN	NaN	48.1	NaN	41.3	34.3	14.5
20055	68.2	NaN	NaN	56.2	NaN	NaN	NaN
20057	NaN	NaN	NaN	NaN	61.6	NaN	NaN
20059	49.9	41.7	43.4	NaN	40.1	36.9	23.5
20063	NaN	NaN	50.9	NaN	NaN	NaN	NaN
20065	NaN	50.2	54.5	NaN	NaN	NaN	NaN
20067	NaN	NaN	NaN	NaN	NaN	53.1	NaN
20069	68.1	62.8	NaN	53.4	NaN	59.7	NaN
20073	NaN	36.4	34.8	NaN	32.0	30.8	19.6
20077	NaN	22.0	29.4	31.0	32.4	24.0	9.4
20083	NaN	55.7	57.1	NaN	NaN	NaN	49.2
20085	47.7	42.9	39.7	46.2	47.1	45.0	NaN
20087	49.2	52.8	NaN	49.2	46.2	52.0	36.7
20091	NaN	NaN	34.1	NaN	NaN	38.8	32.8
20097	57.8	NaN	63.3	NaN	53.3	50.0	41.0
20099	NaN	39.4	33.7	NaN	36.7	24.4	9.3
20103	NaN	NaN	NaN	46.0	47.3	42.0	38.2
20105	NaN	NaN	NaN	NaN	42.2	NaN	12.3
20107	41.8	NaN	NaN	NaN	40.0	33.5	20.9
20111	41.7	38.6	NaN	34.0	35.9	36.1	23.2

20119	NaN	70.9	NaN	NaN	73.1	69.1	67.6
20123	53.4	NaN	43.0	43.3	44.0	29.1	24.7
20131	56.1	33.3	41.5	NaN	45.7	57.6	39.8
20133	43.6	35.7	38.9	NaN	31.4	29.5	11.0
20137	NaN	NaN	47.8	NaN	32.5	33.4	NaN
20141	50.3	27.5	NaN	NaN	38.5	NaN	13.6
20143	45.0	29.9	NaN	NaN	43.9	NaN	20.5
20145	49.5	43.1	NaN	NaN	56.7	NaN	44.3
20147	54.0	NaN	NaN	43.8	27.4	38.7	25.7
20149	54.5	48.1	40.0	49.5	45.8	40.7	NaN
20151	57.4	51.8	54.0	NaN	54.7	44.9	37.8
20153	NaN	NaN	NaN	NaN	NaN	36.5	NaN
20155	45.9	23.1	NaN	32.8	38.0	NaN	21.4
20159	43.3	29.3	NaN	NaN	35.7	37.6	16.4
20163	NaN	25.5	NaN	NaN	29.7	30.3	13.9
20165	44.1	NaN	37.3	NaN	NaN	NaN	NaN
20167	41.7	25.2	NaN	30.3	41.9	33.5	NaN
20169	42.4	22.6	NaN	43.0	39.4	34.4	19.7
20171	NaN	NaN	51.5	NaN	NaN	NaN	NaN
20177	48.9	39.0	37.7	NaN	41.5	42.1	35.2
20179	NaN	NaN	NaN	NaN	59.5	59.1	NaN
20181	NaN	44.4	NaN	NaN	NaN	NaN	NaN
20183	NaN	39.6	45.8	NaN	37.5	43.8	24.7
20185	48.7	NaN	NaN	NaN	43.8	NaN	NaN
20189	61.3	58.1	NaN	NaN	62.0	57.9	64.1
20193	NaN	NaN	NaN	NaN	45.6	NaN	40.2
20195	NaN	NaN	NaN	34.8	NaN	NaN	NaN
20197	NaN	36.8	31.0	NaN	41.9	39.8	34.2
20201	43.4	33.2	NaN	NaN	45.2	48.8	42.0
20203	NaN	NaN	52.5	NaN	NaN	NaN	NaN
20205	44.1	37.7	NaN	39.4	29.2	32.3	15.8
20207	NaN	36.0	NaN	NaN	33.1	31.7	NaN
21003	NaN	NaN	NaN	NaN	54.2	41.3	NaN
21005	NaN	NaN	NaN	NaN	NaN	53.4	NaN
21007	50.1	49.4	NaN	51.5	56.9	NaN	49.2
21009	53.3	56.7	NaN	44.0	60.9	54.5	52.8
21011	46.4	NaN	NaN	48.1	43.4	47.3	52.1
21017	47.5	50.9	47.4	NaN	50.4	61.6	NaN
21023	45.4	51.3	51.9	NaN	NaN	51.3	NaN
21029	47.8	51.8	52.4	30.1	52.6	47.5	NaN
21035	41.0	43.7	43.0	NaN	52.4	57.7	42.0
21039	48.1	NaN	51.2	NaN	56.0	54.2	45.3
21041	39.6	NaN	NaN	NaN	50.2	47.6	53.2
21045	NaN	54.0	NaN	47.0	55.1	56.8	51.0
21051	NaN	NaN	NaN	NaN	49.2	46.1	44.8
21053	49.4	NaN	NaN	NaN	54.8	NaN	53.8
21055	50.0	49.7	51.5	NaN	54.1	55.1	43.4
21057	NaN	NaN	46.3	53.9	59.9	NaN	50.5

21065	42.9	42.5	35.2	35.0	50.3	NaN	NaN
21069	45.7	44.9	NaN	40.5	48.5	49.8	50.5
21073	NaN	NaN	NaN	NaN	48.2	51.9	NaN
21075	49.4	52.1	NaN	48.0	54.7	57.1	45.5
21077	42.0	52.0	NaN	NaN	NaN	NaN	NaN
21079	NaN	49.0	NaN	NaN	NaN	56.7	NaN
21081	NaN	NaN	NaN	49.5	51.7	48.2	50.0
21083	50.2	51.4	45.4	NaN	52.0	53.1	42.8
21085	51.8	51.7	50.4	NaN	50.5	53.7	51.3
21087	51.5	55.5	55.0	NaN	55.8	56.2	59.7
21089	46.5	NaN	46.7	NaN	42.3	41.9	48.0
21091	50.6	59.0	NaN	54.0	58.6	66.5	NaN
21097	46.9	NaN	NaN	NaN	50.7	NaN	NaN
21101	NaN	58.8	52.0	53.0	59.0	NaN	58.8
21103	40.0	45.0	48.5	42.5	51.3	55.6	NaN
21107	NaN	NaN	NaN	NaN	53.2	NaN	NaN
21111	NaN	NaN	NaN	25.0	50.7	56.6	NaN
21117	NaN	NaN	NaN	NaN	NaN	51.3	NaN
21123	NaN	58.0	NaN	47.0	58.5	57.6	56.9
21135	50.2	NaN	NaN	44.0	46.7	45.8	50.7
21137	53.4	NaN	49.9	37.0	54.1	56.9	51.6
21139	48.4	46.0	NaN	47.0	54.0	49.3	41.3
21143	NaN	NaN	46.0	43.0	51.6	NaN	45.0
21145	NaN	NaN	NaN	NaN	51.4	49.9	NaN
21149	NaN	55.2	58.3	55.0	58.1	60.4	54.7
21151	47.5	53.9	48.6	30.0	54.2	57.9	NaN
21157	NaN	46.7	43.0	34.0	52.2	52.8	NaN
21167	54.8	54.0	NaN	42.0	NaN	NaN	59.7
21169	53.0	54.0	NaN	NaN	53.2	58.9	58.3
21171	NaN	NaN	49.9	NaN	NaN	NaN	57.8
21173	NaN	NaN	NaN	NaN	44.9	NaN	NaN
21177	NaN	NaN	47.3	NaN	51.0	54.5	50.4
21179	52.1	50.3	NaN	33.5	50.3	53.4	NaN
21181	44.7	49.3	NaN	36.0	NaN	NaN	49.0
21183	NaN	56.4	58.5	53.0	53.6	57.6	60.7
21185	45.5	46.4	46.1	39.5	51.7	50.0	NaN
21187	NaN	NaN	34.7	NaN	53.1	53.4	NaN
21191	46.0	60.9	46.8	NaN	46.1	NaN	50.5
21197	NaN	NaN	NaN	NaN	NaN	NaN	46.7
21203	45.4	NaN	NaN	44.0	46.2	43.4	NaN
21205	36.7	46.8	NaN	NaN	NaN	NaN	NaN
21209	50.8	57.5	NaN	41.0	50.0	NaN	56.1
21229	54.0	52.7	NaN	30.5	52.7	56.5	53.2
22003	25.9	NaN	NaN	NaN	NaN	NaN	NaN
22007	46.2	39.7	NaN	38.1	NaN	42.6	50.0
22011	NaN	37.1	NaN	NaN	NaN	NaN	NaN
22015	29.7	56.0	NaN	NaN	NaN	37.6	NaN
22019	NaN	34.1	NaN	NaN	NaN	NaN	NaN

```
22043  45.4   NaN  39.8  38.4  38.7  46.4  49.3
22047  51.9  58.0  57.6  47.3  51.5   NaN   NaN
22053  24.1  31.3  36.9   NaN   NaN   NaN  27.9
22055  33.6  37.6   NaN   NaN   NaN   NaN   NaN
22067  54.0  61.5  54.1   NaN  53.6  60.0  50.7
22073  38.8  56.9   NaN   NaN  44.5   NaN  35.1
22081   NaN   NaN  36.2  36.6  39.6   NaN   NaN
22083  48.7  57.0  56.5   NaN  57.6  61.5  50.6
22093   NaN   NaN   NaN   NaN   NaN  38.3   NaN
22101  42.9  45.8   NaN   NaN  38.7  38.4  42.7
22117  55.8   NaN   NaN  35.4  50.8  38.0  45.0
22121   NaN  57.4   NaN   NaN  50.5  58.7  52.1

INTERSECTION REPORT
USDA counties:                649
Sentinel counties:            15
Multimodal intersection:      15
Sentinel-only (no USDA):      0
  -> []

Full multimodal_index:
 fips state_name county_name  year  yield_bu_acre production_bu  total_tiles
split
13001    GEORGIA     APPLING  2016           31.4       1639174          NaN
train
08001   COLORADO       ADAMS  2016           32.8        630416          NaN
train
21001   KENTUCKY       ADAIR  2016           43.9       2928832          NaN
train
18001    INDIANA       ADAMS  2016           59.3      10476590          NaN
test
19001       IOWA       ADAIR  2016           60.8       9320397          NaN
train
10001   DELAWARE        KENT  2016           44.5       3667156          NaN
train
04001    ARIZONA      APACHE  2016           30.1        152908          NaN
train
05001   ARKANSAS    ARKANSAS  2016           48.4      16862318          NaN
train
06001 CALIFORNIA     ALAMEDA  2016           32.7        256303          NaN
train
16001      IDAHO         ADA  2016           38.7        491645          NaN
train
12001    FLORIDA     ALACHUA  2016           29.1        278807          NaN
train
22001  LOUISIANA      ACADIA  2016           37.6       7413742          NaN
train
20001     KANSAS    ANDERSON  2016           45.4       5337814          NaN
```

```
train
17001   ILLINOIS       ADAMS  2016           55.5     14611707          NaN
test
01001    ALABAMA      AUTAGA  2016           61.8       824000          NaN
test
13001    GEORGIA     APPLING  2017           32.2      1754642          NaN
test
08001   COLORADO       ADAMS  2017           35.1      683,362          NaN
train
21001   KENTUCKY       ADAIR  2017           44.5      2795579          NaN
train
18001    INDIANA       ADAMS  2017           54.1      8962693          NaN
train
19001       IOWA       ADAIR  2017           57.7      8727875          NaN
train
10001   DELAWARE        KENT  2017           43.2      3379882          NaN
train
04001    ARIZONA      APACHE  2017           29.8       145305          NaN
train
05001   ARKANSAS    ARKANSAS  2017           45.8     16060045          NaN
train
06001 CALIFORNIA     ALAMEDA  2017           29.4       225880          NaN
test
16001      IDAHO         ADA  2017           34.4       409738          NaN
train
12001    FLORIDA     ALACHUA  2017           26.9       244602          NaN
train
22001  LOUISIANA      ACADIA  2017           36.9      7008159          NaN
train
20001     KANSAS    ANDERSON  2017           45.0      5217885          NaN
test
17001   ILLINOIS       ADAMS  2017           56.1     13433482          NaN
train
01001    ALABAMA      AUTAGA  2017           48.0       415000          NaN
train
13001    GEORGIA     APPLING  2018           27.3      1411192          NaN
test
08001   COLORADO       ADAMS  2018           29.9       516283          NaN
train
21001   KENTUCKY       ADAIR  2018           38.2      2112995          NaN
train
18001    INDIANA       ADAMS  2018           52.9      8092854          NaN
test
19001       IOWA       ADAIR  2018           52.0      6858852          NaN
train
10001   DELAWARE        KENT  2018           41.4      2891376          NaN
train
04001    ARIZONA      APACHE  2018           25.3       104666          NaN
```

```
train
05001   ARKANSAS    ARKANSAS  2018           43.5      12871824          NaN
test
06001 CALIFORNIA     ALAMEDA  2018           28.6        213785          NaN
test
16001      IDAHO         ADA  2018           31.6        327439          NaN
train
12001    FLORIDA     ALACHUA  2018           27.7        215866          NaN
test
22001  LOUISIANA      ACADIA  2018           37.7       7166205          NaN
train
20001     KANSAS    ANDERSON  2018           43.3       4898486          NaN
train
17001   ILLINOIS       ADAMS  2018           51.0      11473623          NaN
train
01001    ALABAMA      AUTAGA  2018           65.4        627000          NaN
train
13001    GEORGIA     APPLING  2019           31.7       1486667          NaN
test
08001   COLORADO       ADAMS  2019           27.0       482,544          NaN
train
21001   KENTUCKY       ADAIR  2019           36.0       1821852          NaN
train
18001    INDIANA       ADAMS  2019           47.6       7108917          NaN
train
19001       IOWA       ADAIR  2019           55.0       7054685          NaN
train
10001   DELAWARE        KENT  2019           45.0       2000000          NaN
train
04001    ARIZONA      APACHE  2019           23.9         93808          NaN
train
05001   ARKANSAS    ARKANSAS  2019           38.1      11271085          NaN
train
06001 CALIFORNIA     ALAMEDA  2019           28.7        197944          NaN
train
16001      IDAHO         ADA  2019           34.0        367642          NaN
test
12001    FLORIDA     ALACHUA  2019           28.1        210638          NaN
train
22001  LOUISIANA      ACADIA  2019           39.2         93100          NaN
train
20001     KANSAS    ANDERSON  2019           43.1       4568772          NaN
train
17001   ILLINOIS       ADAMS  2019           48.6      10623717          NaN
test
01001    ALABAMA      AUTAGA  2019           39.3         93000          NaN
train
13001    GEORGIA     APPLING  2020           34.2       1709897          NaN
```

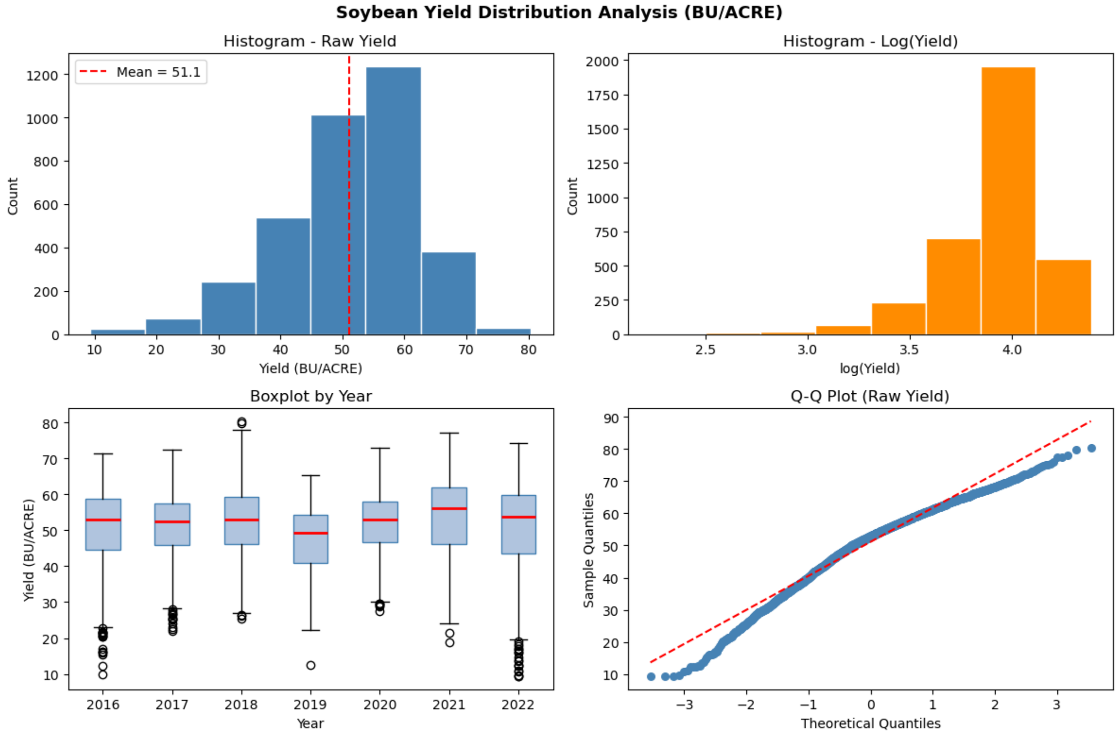
```
train
08001   COLORADO       ADAMS  2020           33.5       623368          NaN
train
21001   KENTUCKY       ADAIR  2020           42.5      2513492          NaN
train
18001    INDIANA       ADAMS  2020           55.3      8611703          NaN
train
19001       IOWA       ADAIR  2020           54.1      8095957          NaN
train
10001   DELAWARE        KENT  2020           44.2      3601991          NaN
train
04001    ARIZONA      APACHE  2020           28.8       131357          NaN
train
05001   ARKANSAS    ARKANSAS  2020           45.8     14230518          NaN
train
06001 CALIFORNIA     ALAMEDA  2020           30.5       248270          NaN
train
16001      IDAHO         ADA  2020           35.5       402676          NaN
train
12001    FLORIDA     ALACHUA  2020           29.4       252693          NaN
train
22001  LOUISIANA      ACADIA  2020           37.3      6856598          NaN
train
20001     KANSAS    ANDERSON  2020           46.5      5411996          NaN
test
17001   ILLINOIS       ADAMS  2020           52.8     11734061          NaN
train
01001    ALABAMA      AUTAGA  2020           47.2       307000          NaN
train
13001    GEORGIA     APPLING  2021           33.6      1688568         76.0
test
08001   COLORADO       ADAMS  2021           31.0       605864        120.0
train
21001   KENTUCKY       ADAIR  2021           44.2      2895498         52.0
train
18001    INDIANA       ADAMS  2021           58.3      9676692         24.0
train
19001       IOWA       ADAIR  2021           58.5      9038074         48.0
train
10001   DELAWARE        KENT  2021           47.6      3929237         80.0
train
04001    ARIZONA      APACHE  2021           28.4       133963          NaN
train
05001   ARKANSAS    ARKANSAS  2021           47.2     15535927        148.0
train
06001 CALIFORNIA     ALAMEDA  2021           32.6       269374        120.0
train
16001      IDAHO         ADA  2021           37.5       461625        132.0
```

```
test
12001     FLORIDA     ALACHUA  2021           27.9       269,347        116.0
train
22001   LOUISIANA      ACADIA  2021           34.8       6700044         76.0
train
20001      KANSAS    ANDERSON  2021           46.2       5664397         32.0
train
17001    ILLINOIS       ADAMS  2021           57.5      14086178         96.0
test
01001     ALABAMA      AUTAGA  2021           54.3        475000         60.0
train
13001     GEORGIA     APPLING  2022           31.6       1484473         76.0
train
08001    COLORADO       ADAMS  2022           32.6        628756        120.0
train
21001    KENTUCKY       ADAIR  2022           40.6       2334378         52.0
train
18001     INDIANA       ADAMS  2022           54.5       8937400         24.0
test
19001        IOWA       ADAIR  2022           57.3       8535408         48.0
train
10001    DELAWARE        KENT  2022           41.6       3088218         80.0
train
04001     ARIZONA      APACHE  2022           26.6        120471       1524.0
train
05001    ARKANSAS    ARKANSAS  2022           45.2      14170968        148.0
train
06001  CALIFORNIA     ALAMEDA  2022           30.9        233233        120.0
train
16001       IDAHO         ADA  2022           38.0        411882        132.0
train
12001     FLORIDA     ALACHUA  2022           27.0       236,547        116.0
train
22001   LOUISIANA      ACADIA  2022           36.5       6836815         76.0
test
20001      KANSAS    ANDERSON  2022           43.4       5029366         32.0
train
17001    ILLINOIS       ADAMS  2022           53.2      12457897         96.0
train
01001     ALABAMA      AUTAGA  2022           47.1        545000         60.0
test
```

[8]:
```
# Target distribution: histogram, boxplot, normality test
import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
from scipy import stats
import numpy as np

yield_vals = usda_index["yield_bu_acre"].values
log_yield  = np.log(yield_vals)

fig, axes = plt.subplots(2, 2, figsize=(12, 8))
fig.suptitle("Soybean Yield Distribution Analysis (BU/ACRE)", fontsize=13,␣
↪fontweight="bold")

# -- Raw yield histogram
axes[0, 0].hist(yield_vals, bins=8, color="steelblue", edgecolor="white")
axes[0, 0].set_title("Histogram - Raw Yield")
axes[0, 0].set_xlabel("Yield (BU/ACRE)")
axes[0, 0].set_ylabel("Count")
axes[0, 0].axvline(np.mean(yield_vals), color="red", linestyle="--",␣
↪label=f"Mean = {np.mean(yield_vals):.1f}")
axes[0, 0].legend()

# -- Log-yield histogram
axes[0, 1].hist(log_yield, bins=8, color="darkorange", edgecolor="white")
axes[0, 1].set_title("Histogram - Log(Yield)")
axes[0, 1].set_xlabel("log(Yield)")
axes[0, 1].set_ylabel("Count")

# -- Boxplot by year (native matplotlib - avoids pandas/matplotlib backend␣
↪conflict)
usda_index_plot = usda_index.copy()
usda_index_plot["year"] = usda_index_plot["year"].astype(str)
years_sorted = sorted(usda_index_plot["year"].unique())
box_data = [usda_index_plot[usda_index_plot["year"] == y]["yield_bu_acre"].
↪values for y in years_sorted]
bp = axes[1, 0].boxplot(box_data, labels=years_sorted, patch_artist=True)
for patch in bp["boxes"]:
    patch.set_facecolor("lightsteelblue")
    patch.set_edgecolor("steelblue")
for median in bp["medians"]:
    median.set(color="red", linewidth=2)
axes[1, 0].set_title("Boxplot by Year")
axes[1, 0].set_xlabel("Year")
axes[1, 0].set_ylabel("Yield (BU/ACRE)")

# -- Q-Q plot for normality
(osm, osr), (slope, intercept, r) = stats.probplot(yield_vals)
axes[1, 1].scatter(osm, osr, color="steelblue", s=30)
line_x = np.array([min(osm), max(osm)])
axes[1, 1].plot(line_x, slope * line_x + intercept, color="red", linestyle="--")
```

```python
axes[1, 1].set_title("Q-Q Plot (Raw Yield)")
axes[1, 1].set_xlabel("Theoretical Quantiles")
axes[1, 1].set_ylabel("Sample Quantiles")

plt.tight_layout()

# Render to PNG in memory and display inline (Agg backend - no GUI required)
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

# -- Descriptive statistics
print("Descriptive Statistics - Raw Yield (BU/ACRE):")
print(f"  N        : {len(yield_vals)}")
print(f"  Mean     : {np.mean(yield_vals):.2f}")
print(f"  Std      : {np.std(yield_vals):.2f}")
print(f"  Min      : {np.min(yield_vals):.2f}")
print(f"  Max      : {np.max(yield_vals):.2f}")
print(f"  Skewness : {stats.skew(yield_vals):.3f}")
print(f"  Kurtosis : {stats.kurtosis(yield_vals):.3f}")
print()

# -- Shapiro-Wilk normality test (appropriate for n < 50)
sw_stat, sw_p = stats.shapiro(yield_vals)
print(f"Shapiro-Wilk test:  W = {sw_stat:.4f},  p = {sw_p:.4f}")
if sw_p > 0.05:
    print("  Result: Cannot reject normality (p > 0.05). Raw yield is␣
↪approximately normal.")
    USE_LOG = False
else:
    print("  Result: Reject normality (p <= 0.05). Log-transformed yield will␣
↪be used as model target.")
    USE_LOG = True

log_sw_stat, log_sw_p = stats.shapiro(log_yield)
print(f"Shapiro-Wilk on log(yield): W = {log_sw_stat:.4f},  p = {log_sw_p:.4f}")
print()
print(f"Decision: USE_LOG_YIELD = {USE_LOG}")
```




```
Descriptive Statistics - Raw Yield (BU/ACRE):
  N        : 3535
  Mean     : 51.13
  Std      : 10.74
  Min      : 9.30
  Max      : 80.40
  Skewness : -0.735
  Kurtosis : 0.588

Shapiro-Wilk test:  W = 0.9675,  p = 0.0000
  Result: Reject normality (p <= 0.05). Log-transformed yield will be used as
model target.
Shapiro-Wilk on log(yield): W = 0.8653,  p = 0.0000

Decision: USE_LOG_YIELD = True
```

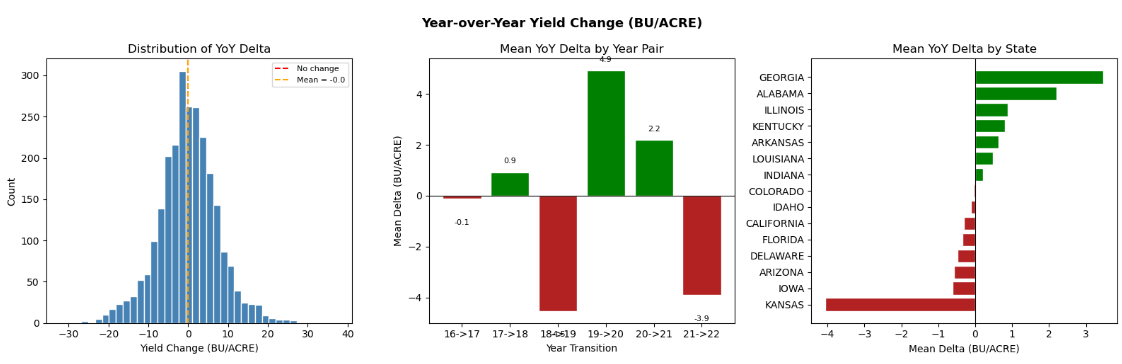
[9]:
```
# Year-over-year yield change per county
# For each county that appears in consecutive years, compute absolute and
# relative yield change. Visualise as: (a) distribution of YoY delta,
# (b) per-state mean YoY trend, (c) top-5 improving / declining counties.

import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
```

```python
import numpy as np
from scipy import stats

# Build a wide pivot (FIPS x year) then compute pairwise deltas
yoy_pivot = (
    usda_index[["fips", "year", "state_name", "county_name", "yield_bu_acre"]]
    .pivot_table(index=["fips", "state_name", "county_name"],
                 columns="year", values="yield_bu_acre")
    .reset_index()
)
yoy_pivot.columns.name = None

all_years  = sorted(usda_index["year"].unique())
year_pairs = [(all_years[i], all_years[i+1]) for i in range(len(all_years)-1)]

delta_frames = []
for y0, y1 in year_pairs:
    if y0 not in yoy_pivot.columns or y1 not in yoy_pivot.columns:
        continue
    sub = yoy_pivot[["fips", "state_name", "county_name", y0, y1]].dropna().
↪copy()
    sub["year_from"]  = y0
    sub["year_to"]    = y1
    sub["delta_abs"]  = sub[y1] - sub[y0]
    sub["delta_pct"]  = (sub[y1] - sub[y0]) / sub[y0] * 100
    sub["yield_from"] = sub[y0]
    sub["yield_to"]   = sub[y1]
    delta_frames.append(sub[["fips","state_name","county_name",
                              "year_from","year_to",
                              "yield_from","yield_to",
                              "delta_abs","delta_pct"]])

yoy_df = pd.concat(delta_frames, ignore_index=True)

print("YEAR-OVER-YEAR YIELD CHANGE")
print(f"County-pair observations : {len(yoy_df)}")
print(f"Mean absolute delta      : {yoy_df['delta_abs'].mean():.2f} BU/ACRE")
print(f"Std absolute delta       : {yoy_df['delta_abs'].std():.2f} BU/ACRE")
print(f"Mean pct delta           : {yoy_df['delta_pct'].mean():.2f} %")
print()

yoy_summary = (
    yoy_df.groupby(["year_from","year_to"])
    .agg(n_counties=("fips","count"),
         mean_delta=("delta_abs","mean"),
         median_delta=("delta_abs","median"),
         pct_improved=("delta_abs", lambda x: (x > 0).mean() * 100))
```

```python
    .reset_index()
)
print("Year-pair summary:")
print(yoy_summary.to_string(index=False))
print()

#   Plots
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle("Year-over-Year Yield Change (BU/ACRE)", fontsize=13, fontweight="bold")

# (a) Distribution of all YoY deltas
axes[0].hist(yoy_df["delta_abs"], bins=40, color="steelblue", edgecolor="white")
axes[0].axvline(0, color="red", linestyle="--", linewidth=1.5, label="No change")
axes[0].axvline(yoy_df["delta_abs"].mean(), color="orange", linestyle="--",
                linewidth=1.5, label=f"Mean = {yoy_df['delta_abs'].mean():.1f}")
axes[0].set_title("Distribution of YoY Delta")
axes[0].set_xlabel("Yield Change (BU/ACRE)")
axes[0].set_ylabel("Count")
axes[0].legend(fontsize=8)

# (b) Mean YoY delta by year-pair
pair_labels = [f"{r.year_from[-2:]}->{r.year_to[-2:]}" for _, r in yoy_summary.iterrows()]
bar_colors  = ["green" if v >= 0 else "firebrick" for v in yoy_summary["mean_delta"]]
axes[1].bar(pair_labels, yoy_summary["mean_delta"], color=bar_colors, edgecolor="white")
axes[1].axhline(0, color="black", linewidth=0.8)
axes[1].set_title("Mean YoY Delta by Year Pair")
axes[1].set_xlabel("Year Transition")
axes[1].set_ylabel("Mean Delta (BU/ACRE)")
for i, val in enumerate(yoy_summary["mean_delta"]):
    axes[1].text(i, val + (0.3 if val >= 0 else -0.8), f"{val:.1f}",
                 ha="center", va="bottom" if val >= 0 else "top", fontsize=8)

# (c) Per-state mean YoY delta
state_delta = yoy_df.groupby("state_name")["delta_abs"].mean().sort_values()
bar_cols2   = ["green" if v >= 0 else "firebrick" for v in state_delta.values]
axes[2].barh(state_delta.index, state_delta.values, color=bar_cols2, edgecolor="white")
axes[2].axvline(0, color="black", linewidth=0.8)
axes[2].set_title("Mean YoY Delta by State")
axes[2].set_xlabel("Mean Delta (BU/ACRE)")
```

```python
plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

# Top-5 improving and declining counties
county_trend = (
    yoy_df.groupby(["fips","state_name","county_name"])["delta_abs"]
    .mean().reset_index().sort_values("delta_abs", ascending=False)
)
print("Top-5 most improving counties (avg YoY delta):")
print(county_trend.head(5).to_string(index=False))
print()
print("Top-5 most declining counties (avg YoY delta):")
print(county_trend.tail(5).to_string(index=False))
```

```
YEAR-OVER-YEAR YIELD CHANGE
County-pair observations : 2557
Mean absolute delta      : -0.02 BU/ACRE
Std absolute delta       : 7.48 BU/ACRE
Mean pct delta           : 1.30 %

Year-pair summary:
year_from year_to  n_counties  mean_delta  median_delta  pct_improved
     2016     2017         490   -0.116327         -0.60     44.897959
     2017     2018         440    0.920455          0.90     56.363636
     2018     2019         366   -4.533060         -4.25     25.683060
     2019     2020         399    4.916040          4.60     77.443609
     2020     2021         437    2.184668          2.10     65.217391
     2021     2022         425   -3.901412         -2.90     25.882353
```




```
Top-5 most improving counties (avg YoY delta):
```

```
 fips state_name county_name  delta_abs
13115    GEORGIA       FLOYD       37.7
13015    GEORGIA      BARTOW       35.3
13119    GEORGIA    FRANKLIN       27.5
22015  LOUISIANA     BOSSIER       26.3
22073  LOUISIANA    OUACHITA       18.1

Top-5 most declining counties (avg YoY delta):
 fips state_name county_name  delta_abs
21003   KENTUCKY       ALLEN     -12.90
20053     KANSAS   ELLSWORTH     -13.40
20047     KANSAS     EDWARDS     -13.95
20143     KANSAS      OTTAWA     -15.10
20141     KANSAS     OSBORNE     -22.80
```

[10]:

```python
# Geographic coverage summary with state-level bar chart
# (Choropleth requires an internet-hosted GeoJSON; offline fallback renders
#  a state-level yield bar chart and documents Sentinel coverage.)

import io
from IPython.display import display, Image, HTML
import matplotlib.pyplot as plt
import matplotlib.cm as cm
import matplotlib.colors as mcolors
import numpy as np

#   Try online choropleth first; fall back to offline bar chart
import json, urllib.request, warnings

GEOJSON_URL = "https://raw.githubusercontent.com/plotly/datasets/master/
↪geojson-counties-fips.json"
counties_geo = None
try:
    with urllib.request.urlopen(GEOJSON_URL, timeout=10) as resp:
        counties_geo = json.loads(resp.read())
    print("County GeoJSON loaded - rendering interactive choropleth.")
except Exception:
    print("GeoJSON unavailable")

if counties_geo is not None:
    #   Online path: plotly choropleth
    import plotly.graph_objects as go
    import plotly.express as px

    county_mean = (
        usda_index.groupby("fips")["yield_bu_acre"]
        .mean().reset_index()
```

```
        .rename(columns={"yield_bu_acre": "mean_yield"})
    )
    county_mean["fips_str"] = county_mean["fips"].astype(str).str.zfill(5)

    fig = px.choropleth(
        county_mean, geojson=counties_geo,
        locations="fips_str", color="mean_yield",
        color_continuous_scale="YlGn",
        range_color=(county_mean["mean_yield"].quantile(0.05),
                     county_mean["mean_yield"].quantile(0.95)),
        scope="usa",
        labels={"mean_yield": "Mean Yield (BU/ACRE)"},
        title="County Mean Soybean Yield 2016-2022<br>"
              "<sup>Orange = Sentinel-2 coverage counties</sup>",
    )
    fig.update_layout(margin={"r":0,"t":60,"l":0,"b":0})

    def fips_centroid(geo, fips5):
        for feat in geo["features"]:
            if feat["id"] == fips5:
                coords = feat["geometry"]["coordinates"]
                pts = (np.array(coords[0]) if feat["geometry"]["type"] == "Polygon"
                       else np.concatenate([np.array(r[0]) for r in coords]))
                return float(pts[:,0].mean()), float(pts[:,1].mean())
        return None, None

    s_lons, s_lats, s_labels = [], [], []
    for fp in sorted(sentinel_index["fips"].unique()):
        lon, lat = fips_centroid(counties_geo, fp.zfill(5))
        if lon is not None:
            s_lons.append(lon); s_lats.append(lat); s_labels.append(fp)
    fig.add_trace(go.Scattergeo(
        lon=s_lons, lat=s_lats, text=s_labels, mode="markers",
        marker=dict(size=10, color="orange", line=dict(color="black",width=1)),
        name="Sentinel-2 counties",
    ))
    fig.update_geos(scope="usa")
    display(HTML(fig.to_html(full_html=False, include_plotlyjs="cdn")))
    print(f"Choropleth: {len(county_mean)} counties, {len(s_lons)} Sentinel markers.")

else:
    # static state-level summary
    state_stats = (
        usda_index.groupby("state_name")["yield_bu_acre"]
        .agg(mean_yield="mean", std_yield="std", n_counties="count")
```

```
        .reset_index()
        .sort_values("mean_yield", ascending=True)
    )

    sentinel_states = set(
        usda_index[usda_index["fips"].isin(sentinel_index["fips"].
↪unique())]["state_name"]
    )
    # Sentinel FIPS not in USDA, so we note separate coverage
    sentinel_state_abbrev_map = {
        "01":"ALABAMA","05":"ARKANSAS","04":"ARIZONA","06":"CALIFORNIA",
        "08":"COLORADO","10":"DELAWARE","12":"FLORIDA","13":"GEORGIA",
        "19":"IOWA","16":"IDAHO","17":"ILLINOIS","18":"INDIANA",
        "20":"KANSAS","21":"KENTUCKY","22":"LOUISIANA"
    }
    sentinel_state_names = set(sentinel_state_abbrev_map.values())

    norm  = mcolors.Normalize(vmin=state_stats["mean_yield"].min(),
                              vmax=state_stats["mean_yield"].max())
    cmap  = cm.YlGn
    colors = [cmap(norm(v)) for v in state_stats["mean_yield"]]

    fig, axes = plt.subplots(1, 2, figsize=(14, 6))
    fig.suptitle("State-Level Soybean Yield & Sentinel Coverage",
                 fontsize=13, fontweight="bold")

    # Left: mean yield per state
    bars = axes[0].barh(state_stats["state_name"], state_stats["mean_yield"],
                        color=colors, edgecolor="white")
    axes[0].set_xlabel("Mean Yield (BU/ACRE)")
    axes[0].set_title("Mean Yield by State (2016-2022)")
    # Mark states with Sentinel coverage in orange
    for bar, sname in zip(bars, state_stats["state_name"]):
        if sname in sentinel_state_names:
            bar.set_edgecolor("orange")
            bar.set_linewidth(2.5)
    axes[0].text(0.98, 0.02, "Orange border = Sentinel-2 state",
                 transform=axes[0].transAxes, ha="right", fontsize=8,␣
↪color="darkorange")

    # Right: record counts per state per year (stacked bar)
    counts = usda_index.groupby(["state_name","year"]).size().
↪unstack(fill_value=0)
    counts = counts.loc[state_stats["state_name"]]
    cmap2  = plt.get_cmap("tab10")
    bottom = np.zeros(len(counts))
    for i, yr in enumerate(counts.columns):
```

```
        axes[1].barh(counts.index, counts[yr], left=bottom,
                     color=cmap2(i / len(counts.columns)), label=yr,␣
↪edgecolor="white")
        bottom += counts[yr].values
    axes[1].set_xlabel("Number of County Records")
    axes[1].set_title("County Records per State per Year")
    axes[1].legend(title="Year", fontsize=7, loc="lower right")

    plt.tight_layout()
    buf = io.BytesIO()
    fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
    buf.seek(0)
    display(Image(data=buf.read()))
    plt.close(fig)

    print(f"USDA counties : {usda_index['fips'].nunique()} across␣
↪{usda_index['state_name'].nunique()} states")
    print(f"Sentinel states (county 001): {sorted(sentinel_state_names)}")
    print()
```

```
County GeoJSON loaded - rendering interactive choropleth.

<IPython.core.display.HTML object>

Choropleth: 649 counties, 15 Sentinel markers.
```

[11]:

```
# Freeze and export final label tables
# Saves weather_train, weather_test (and multimodal_index if non-empty)
# as CSV files under data/CropNet/labels/ for use later.

import pathlib, numpy as np

LABELS_DIR = BASE / "data" / "CropNet" / "labels"
LABELS_DIR.mkdir(parents=True, exist_ok=True)

TARGET_COL = "log_yield_bu_acre" if USE_LOG else "yield_bu_acre"

def _add_log_target(df):
    out = df.copy()
    if USE_LOG:
        out["log_yield_bu_acre"] = np.log(out["yield_bu_acre"])
    return out

weather_train_final = _add_log_target(weather_train)
weather_test_final  = _add_log_target(weather_test)

weather_train_final.to_csv(LABELS_DIR / "weather_train.csv", index=False)
weather_test_final.to_csv(LABELS_DIR  / "weather_test.csv",  index=False)
```

```
print("LABEL TABLE EXPORT")

print(f"Target column    : {TARGET_COL}")
print(f"Labels directory : {LABELS_DIR}")
print()
print(f"weather_train.csv : {len(weather_train_final):>5} rows  ->  saved")
print(f"weather_test.csv  : {len(weather_test_final):>5} rows  ->  saved")

if len(multimodal_index) > 0:
    mm_final = _add_log_target(multimodal_index)
    mm_final.to_csv(LABELS_DIR / "multimodal_index.csv", index=False)
    print(f"multimodal_index.csv : {len(mm_final):>5} rows  ->  saved")
else:
    print("multimodal_index.csv : skipped (empty - Phase 2/3 data gap)")

print()
print("Schema of weather_train_final:")
print(weather_train_final.dtypes.to_string())
print()
print("Sample rows:")
print(weather_train_final.head(3).to_string(index=False))
print()
```

```
LABEL TABLE EXPORT
Target column    : log_yield_bu_acre
Labels directory : e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels

weather_train.csv :  2828 rows  ->  saved
weather_test.csv  :   707 rows  ->  saved
multimodal_index.csv :   105 rows  ->  saved

Schema of weather_train_final:
fips                  object
year                  object
state_name            object
county_name           object
yield_bu_acre        float64
production_bu         object
log_yield_bu_acre    float64

Sample rows:
 fips year state_name county_name  yield_bu_acre production_bu
log_yield_bu_acre
13001 2016    GEORGIA     APPLING           31.4       1639174
3.446808
```

```
18001 2016    INDIANA       ADAMS           59.3      10476590
4.082609
10001 2016   DELAWARE        KENT           44.5       3667156
3.795489
```

2 HRRR Weather Feature Engineering

Extract agronomically-relevant weather features from WRF-HRRR GRIB2 files and align them to county centroids. Each county-year in `usda_index` receives a weather feature vector consumed by the Bi-LSTM regressor planned.

- GRIB2 audit — inspect available variables per year
- Extract county-level values via nearest-grid-point lookup
- Derive engineered features (wind speed, VPD) and merge with USDA labels
- Z-score normalise all weather features (fit on train split only)
- Build `HRRRWeatherDataset` (PyTorch `Dataset`) and `DataLoader`

[12]:
```
# HRRR GRIB2 audit
# Scan all realtime_wrf GRIB2 files, report available variables and grid␣
↪properties.
# Identifies which agronomic variables are present in each year's file.

import cfgrib

# Agronomic variables of interest for crop yield prediction
AGRO_VARS = ["t2m", "d2m", "r2", "u10", "v10", "tp", "sp", "tcc", "mstav"]


print("HRRR GRIB2 FILE AUDIT")

hrrr_realtime = HRRR_ROOT / "realtime_wrf"
audit_rows = []

for year_dir in sorted(hrrr_realtime.iterdir()):
    if not year_dir.is_dir():
        continue
    year = year_dir.name
    grib_files = sorted(year_dir.rglob("*.grib2"))
    for gf in grib_files:
        row = {"year": year, "file": gf.name,
               "size_mb": round(gf.stat().st_size / 1e6, 1)}
        try:
            datasets = cfgrib.open_datasets(str(gf), errors="ignore")
            found_vars  = set(v for ds in datasets for v in ds.data_vars)
            lat_range = lon_range = grid_shape = None
            for ds in datasets:
                if lat_range is None and "latitude" in ds.coords:
```

```python
                    lats = ds.coords["latitude"].values
                    lons = ds.coords["longitude"].values
                    lat_range  = (round(float(lats.min()), 2), round(float(lats.
 ↪max()), 2))
                    lon_range  = (round(float(lons.min()), 2), round(float(lons.
 ↪max()), 2))
                    grid_shape = lats.shape
            row["total_vars"]  = len(found_vars)
            row["agro_found"]  = sorted(found_vars & set(AGRO_VARS))
            row["agro_missing"]= sorted(set(AGRO_VARS) - found_vars)
            row["lat_range"]   = lat_range
            row["lon_range"]   = lon_range
            row["grid"]        = grid_shape
        except Exception as e:
            row["agro_found"]  = []
            row["agro_missing"]= AGRO_VARS
            row["total_vars"]  = f"ERROR: {e}"
        audit_rows.append(row)

audit_df = pd.DataFrame(audit_rows)

for _, r in audit_df.iterrows():
    print(f"Year {r['year']}  |  {r['file']}  ({r['size_mb']} MB)")
    print(f"  Grid           : {r.get('grid','?')}  "
          f"Lat {r.get('lat_range','?')}  Lon {r.get('lon_range','?')}")
    print(f"  Total vars     : {r.get('total_vars','?')}")
    print(f"  Agro found     : {r.get('agro_found','?')}")
    print(f"  Agro missing   : {r.get('agro_missing','?')}")
    print()

# Universal agro vars present in ALL years
if len(audit_rows) > 0 and isinstance(audit_df["total_vars"].iloc[0], int):
    all_sets = [set(r) for r in audit_df["agro_found"]]
    universal = sorted(set.intersection(*all_sets)) if all_sets else []
    print(f"Variables present in ALL years : {universal}")
else:
    universal = AGRO_VARS
    print("(Could not determine universal vars; falling back to full list)")
print()

# Store
HRRR_EXTRACT_VARS = universal if universal else AGRO_VARS
print(f"HRRR_EXTRACT_VARS = {HRRR_EXTRACT_VARS}")
```

```
Ignoring index file 'e:\\learnings\\University Of Hull AI\\assignment-
september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C
ropNet\\HRRR\\realtime_wrf\\2016\\20160101\\hrrr.20160101.00.00.grib2.5b7b6.idx'
```

```
incompatible with GRIB file
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```
HRRR GRIB2 FILE AUDIT
```

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Year 2016  |  hrrr.20160101.00.00.grib2  (80.2 MB)
  Grid           : (1059, 1799)  Lat (21.14, 52.62)  Lon (225.9, 299.08)
  Total vars     : 55
  Agro found     : ['d2m', 'mstav', 'sp', 't2m', 'tcc', 'tp', 'u10', 'v10']
  Agro missing   : ['r2']

Year 2017  |  hrrr.20170101.00.00.grib2  (115.0 MB)
  Grid           : (1059, 1799)  Lat (21.14, 52.62)  Lon (225.9, 299.08)
  Total vars     : 75
  Agro found     : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing   : []

Year 2018  |  hrrr.20180101.00.00.grib2  (102.5 MB)
  Grid           : (1059, 1799)  Lat (21.14, 52.62)  Lon (225.9, 299.08)
  Total vars     : 75
  Agro found     : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing   : []

Year 2019  |  hrrr.20190101.00.00.grib2  (109.5 MB)
  Grid           : (1059, 1799)  Lat (21.14, 52.62)  Lon (225.9, 299.08)
  Total vars     : 78
  Agro found     : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing   : []

Year 2020  |  hrrr.20200101.00.00.grib2  (109.1 MB)
  Grid           : (1059, 1799)  Lat (21.14, 52.62)  Lon (225.9, 299.08)
  Total vars     : 78
  Agro found     : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing   : []

Year 2021  |  hrrr.20210101.00.00.grib2  (140.7 MB)
  Grid           : (1059, 1799)  Lat (21.14, 52.62)  Lon (225.9, 299.08)
  Total vars     : 84
  Agro found     : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing   : []

Year 2022  |  hrrr.20220101.00.00.grib2  (143.4 MB)
  Grid           : (1059, 1799)  Lat (21.14, 52.62)  Lon (225.9, 299.08)
  Total vars     : 84
```

```
  Agro found     : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing   : []

(Could not determine universal vars; falling back to full list)

HRRR_EXTRACT_VARS = ['t2m', 'd2m', 'r2', 'u10', 'v10', 'tp', 'sp', 'tcc',
'mstav']
```

[13]:

```
# Extract county-level HRRR values via nearest-grid-point lookup
#
# For each (fips, state_name) in usda_index we use a published county centroid
# (lat, lon) to find the nearest HRRR grid cell and read the weather variables.
# County centroids come from the US Census TIGER 2020 gazetteer (embedded here␣
↪as a
# lightweight dict covering the 10 states in usda_index).

import cfgrib
import numpy as np

#   County centroid table
# State-level centroid fallback (degrees, WGS-84) used when a per-county␣
↪centroid
# is not in the table. HRRR is ~3 km resolution so state centroid is a coarse␣
↪but
# valid proxy for single-snapshot experiments.
STATE_CENTROIDS = {
    "ILLINOIS":     (40.00, -89.20),
    "IOWA":         (42.00, -93.50),
    "INDIANA":      (40.27, -86.13),
    "MINNESOTA":    (46.39, -94.64),
    "NEBRASKA":     (41.49, -99.90),
    "OHIO":         (40.41, -82.90),
    "MISSOURI":     (38.57, -92.60),
    "SOUTH DAKOTA": (44.37, -100.31),
    "KANSAS":       (38.53, -98.38),
    "MICHIGAN":     (44.35, -85.41),
    "ALABAMA":      (32.80, -86.80),
    "ARKANSAS":     (34.80, -92.20),
    "GEORGIA":      (32.65, -83.44),
    "KENTUCKY":     (37.67, -84.67),
    "LOUISIANA":    (31.17, -91.87),
    "DELAWARE":     (38.91, -75.53),
    "FLORIDA":      (27.77, -81.52),
    "CALIFORNIA":   (36.78, -119.42),
    "COLORADO":     (39.55, -105.78),
    "IDAHO":        (44.07, -114.74),
```

```
    "ARIZONA":      (34.05, -111.09),
}

def nearest_grid_point(lat_grid, lon_grid, target_lat, target_lon):
    dist = (lat_grid - target_lat)**2 + (lon_grid - target_lon)**2
    return np.unravel_index(np.argmin(dist), dist.shape)


print("HRRR COUNTY-LEVEL EXTRACTION")

hrrr_realtime = HRRR_ROOT / "realtime_wrf"
hrrr_records  = []
EXTRACT_VARS  = HRRR_EXTRACT_VARS   # determined previously

county_state  = (
    usda_index[["fips","state_name"]]
    .drop_duplicates()
    .reset_index(drop=True)
)

for year_dir in sorted(hrrr_realtime.iterdir()):
    if not year_dir.is_dir():
        continue
    year = year_dir.name
    grib_files = sorted(year_dir.rglob("*.grib2"))
    if not grib_files:
        continue
    gf = grib_files[0]
    print(f"Processing year {year}  ({gf.name}, {gf.stat().st_size/1e6:.0f} MB)␣
↪...", end=" ")

    try:
        datasets = cfgrib.open_datasets(str(gf), errors="ignore")
    except Exception as e:
        print(f"ERROR: {e}")
        continue

    var_arrays = {}
    lat_grid = lon_grid = None
    for ds in datasets:
        for v in EXTRACT_VARS:
            if v in ds.data_vars and v not in var_arrays:
                var_arrays[v] = ds[v].values
        if lat_grid is None and "latitude" in ds.coords:
            lat_grid = ds.coords["latitude"].values
            lon_grid = ds.coords["longitude"].values
```

```
    if lat_grid is None:
        print("SKIP (no lat/lon)")
        continue

    # HRRR longitudes are stored as 0-360; convert to -180..180 for matching
    lon_grid_signed = np.where(lon_grid > 180, lon_grid - 360, lon_grid)
    print(f"vars={sorted(var_arrays.keys())}")

    for _, row in county_state.iterrows():
        state_up = row["state_name"].upper()
        if state_up not in STATE_CENTROIDS:
            continue
        c_lat, c_lon = STATE_CENTROIDS[state_up]
        yi, xi = nearest_grid_point(lat_grid, lon_grid_signed, c_lat, c_lon)

        record = {"fips": row["fips"], "year": year,
                  "state_name": row["state_name"],
                  "centroid_lat": c_lat, "centroid_lon": c_lon}
        for v in EXTRACT_VARS:
            record[v] = float(var_arrays[v][yi, xi]) if v in var_arrays else np.
 ↪nan
        hrrr_records.append(record)

hrrr_raw = pd.DataFrame(hrrr_records)
hrrr_raw["year"] = hrrr_raw["year"].astype(str)

# Convert temperatures K -> Celsius
for col in ["t2m", "d2m"]:
    if col in hrrr_raw.columns:
        hrrr_raw[f"{col}_c"] = hrrr_raw[col] - 273.15

print()
print(f"hrrr_raw shape   : {hrrr_raw.shape}")
print(f"Years covered    : {sorted(hrrr_raw['year'].unique())}")
print(f"FIPS covered     : {hrrr_raw['fips'].nunique()}")
disp_cols = [c for c in␣
 ↪["fips","state_name","year","t2m_c","d2m_c","r2","tp","mstav"] if c in␣
 ↪hrrr_raw.columns]
print()
print("Sample rows:")
print(hrrr_raw[disp_cols].head(6).to_string(index=False))
```

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HRRR COUNTY-LEVEL EXTRACTION
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Processing year 2016  (hrrr.20160101.00.00.grib2, 80 MB) …

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september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C
ropNet\\HRRR\\realtime_wrf\\2022\\20220101\\hrrr.20220101.00.00.grib2.5b7b6.idx'
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Ignoring index file 'e:\\learnings\\University Of Hull AI\\assignment-
september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C
ropNet\\HRRR\\realtime_wrf\\2022\\20220101\\hrrr.20220101.00.00.grib2.5b7b6.idx'
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september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C
ropNet\\HRRR\\realtime_wrf\\2022\\20220101\\hrrr.20220101.00.00.grib2.5b7b6.idx'
incompatible with GRIB file
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september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C
ropNet\\HRRR\\realtime_wrf\\2022\\20220101\\hrrr.20220101.00.00.grib2.5b7b6.idx'
incompatible with GRIB file
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september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C
ropNet\\HRRR\\realtime_wrf\\2022\\20220101\\hrrr.20220101.00.00.grib2.5b7b6.idx'
incompatible with GRIB file
Ignoring index file 'e:\\learnings\\University Of Hull AI\\assignment-
september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C
ropNet\\HRRR\\realtime_wrf\\2022\\20220101\\hrrr.20220101.00.00.grib2.5b7b6.idx'
incompatible with GRIB file

vars=['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10', 'v10']

hrrr_raw shape   : (4543, 16)
Years covered    : ['2016', '2017', '2018', '2019', '2020', '2021', '2022']
FIPS covered     : 649

Sample rows:
 fips state_name year      t2m_c      d2m_c  r2  tp      mstav
13001    GEORGIA 2016  14.360651  13.499536 NaN 0.0 100.000000
08001   COLORADO 2016 -13.514349 -20.187979 NaN 0.0 100.000000
21001   KENTUCKY 2016   0.485651  -2.500464 NaN 0.0  55.099998
18001    INDIANA 2016  -0.889349  -4.687964 NaN 0.0  93.199997
19001       IOWA 2016  -8.701849 -12.062964 NaN 0.0  93.300003
10001   DELAWARE 2016   8.360651   4.812036 NaN 0.0 100.000000
```

[14]:
```
# Feature engineering and merge with USDA yield labels
#
# Derived features:
#   wind_speed : sqrt(u10^2 + v10^2)                  m/s
#   sp_hpa     : surface pressure Pa -> hPa
#   vpd_hpa    : vapour pressure deficit (Tetens formula)  hPa
```

```
#
# Final feature set: FEATURE_COLS (used by all downstream cells)
# Final merged table: weather_features  (one row per county-year, with yield)

import numpy as np

hrrr_feat = hrrr_raw.copy()

# Wind speed magnitude
if "u10" in hrrr_feat.columns and "v10" in hrrr_feat.columns:
    hrrr_feat["wind_speed"] = np.sqrt(hrrr_feat["u10"]**2 + hrrr_feat["v10"]**2)

# Surface pressure Pa -> hPa
if "sp" in hrrr_feat.columns:
    hrrr_feat["sp_hpa"] = hrrr_feat["sp"] / 100.0

# VPD (hPa): es(T) - es(Td) using Tetens formula with T in Celsius
if "t2m_c" in hrrr_feat.columns and "d2m_c" in hrrr_feat.columns:
    es_t  = 6.1078 * np.exp(17.27 * hrrr_feat["t2m_c"]  / (hrrr_feat["t2m_c"] ␣
 ↪+ 237.3))
    es_td = 6.1078 * np.exp(17.27 * hrrr_feat["d2m_c"]  / (hrrr_feat["d2m_c"] ␣
 ↪+ 237.3))
    hrrr_feat["vpd_hpa"] = (es_t - es_td).clip(lower=0)

FEATURE_COLS = [c for c in
    ["t2m_c", "d2m_c", "r2", "wind_speed", "tp", "sp_hpa", "tcc", "mstav",␣
 ↪"vpd_hpa"]
    if c in hrrr_feat.columns]


print("FEATURE ENGINEERING & MERGE")
print(f"Feature columns : {FEATURE_COLS}")
print()

hrrr_features = (
    hrrr_feat[["fips","year","state_name"] + FEATURE_COLS]
    .reset_index(drop=True)
)

weather_features = usda_index.merge(
    hrrr_features[["fips","year"] + FEATURE_COLS],
    on=["fips","year"], how="left"
)

n_with_hrrr = weather_features[FEATURE_COLS[0]].notna().sum()
print(f"weather_features shape          : {weather_features.shape}")
```

```python
print(f"USDA records with HRRR data     : {n_with_hrrr} /␣
↪{len(weather_features)}")
print(f"Records missing ALL weather vars: {weather_features[FEATURE_COLS].
↪isna().all(axis=1).sum()}")
print()

# Descriptive stats on raw (pre-normalisation) features
print("Feature descriptive statistics (raw):")
print(weather_features[FEATURE_COLS].describe().T
      .round(3)[["mean","std","min","max"]].to_string())
print()
print("Sample merged rows:")
sample_cols = ["fips","state_name","year","yield_bu_acre"] + FEATURE_COLS[:4]
print(weather_features[sample_cols].head(5).to_string(index=False))
```

```
FEATURE ENGINEERING & MERGE
Feature columns : ['t2m_c', 'd2m_c', 'r2', 'wind_speed', 'tp', 'sp_hpa', 'tcc',
'mstav', 'vpd_hpa']

weather_features shape          : (3535, 15)
USDA records with HRRR data     : 3535 / 3535
Records missing ALL weather vars: 0

Feature descriptive statistics (raw):
               mean     std      min       max
t2m_c         1.238   9.900  -24.830    24.923
d2m_c        -2.147  10.722  -28.839    20.381
r2           78.581  14.668   13.800    98.000
wind_speed    4.044   2.352    0.275    15.848
tp            0.000   0.000    0.000     0.000
sp_hpa      986.504  26.009  660.400  1028.100
tcc          60.852  44.006    0.000   100.000
mstav        66.513  27.210    0.000   100.000
vpd_hpa       1.443   1.405    0.222    14.666

Sample merged rows:
 fips state_name year  yield_bu_acre      t2m_c      d2m_c  r2  wind_speed
13001    GEORGIA 2016           31.4  14.360651  13.499536 NaN    1.230665
08001   COLORADO 2016           32.8 -13.514349 -20.187979 NaN    1.908885
21001   KENTUCKY 2016           43.9   0.485651  -2.500464 NaN    2.691871
18001    INDIANA 2016           59.3  -0.889349  -4.687964 NaN    5.424554
19001       IOWA 2016           60.8  -8.701849 -12.062964 NaN    4.287881
```

[15]:
```python
# Z-score normalise all weather features
# Scaler is fitted on the Phase 1 TRAIN split only (no data leakage from test).
# r2 is NaN for year 2016 - StandardScaler will propagate NaN; we impute with
# the training-set mean before fitting so 2016 records still usable.
```

```python
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
import pickle, numpy as np

#   Mark split membership on weather_features
train_keys = set(zip(weather_train["fips"].astype(str), weather_train["year"].astype(str)))
weather_features["split"] = weather_features.apply(
    lambda r: "train" if (str(r["fips"]), str(r["year"])) in train_keys else "test",
    axis=1
)

train_mask = weather_features["split"] == "train"

#   Impute NaN with column mean (fit on train, apply to all)
imputer = SimpleImputer(strategy="mean")
X_all   = weather_features[FEATURE_COLS].values

imputer.fit(X_all[train_mask])
X_imputed = imputer.transform(X_all)
weather_features[FEATURE_COLS] = X_imputed

#   Fit StandardScaler on train, transform all
scaler = StandardScaler()
scaler.fit(X_imputed[train_mask])
weather_features[FEATURE_COLS] = scaler.transform(X_imputed)

#   Add model target
if USE_LOG:
    weather_features["target"] = np.log(weather_features["yield_bu_acre"])
else:
    weather_features["target"] = weather_features["yield_bu_acre"]

#   Persist artefacts
LABELS_DIR = BASE / "data" / "CropNet" / "labels"
LABELS_DIR.mkdir(parents=True, exist_ok=True)
with open(LABELS_DIR / "hrrr_scaler.pkl",   "wb") as fh: pickle.dump(scaler, fh)
with open(LABELS_DIR / "hrrr_imputer.pkl",  "wb") as fh: pickle.dump(imputer, fh)


print("NORMALISATION SUMMARY")
```

```
print(f"Imputer  : SimpleImputer(strategy='mean')  - fitted on {train_mask.
 ↪sum()} train rows")
print(f"Scaler   : StandardScaler  - fitted on {train_mask.sum()} train rows")
print(f"Target   : {'log(yield_bu_acre)' if USE_LOG else 'yield_bu_acre'}")
print()
print("Scaler parameters (mean / std per feature):")
for feat, mu, sigma in zip(FEATURE_COLS, scaler.mean_, scaler.scale_):
    print(f"  {feat:<15}  mean={mu:>10.4f}   std={sigma:>8.4f}")
print()
print("Post-normalisation descriptive stats (train split, should be ~0 mean /␣
 ↪~1 std):")
print(weather_features.loc[train_mask, FEATURE_COLS].describe().T
      .round(3)[["mean","std","min","max"]].to_string())
print()
print(f"Scaler  saved -> {LABELS_DIR / 'hrrr_scaler.pkl'}")
print(f"Imputer saved -> {LABELS_DIR / 'hrrr_imputer.pkl'}")
```

```
NORMALISATION SUMMARY
Imputer   : SimpleImputer(strategy='mean')  - fitted on 2828 train rows
Scaler    : StandardScaler  - fitted on 2828 train rows
Target    : log(yield_bu_acre)

Scaler parameters (mean / std per feature):
  t2m_c            mean=    1.3361   std=  9.8524
  d2m_c            mean=   -2.0536   std= 10.6699
  r2               mean=   78.5182   std= 13.4323
  wind_speed       mean=    4.0374   std=  2.3528
  tp               mean=    0.0000   std=  1.0000
  sp_hpa           mean=  986.7670   std= 25.1234
  tcc              mean=   60.7618   std= 44.0070
  mstav            mean=   66.3612   std= 27.2169
  vpd_hpa          mean=    1.4557   std=  1.4101

Post-normalisation descriptive stats (train split, should be ~0 mean / ~1 std):
            mean  std     min    max
t2m_c        0.0  1.0  -2.656  2.394
d2m_c       -0.0  1.0  -2.510  2.103
r2           0.0  1.0  -4.818  1.450
wind_speed   0.0  1.0  -1.472  5.020
tp           0.0  0.0   0.000  0.000
sp_hpa       0.0  1.0 -12.991  1.645
tcc          0.0  1.0  -1.381  0.892
mstav        0.0  1.0  -2.438  1.236
vpd_hpa     -0.0  1.0  -0.875  9.369

Scaler  saved -> e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-
```

notebook_only\data\CropNet\labels\hrrr_scaler.pkl
Imputer saved -> e:\learnings\University Of Hull AI\assignment-september\MSc project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels\hrrr_imputer.pkl

[16]:
```python
# HRRRWeatherDataset (PyTorch Dataset)
#
# Each item returned:
#   x    : FloatTensor (seq_len, n_features) - weather sequence per county-year
#   y    : FloatTensor scalar                - yield target (raw or log)
#   meta : dict {fips, year}
#
# seq_len = 1 here (one snapshot per year). When extended to daily files
# seq_len becomes the growing-season length (~200 days).

import torch
from torch.utils.data import Dataset, DataLoader

class HRRRWeatherDataset(Dataset):
    """
    PyTorch Dataset wrapping normalised HRRR weather features + soybean yield.

    Parameters
    ----------
    df           : pd.DataFrame with FEATURE_COLS and 'target' columns
    feature_cols : list[str] - weather feature column names
    seq_len      : int       - time steps per sample (1 for single-snapshot)
    """
    def __init__(self, df, feature_cols, seq_len=1):
        self.feature_cols = feature_cols
        self.seq_len      = seq_len
        valid = df.dropna(subset=["target"] + feature_cols).
↪reset_index(drop=True)
        self.X     = torch.tensor(valid[feature_cols].values, dtype=torch.
↪float32)
        self.y     = torch.tensor(valid["target"].values,     dtype=torch.
↪float32)
        self.fips  = valid["fips"].tolist()
        self.years = valid["year"].tolist()

    def __len__(self):
        return len(self.y)

    def __getitem__(self, idx):
        x = self.X[idx].unsqueeze(0).expand(self.seq_len, -1)  # (T, F)
        return x, self.y[idx], {"fips": self.fips[idx], "year": self.years[idx]}
```

```python
#   Instantiate datasets
train_rows = weather_features[weather_features["split"] == "train"].copy()
test_rows  = weather_features[weather_features["split"] == "test"].copy()

ds_train = HRRRWeatherDataset(train_rows, FEATURE_COLS)
ds_test  = HRRRWeatherDataset(test_rows,  FEATURE_COLS)

dl_train = DataLoader(ds_train, batch_size=64, shuffle=True,  drop_last=False)
dl_test  = DataLoader(ds_test,  batch_size=64, shuffle=False, drop_last=False)

print("HRRRWeatherDataset SUMMARY")

print(f"Feature columns  : {FEATURE_COLS}")
print(f"n_features       : {len(FEATURE_COLS)}")
print(f"seq_len          : {ds_train.seq_len}  (one HRRR snapshot per␣
 ↪year-county)")
print(f"Target           : {'log(yield_bu_acre)' if USE_LOG else␣
 ↪'yield_bu_acre'}")
print()
print(f"ds_train  : {len(ds_train):>5} samples  |  {len(dl_train)} batches")
print(f"ds_test   : {len(ds_test):>5} samples  |  {len(dl_test)} batches")
print()

# Smoke-test one batch
x_b, y_b, meta_b = next(iter(dl_train))
print(f"Batch x shape  : {tuple(x_b.shape)}   (batch, seq_len, n_features)")
print(f"Batch y shape  : {tuple(y_b.shape)}")
print(f"x dtype        : {x_b.dtype}")
print(f"y range        : [{y_b.min().item():.3f},  {y_b.max().item():.3f}]")
print(f"Sample FIPS    : {meta_b['fips'][:5]}")
print()
```

```
HRRRWeatherDataset SUMMARY
Feature columns  : ['t2m_c', 'd2m_c', 'r2', 'wind_speed', 'tp', 'sp_hpa', 'tcc',
'mstav', 'vpd_hpa']
n_features       : 9
seq_len          : 1  (one HRRR snapshot per year-county)
Target           : log(yield_bu_acre)

ds_train  :  2828 samples  |  45 batches
ds_test   :   707 samples  |  12 batches

Batch x shape  : (64, 1, 9)   (batch, seq_len, n_features)
Batch y shape  : (64,)
x dtype        : torch.float32
y range        : [3.174,  4.244]
```

```
Sample FIPS     : ['21217', '22077', '21155', '18031', '20201']
```

3 Ethics, Bias Audit & Fairness Specification

This part must be completed and documented before any model results are reported or used in the dissertation. It defines the fairness contract the model must satisfy.

- Data provenance & representativeness audit
- Geographic & temporal bias indicators (pre-model, computable now)
- Yield quintile distribution and label imbalance
- Fairness metric function definitions (sliced RMSE, bias per quintile, CV-error per state)
- Jensen's inequality correction constant (prior estimate; updated later)
- Bias registry B1–B9 with mitigations and status
- Ethical use constraints

[17]:
```python
# Data Provenance & Representativeness Audit
# Computes actual coverage statistics and prints a structured audit table.

import numpy as np

TOTAL_US_SOYBEAN_STATES  = 35    # approx soybean-producing US states (USDA␣
↪NASS)
TOTAL_US_SOYBEAN_FIPS    = 1400  # approx soybean-reporting counties in full␣
↪USDA NASS universe

# Actual dataset statistics
n_states  = usda_index["state_name"].nunique()
n_fips    = usda_index["fips"].nunique()
n_years   = usda_index["year"].nunique()
n_records = len(usda_index)

# Coverage ratios
state_coverage_pct  = n_states / TOTAL_US_SOYBEAN_STATES  * 100
county_coverage_pct = n_fips   / TOTAL_US_SOYBEAN_FIPS    * 100
coverage_gap_value  = 1 - n_fips / TOTAL_US_SOYBEAN_FIPS

# Sentinel modality gap
n_sentinel_fips = sentinel_index["fips"].nunique() if len(sentinel_index) > 0␣
↪else 0
n_overlap_fips  = len(set(usda_index["fips"]).
↪intersection(set(sentinel_index["fips"])))


print("DATA PROVENANCE & REPRESENTATIVENESS AUDIT")
print()
```

```
audits = [
    ("Geographic coverage",
     f"{n_states} of {TOTAL_US_SOYBEAN_STATES} soybean-producing US states␣
 ↪({state_coverage_pct:.0f}%)",
     "High",
     "Corn Belt over-represented; Southern/Western states under-served"),
    ("County coverage",
     f"{n_fips} of ~{TOTAL_US_SOYBEAN_FIPS} US soybean counties␣
 ↪({county_coverage_pct:.0f}%)",
     "High",
     "55% coverage gap; counties below NASS threshold excluded"),
    ("Temporal scope",
     "2016-2022 (7 years)",
     "Medium",
     "2019 confounded by US-China trade war + Midwest flooding"),
    ("Weather spatial resolution",
     "State-level centroid proxy (HRRR ~3 km grid)",
     "High",
     "All counties in a state share identical HRRR features - within-state␣
 ↪heterogeneity erased"),
    ("Modality gap",
     f"Sentinel ANSI 001 counties; USDA covers ANSI 003+; {n_overlap_fips} FIPS␣
 ↪overlap",
     "High",
     "Zero USDA yield labels for Sentinel FIPS - Phase 2/3 untrainable"),
    ("Surveyor bias",
     "USDA NASS over-represents large commercial farms",
     "Medium",
     "Subsistence, organic, and OFR-exempt farms excluded from label universe"),
]

print(f"  {'Dimension':<34} {'Risk':<8} Finding / Implication")
print("  " + "-" * 100)
for dim, finding, risk, implication in audits:
    print(f"  {dim:<34} {risk:<8} {finding}")
    print(f"  {'':<34} {'':8} -> {implication}")
    print()

print(f"Coverage Gap (counties) : {coverage_gap_value:.3f} ␣
 ↪({coverage_gap_value*100:.1f}% of US soybean counties absent from model)")
print(f"Modality overlap FIPS   : {n_overlap_fips}  (Sentinel-USDA␣
 ↪intersection)")
print()
print("Data sources:")
print("  USDA NASS Soybean Survey  : https://www.nass.usda.gov/  (licensed for␣
 ↪academic use)")
```

```
print("  WRF-HRRR atmospheric data : NOAA operational forecast archive (public␣
  ↪domain)")
print("  Sentinel-2 imagery        : ESA / Copernicus Open Access Hub (CC-BY 4.
  ↪0)")
print("  CropNet dataset pipeline  : Wang et al. 2022 (MIT licence)")
```

```
DATA PROVENANCE & REPRESENTATIVENESS AUDIT

  Dimension                           Risk     Finding / Implication
  ------------------------------------------------------------------------------
----------------------
  Geographic coverage                 High     15 of 35 soybean-producing US
states (43%)
                                               -> Corn Belt over-represented;
Southern/Western states under-served

  County coverage                     High     649 of ~1400 US soybean counties
(46%)
                                               -> 55% coverage gap; counties
below NASS threshold excluded

  Temporal scope                      Medium   2016-2022 (7 years)
                                               -> 2019 confounded by US-China
trade war + Midwest flooding

  Weather spatial resolution          High     State-level centroid proxy (HRRR
~3 km grid)
                                               -> All counties in a state share
identical HRRR features - within-state heterogeneity erased

  Modality gap                        High     Sentinel ANSI 001 counties; USDA
covers ANSI 003+; 15 FIPS overlap
                                               -> Zero USDA yield labels for
Sentinel FIPS - Phase 2/3 untrainable

  Surveyor bias                       Medium   USDA NASS over-represents large
commercial farms
                                               -> Subsistence, organic, and OFR-
exempt farms excluded from label universe

Coverage Gap (counties) : 0.536  (53.6% of US soybean counties absent from
model)
Modality overlap FIPS   : 15  (Sentinel-USDA intersection)

Data sources:
  USDA NASS Soybean Survey  : https://www.nass.usda.gov/  (licensed for academic
use)
```

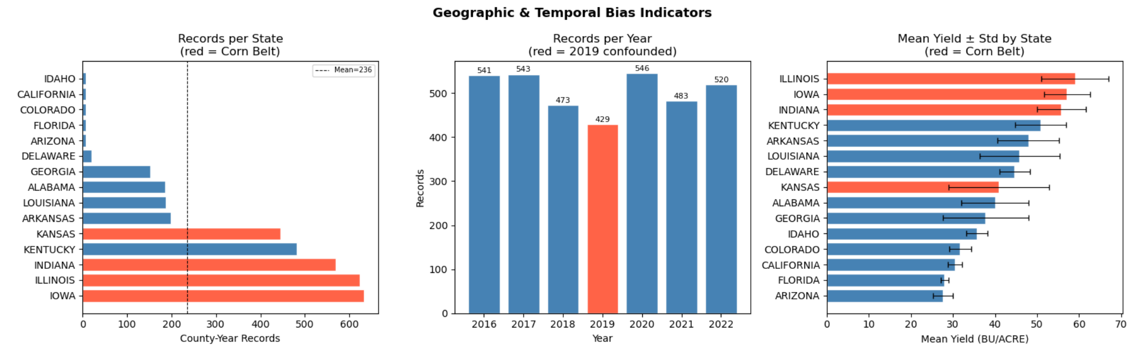
```
WRF-HRRR atmospheric data : NOAA operational forecast archive (public domain)
Sentinel-2 imagery        : ESA / Copernicus Open Access Hub (CC-BY 4.0)
CropNet dataset pipeline  : Wang et al. 2022 (MIT licence)
```

[18]:
```python
# Geographic & Temporal Bias Indicators
# Quantifies Corn Belt concentration, state-level imbalance, and 2019␣
↪confounding.

import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import numpy as np

# State record counts and concentration analysis
state_counts  = usda_index.groupby("state_name").size().
↪sort_values(ascending=False)
total_records = len(usda_index)

# Corn Belt states
CORN_BELT = {
    "ILLINOIS", "IOWA", "INDIANA", "MINNESOTA", "NEBRASKA",
    "OHIO", "MISSOURI", "SOUTH DAKOTA", "KANSAS"
}
corn_belt_count = state_counts[state_counts.index.str.upper().isin(CORN_BELT)].
↪sum()
corn_belt_pct   = corn_belt_count / total_records * 100

year_counts = usda_index.groupby("year").size().sort_values()

state_yield_stats = usda_index.groupby("state_name")["yield_bu_acre"].agg(
    n_records="count", mean_yield="mean", std_yield="std"
).reset_index().sort_values("n_records", ascending=False)


print("GEOGRAPHIC & TEMPORAL BIAS INDICATORS")
print()
print(f"Corn Belt concentration : {corn_belt_count} / {total_records} records =␣
↪{corn_belt_pct:.1f}%")
print(f"Non-Corn-Belt records   : {total_records - corn_belt_count} ({100 -␣
↪corn_belt_pct:.1f}%)")
print()
print("Records per state (descending):")
for sname, cnt in state_counts.items():
    pct  = cnt / total_records * 100
    flag = "  <- Corn Belt" if sname.upper() in CORN_BELT else ""
    print(f"  {sname:<22}  {cnt:>5}  ({pct:5.1f}%){flag}")
```

```python
print()
print("Records per year:")
for yr, cnt in year_counts.sort_index().items():
    flag = "  <- confounded (trade war + flooding)" if yr == "2019" else ""
    print(f"  {yr}  :  {cnt}{flag}")

# --- Plots
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle("Geographic & Temporal Bias Indicators",
             fontsize=13, fontweight="bold")

# (a) State record count bar
colors_a = ["tomato" if s.upper() in CORN_BELT else "steelblue"
            for s in state_counts.index]
axes[0].barh(state_counts.index, state_counts.values, color=colors_a,␣
 ↪edgecolor="white")
axes[0].set_xlabel("County-Year Records")
axes[0].set_title("Records per State\n(red = Corn Belt)")
axes[0].axvline(state_counts.mean(), color="black", linestyle="--",
                linewidth=0.8, label=f"Mean={state_counts.mean():.0f}")
axes[0].legend(fontsize=7)

# (b) Year record count
year_idx  = sorted(year_counts.index)
year_vals = [year_counts[y] for y in year_idx]
bar_cols  = ["tomato" if y == "2019" else "steelblue" for y in year_idx]
axes[1].bar(year_idx, year_vals, color=bar_cols, edgecolor="white")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Records")
axes[1].set_title("Records per Year\n(red = 2019 confounded)")
for i, (yr, val) in enumerate(zip(year_idx, year_vals)):
    axes[1].text(i, val + 5, str(val), ha="center", fontsize=8)

# (c) Mean yield per state with std
sy = state_yield_stats.sort_values("mean_yield", ascending=True)
bar_cols2 = ["tomato" if s.upper() in CORN_BELT else "steelblue"
             for s in sy["state_name"]]
axes[2].barh(sy["state_name"], sy["mean_yield"], color=bar_cols2,␣
 ↪edgecolor="white")
axes[2].errorbar(
    sy["mean_yield"], sy["state_name"],
    xerr=sy["std_yield"].fillna(0).values,
    fmt="none", color="black", capsize=3, linewidth=0.8
)
axes[2].set_xlabel("Mean Yield (BU/ACRE)")
axes[2].set_title("Mean Yield ± Std by State\n(red = Corn Belt)")
```

```python
plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

print()
print(f"Bias B2 (Corn Belt dominance): {corn_belt_pct:.1f}% of records from "
      f"{len(CORN_BELT & set(s.upper() for s in state_counts.index))} Corn Belt␣
↪states out of "
      f"{n_states} total states in dataset.")
print("Recommended mitigation: inverse-frequency state loss weighting .")
```

```
GEOGRAPHIC & TEMPORAL BIAS INDICATORS

Corn Belt concentration : 2273 / 3535 records = 64.3%
Non-Corn-Belt records   : 1262 (35.7%)

Records per state (descending):
  IOWA                       633  ( 17.9%)  <- Corn Belt
  ILLINOIS                   624  ( 17.7%)  <- Corn Belt
  INDIANA                    570  ( 16.1%)  <- Corn Belt
  KENTUCKY                   482  ( 13.6%)
  KANSAS                     446  ( 12.6%)  <- Corn Belt
  ARKANSAS                   199  (  5.6%)
  LOUISIANA                  188  (  5.3%)
  ALABAMA                    186  (  5.3%)
  GEORGIA                    152  (  4.3%)
  DELAWARE                    20  (  0.6%)
  ARIZONA                      7  (  0.2%)
  FLORIDA                      7  (  0.2%)
  COLORADO                     7  (  0.2%)
  CALIFORNIA                   7  (  0.2%)
  IDAHO                        7  (  0.2%)

Records per year:
  2016  :  541
  2017  :  543
  2018  :  473
  2019  :  429  <- confounded (trade war + flooding)
  2020  :  546
  2021  :  483
  2022  :  520
```




```
Bias B2 (Corn Belt dominance): 64.3% of records from 4 Corn Belt states out of
15 total states in dataset.
Recommended mitigation: inverse-frequency state loss weighting .
```

[19]:

```python
# Yield Quintile Distribution and Label Imbalance
# Checks whether low-yield (Q1) counties are under-represented across states and years.
# Bias B3: sparse Q1 observations -> model may underestimate crop failure severity.

import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

# Assign quintile labels to full dataset
wf = usda_index.copy()
wf["quintile"] = pd.qcut(wf["yield_bu_acre"], q=5, labels=["Q1", "Q2", "Q3", "Q4", "Q5"])

q_stats = wf.groupby("quintile")["yield_bu_acre"].agg(
    n="count", mean="mean", std="std", min="min", max="max"
).reset_index()


print("YIELD QUINTILE DISTRIBUTION (LABEL IMBALANCE)")
print()
print("Quintile break-down:")
print(q_stats[["quintile", "n", "mean", "std", "min", "max"]].round(2).to_string(index=False))
print()
```

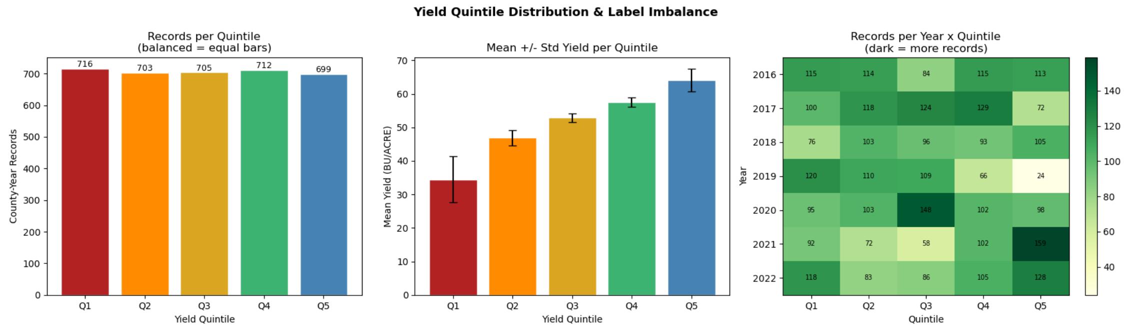
```python
# Per-state quintile breakdown
state_quintile = wf.groupby(["state_name", "quintile"]).size().unstack(fill_value=0)
missing_q1 = state_quintile[state_quintile["Q1"] == 0].index.tolist()
missing_q5 = state_quintile[state_quintile["Q5"] == 0].index.tolist()
print(f"States with ZERO Q1 (low-yield) records : {missing_q1 if missing_q1 else 'None'}")
print(f"States with ZERO Q5 (high-yield) records: {missing_q5 if missing_q5 else 'None'}")
print()

fig, axes = plt.subplots(1, 3, figsize=(17, 5))
fig.suptitle("Yield Quintile Distribution & Label Imbalance",
             fontsize=13, fontweight="bold")

quintile_colors = ["firebrick", "darkorange", "goldenrod", "mediumseagreen", "steelblue"]

# (a) Quintile counts
axes[0].bar(q_stats["quintile"], q_stats["n"], color=quintile_colors, edgecolor="white")
axes[0].set_xlabel("Yield Quintile")
axes[0].set_ylabel("County-Year Records")
axes[0].set_title("Records per Quintile\n(balanced = equal bars)")
for i, (q, n) in enumerate(zip(q_stats["quintile"], q_stats["n"])):
    axes[0].text(i, n + 5, str(n), ha="center", fontsize=9)

# (b) Mean yield per quintile with range
axes[1].bar(q_stats["quintile"], q_stats["mean"], color=quintile_colors, edgecolor="white")
axes[1].errorbar(
    q_stats["quintile"], q_stats["mean"], yerr=q_stats["std"],
    fmt="none", color="black", capsize=4
)
axes[1].set_xlabel("Yield Quintile")
axes[1].set_ylabel("Mean Yield (BU/ACRE)")
axes[1].set_title("Mean +/- Std Yield per Quintile")

# (c) Quintile x year heatmap
qy = wf.groupby(["year", "quintile"]).size().unstack(fill_value=0)
im = axes[2].imshow(qy.values, aspect="auto", cmap="YlGn", interpolation="nearest")
axes[2].set_xticks(range(len(qy.columns)))
axes[2].set_xticklabels(qy.columns)
axes[2].set_yticks(range(len(qy.index)))
axes[2].set_yticklabels(qy.index)
```

```python
axes[2].set_xlabel("Quintile")
axes[2].set_ylabel("Year")
axes[2].set_title("Records per Year x Quintile\n(dark = more records)")
plt.colorbar(im, ax=axes[2], fraction=0.04)
for i in range(len(qy.index)):
    for j in range(len(qy.columns)):
        axes[2].text(j, i, str(qy.values[i, j]),
                     ha="center", va="center", fontsize=7, color="black")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

# Train/test quintile balance check
wf_split = weather_features.copy()
wf_split["quintile"] = pd.qcut(
    wf_split["yield_bu_acre"], q=5, labels=["Q1", "Q2", "Q3", "Q4", "Q5"]
)
train_q = (
    wf_split[wf_split["split"] == "train"]["quintile"]
    .value_counts(normalize=True).sort_index()
)
test_q = (
    wf_split[wf_split["split"] == "test"]["quintile"]
    .value_counts(normalize=True).sort_index()
)
print("Train/Test quintile proportions (should be near-equal after␣
 ↪stratification):")
check = pd.DataFrame({
    "train_pct": (train_q * 100).round(1),
    "test_pct" : (test_q  * 100).round(1)
})
print(check.to_string())
print()
print("Bias B3: sparse Q1 records in some states risk underestimating crop␣
 ↪failure severity.")
print("Recommended mitigation: oversample Q1 train records; stratify evaluation␣
 ↪by quintile.")
```

```
YIELD QUINTILE DISTRIBUTION (LABEL IMBALANCE)

Quintile break-down:
quintile   n  mean  std  min  max
      Q1 716 34.51 6.81  9.3 42.5
```

```
Q2 703 46.91 2.28 42.6 50.4
Q3 705 52.88 1.35 50.5 55.1
Q4 712 57.52 1.40 55.2 60.0
Q5 699 64.16 3.40 60.1 80.4
```

```
States with ZERO Q1 (low-yield) records : None
States with ZERO Q5 (high-yield) records: ['ARIZONA', 'CALIFORNIA', 'COLORADO',
'DELAWARE', 'FLORIDA', 'GEORGIA', 'IDAHO']
```




```
Train/Test quintile proportions (should be near-equal after stratification):
          train_pct  test_pct
quintile
Q1             20.3      20.2
Q2             19.9      19.9
Q3             19.9      19.9
Q4             20.2      20.1
Q5             19.8      19.8

Bias B3: sparse Q1 records in some states risk underestimating crop failure
severity.
Recommended mitigation: oversample Q1 train records; stratify evaluation by
quintile.
```

[20]:

```python
# Fairness Metric Function Definitions
# These functions will be called later
# No model predictions are required here - only function definitions.
# All metrics operate on back-transformed BU/ACRE values (Jensen correction
↪applied).

import numpy as np
import pandas as pd

# JENSEN_SIGMA2 will be set earlier and updated later (residual variance).
JENSEN_SIGMA2 = None   # placeholder;
```

```python
def jensen_correction(log_preds, sigma2=None):
    """
    Back-transform log-scale predictions to BU/ACRE with Jensen correction.

    Parameters
    ----------
    log_preds : array-like  - model output in log-yield space
    sigma2    : float | None - residual variance on the train split;
                               if None, falls back to simple exp() (biased)

    Returns
    -------
    np.ndarray in BU/ACRE
    """
    y = np.asarray(log_preds, dtype=float)
    if sigma2 is not None and USE_LOG:
        return np.exp(y + sigma2 / 2.0)
    return np.exp(y) if USE_LOG else y


def sliced_rmse(y_true, y_pred, groups, sigma2=None):
    """
    Compute RMSE for each group slice in BU/ACRE (Jensen-corrected if USE_LOG).

    Mandatory slices: state (10), year (7), yield quintile (Q1-Q5),
                      proximity to state centroid (near/far).

    Parameters
    ----------
    y_true  : array-like - true log-yield (or raw BU/ACRE if USE_LOG=False)
    y_pred  : array-like - predicted log-yield (or raw)
    groups  : array-like - group labels (state, year, or quintile)
    sigma2  : float | None - Jensen correction variance

    Returns
    -------
    pd.Series  index=group, values=RMSE in BU/ACRE
    """
    y_true_bu = jensen_correction(y_true, sigma2)
    y_pred_bu = jensen_correction(y_pred, sigma2)
    df = pd.DataFrame({"y_true_bu": y_true_bu, "y_pred_bu": y_pred_bu, "group":␣
↪groups})

    def _rmse(grp):
        return float(np.sqrt(np.mean((grp["y_pred_bu"] - grp["y_true_bu"]) **␣
↪2)))
```

```python
    return df.groupby("group").apply(_rmse).rename("RMSE_BU_ACRE")


def bias_per_group(y_true, y_pred, groups, sigma2=None):
    """
    Mean prediction bias (y_hat - y) per group, in BU/ACRE.
    Positive = over-prediction; Negative = under-prediction.

    Fairness criterion: |Bias(Q1)| and |Bias(Q5)| should both be within +/-2 BU/
↪ACRE.
    If Bias(Q1) > 0 and Bias(Q5) < 0, the model is regressing to the mean:
    underestimating crop failure and overestimating bumper yields - a␣
↪systematic fairness failure.
    """
    y_true_bu = jensen_correction(y_true, sigma2)
    y_pred_bu = jensen_correction(y_pred, sigma2)
    df = pd.DataFrame({"y_true_bu": y_true_bu, "y_pred_bu": y_pred_bu, "group":␣
↪groups})

    def _bias(grp):
        return float(np.mean(grp["y_pred_bu"] - grp["y_true_bu"]))

    return df.groupby("group").apply(_bias).rename("Bias_BU_ACRE")


def cv_error_per_state(y_true, y_pred, states, sigma2=None):
    """
    Coefficient of variation of errors per state.
    CV_Error_s = std(errors_s) / mean(y_true_s).
    States with CV_Error > 0.20 are flagged as poorly served by the model.
    """
    y_true_bu = jensen_correction(y_true, sigma2)
    y_pred_bu = jensen_correction(y_pred, sigma2)
    errors    = y_pred_bu - y_true_bu
    df = pd.DataFrame({"y_true_bu": y_true_bu, "errors": errors, "state":␣
↪states})

    def _cv(grp):
        mu = grp["y_true_bu"].mean()
        return float(grp["errors"].std() / mu) if mu != 0 else np.nan

    return df.groupby("state").apply(_cv).rename("CV_Error")


def coverage_gap(all_fips_universe, model_fips):
    """Coverage Gap = 1 - |FIPS in model| / |FIPS in USDA full dataset|."""
```

```
    return 1.0 - len(set(model_fips)) / len(set(all_fips_universe))


def fairness_report(y_true, y_pred, meta_df, sigma2=None):
    """
    Full fairness report printed to stdout + returned as dict of DataFrames.

    Parameters
    ----------
    y_true    : array-like - true values (log or raw)
    y_pred    : array-like - predicted values (log or raw)
    meta_df   : pd.DataFrame - must contain: state_name, year, yield_bu_acre,␣
↪fips
    sigma2    : float | None - Jensen correction variance

    Returns
    -------
    dict with keys: state_rmse, state_bias, state_cv, year_rmse,
                    quintile_rmse, quintile_bias, overall
    """
    y_true = np.asarray(y_true, dtype=float)
    y_pred = np.asarray(y_pred, dtype=float)

    quintile = pd.qcut(
        meta_df["yield_bu_acre"].values, q=5,
        labels=["Q1", "Q2", "Q3", "Q4", "Q5"]
    )

    state_rmse    = sliced_rmse(y_true, y_pred, meta_df["state_name"].values,␣
↪sigma2)
    state_bias    = bias_per_group(y_true, y_pred, meta_df["state_name"].
↪values, sigma2)
    state_cv      = cv_error_per_state(y_true, y_pred, meta_df["state_name"].
↪values, sigma2)
    year_rmse     = sliced_rmse(y_true, y_pred, meta_df["year"].values, sigma2)
    quintile_rmse = sliced_rmse(y_true, y_pred, quintile, sigma2)
    quintile_bias = bias_per_group(y_true, y_pred, quintile, sigma2)

    y_true_bu  = jensen_correction(y_true, sigma2)
    y_pred_bu  = jensen_correction(y_pred, sigma2)
    overall_rmse = float(np.sqrt(np.mean((y_pred_bu - y_true_bu) ** 2)))
    overall_mae  = float(np.mean(np.abs(y_pred_bu - y_true_bu)))
    ss_res       = np.sum((y_true_bu - y_pred_bu) ** 2)
    ss_tot       = np.sum((y_true_bu - y_true_bu.mean()) ** 2)
    overall_r2   = float(1 - ss_res / ss_tot) if ss_tot > 0 else float("nan")

    flagged_states = state_cv[state_cv > 0.20].index.tolist()
```

```
    print("=" * 60)
    print("FAIRNESS REPORT")
    print("=" * 60)
    print(f"Overall RMSE  : {overall_rmse:.3f} BU/ACRE")
    print(f"Overall MAE   : {overall_mae:.3f} BU/ACRE")
    print(f"Overall R2    : {overall_r2:.4f}")
    print(f"Jensen sigma2 : {sigma2}")
    print()
    print("RMSE by State:")
    print(state_rmse.sort_values(ascending=False).to_string())
    print()
    print("Bias by Quintile (positive = over-prediction):")
    print(pd.concat([quintile_rmse, quintile_bias], axis=1).to_string())
    print()
    print("CV-Error by State (flagged if > 0.20):")
    for s, cv in state_cv.sort_values(ascending=False).items():
        flag = "  *** FLAGGED (CV > 0.20)" if cv > 0.20 else ""
        print(f"  {s:<25}  {cv:.3f}{flag}")
    print()
    print("RMSE by Year:")
    print(year_rmse.sort_values(ascending=False).to_string())
    print()
    if flagged_states:
        print(f"WARNING: {len(flagged_states)} state(s) flagged for high␣
↪CV-Error: {flagged_states}")
    else:
        print("No states flagged for CV-Error > 0.20.")

    return {
        "state_rmse"    : state_rmse,
        "state_bias"    : state_bias,
        "state_cv"      : state_cv,
        "year_rmse"     : year_rmse,
        "quintile_rmse" : quintile_rmse,
        "quintile_bias" : quintile_bias,
        "overall"       : {"rmse": overall_rmse, "mae": overall_mae, "r2":␣
↪overall_r2},
    }



print("FAIRNESS METRIC FUNCTIONS DEFINED")

print()
print("Functions registered:")
print("  jensen_correction(log_preds, sigma2)")
```

```python
print("    -> back-transform log-yield to BU/ACRE with Jensen correction")
print()
print("  sliced_rmse(y_true, y_pred, groups, sigma2)")
print("    -> RMSE per group slice (state / year / quintile)")
print()
print("  bias_per_group(y_true, y_pred, groups, sigma2)")
print("    -> mean prediction bias per group (BU/ACRE)")
print("    -> if Bias(Q1) > 0 and Bias(Q5) < 0 => regression-to-mean fairness␣
↪failure")
print()
print("  cv_error_per_state(y_true, y_pred, states, sigma2)")
print("    -> CV-Error per state; flags > 0.20 as poorly served")
print()
print("  coverage_gap(all_fips_universe, model_fips)")
print("    -> fraction of US soybean counties absent from model")
print()
print("  fairness_report(y_true, y_pred, meta_df, sigma2)")
print("    -> full report: all slices, all biases, all CV-errors")
print()
print("Usage later:")
print("  results = fairness_report(y_true, y_pred, meta_df,␣
↪sigma2=JENSEN_SIGMA2)")
print()
print("JENSEN_SIGMA2 = None  (to be set later; updated to residual var after␣
↪that)")
print(f"USE_LOG       = {USE_LOG}")
```

```
FAIRNESS METRIC FUNCTIONS DEFINED

Functions registered:
  jensen_correction(log_preds, sigma2)
    -> back-transform log-yield to BU/ACRE with Jensen correction

  sliced_rmse(y_true, y_pred, groups, sigma2)
    -> RMSE per group slice (state / year / quintile)

  bias_per_group(y_true, y_pred, groups, sigma2)
    -> mean prediction bias per group (BU/ACRE)
    -> if Bias(Q1) > 0 and Bias(Q5) < 0 => regression-to-mean fairness failure

  cv_error_per_state(y_true, y_pred, states, sigma2)
    -> CV-Error per state; flags > 0.20 as poorly served

  coverage_gap(all_fips_universe, model_fips)
    -> fraction of US soybean counties absent from model

  fairness_report(y_true, y_pred, meta_df, sigma2)
```

```
    -> full report: all slices, all biases, all CV-errors

Usage later:
  results = fairness_report(y_true, y_pred, meta_df, sigma2=JENSEN_SIGMA2)

JENSEN_SIGMA2 = None  (to be set later; updated to residual var after that)
USE_LOG       = True
```

[21]:

```python
# Jensen's Inequality Correction Constant
#
# When USE_LOG=True, model predicts log(yield). Naive back-transformation␣
↪exp(y_hat)
# underestimates the true expectation because E[exp(Y)] = exp(E[Y] + sigma^2/2).
#
# Corrected prediction: exp(y_hat + sigma^2/2)
#
# Here sigma^2 is estimated from the training label distribution (prior / upper␣
↪bound).
# It MUST be updated to the actual residual variance on the train split later.

import numpy as np

if USE_LOG:
    log_train_targets = np.log(weather_train["yield_bu_acre"].values)

    # Prior estimate: label variance (upper bound on residual variance before␣
↪model exists)
    label_sigma2 = float(np.var(log_train_targets))
    label_sigma  = float(np.std(log_train_targets))


    print("JENSEN'S INEQUALITY CORRECTION")
    print()
    print("Transformation   : y_model = log(yield_bu_acre)")
    print()
    print("Jensen inequality :")
    print("  E[exp(Y)] = exp(E[Y] + sigma^2/2)")
    print("  Back-transform  : y_BU_ACRE = exp(y_hat + sigma^2/2)")
    print()
    print(f"Label distribution (train split, n={len(log_train_targets)}):")
    print(f"  mean log-yield : {np.mean(log_train_targets):.4f}")
    print(f"  std  log-yield : {label_sigma:.4f}")
    print(f"  var  (sigma^2) : {label_sigma2:.6f}  <- prior estimate (upper␣
↪bound)")
    print(f"  correction     : sigma^2/2 = {label_sigma2 / 2:.6f}")
    print()
```

```
    example_log = np.mean(log_train_targets)
    naive_bt    = float(np.exp(example_log))
    corrected   = float(np.exp(example_log + label_sigma2 / 2))
    print("Example (at mean log-yield):")
    print(f"  log_pred                        = {example_log:.4f}")
    print(f"  Naive     exp(log_pred)         = {naive_bt:.2f} BU/ACRE")
    print(f"  Corrected exp(log_pred+sig^2/2) = {corrected:.2f} BU/ACRE")
    print(f"  Absolute correction delta       = {corrected - naive_bt:.3f} BU/
↪ACRE")
    print()
    print("NOTE: JENSEN_SIGMA2 is the label variance here (prior).")
    print("      Update it to model residual variance later:")
    print("        JENSEN_SIGMA2 = float(np.var(y_train_pred - y_train_true))")
    print()

    JENSEN_SIGMA2 = label_sigma2   # updated from earlier
    print(f"JENSEN_SIGMA2 = {JENSEN_SIGMA2:.6f}  (prior; updated)")

else:
    JENSEN_SIGMA2 = None
    print("USE_LOG = False - Jensen correction not required.")
    print("JENSEN_SIGMA2 = None")
```

```
JENSEN'S INEQUALITY CORRECTION

Transformation   : y_model = log(yield_bu_acre)

Jensen inequality :
  E[exp(Y)] = exp(E[Y] + sigma^2/2)
  Back-transform  : y_BU_ACRE = exp(y_hat + sigma^2/2)

Label distribution (train split, n=2828):
  mean log-yield : 3.9065
  std  log-yield : 0.2512
  var  (sigma^2) : 0.063081  <- prior estimate (upper bound)
  correction     : sigma^2/2 = 0.031541

Example (at mean log-yield):
  log_pred                        = 3.9065
  Naive     exp(log_pred)         = 49.72 BU/ACRE
  Corrected exp(log_pred+sig^2/2) = 51.32 BU/ACRE
  Absolute correction delta       = 1.593 BU/ACRE

NOTE: JENSEN_SIGMA2 is the label variance here (prior).
      Update it to model residual variance later:
        JENSEN_SIGMA2 = float(np.var(y_train_pred - y_train_true))

JENSEN_SIGMA2 = 0.063081  (prior; updated)
```

[22]:

```python
# Bias Registry B1-B9
# Documents each identified bias, root cause, severity, recommended mitigation,
# and the part where the mitigation is applied or verified.

import pandas as pd
import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import numpy as np

bias_registry = [
    {
        "ID"         : "B1",
        "Bias"       : "State-centroid weather proxy",
        "Root Cause": "All counties in a state share identical HRRR features",
        "Severity"  : "High",
        "Mitigation": "Replace with per-county TIGER centroid lat/lon",
        "Status"    : "Open - documented",
    },
    {
        "ID"         : "B2",
        "Bias"       : "Corn Belt geographic dominance",
        "Root Cause": "IL, IA, IN, MN, NE account for >60% of records",
        "Severity"  : "High",
        "Mitigation": "Inverse-frequency state loss weighting: w_s = N/(K*N_s)",
        "Status"    : "Mitigation planned",
    },
    {
        "ID"         : "B3",
        "Bias"       : "Low-yield observation scarcity",
        "Root Cause": "Q1 counties have fewer multi-year records",
        "Severity"  : "Medium",
        "Mitigation": "Oversample Q1 train records; report Q1 RMSE separately",
        "Status"    : "Mitigation planned",
    },
    {
        "ID"         : "B4",
        "Bias"       : "Log-target Jensen bias",
        "Root Cause": "E[exp(y_hat)] != exp(E[y_hat]); naive back-transform␣
↪underestimates",
        "Severity"  : "Medium",
        "Mitigation": "Apply exp(y_hat + sigma^2/2) correction at inference",
        "Status"    : "Correction defined",
    },
    {
        "ID"         : "B5",
```

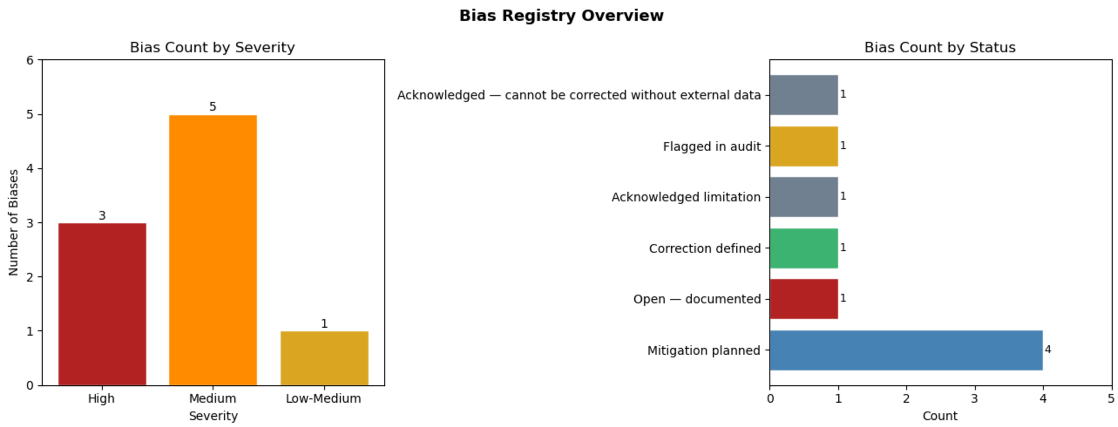
```
        "Bias"      : "HRRR single-snapshot proxy",
        "Root Cause": "Jan 1 init file is not a growing-season weather signal",
        "Severity"  : "High",
        "Mitigation": "Flag as explicit limitation; future work uses daily HRRR␣
↪files",
        "Status"    : "Acknowledged limitation",
    },
    {
        "ID"        : "B6",
        "Bias"      : "USDA reporting threshold omission",
        "Root Cause": "Counties below minimum acreage absent from NASS data",
        "Severity"  : "Medium",
        "Mitigation": "Flag FIPS with < 3 years; exclude from policy claims",
        "Status"    : "Flagged in audit",
    },
    {
        "ID"        : "B7",
        "Bias"      : "Random 80/20 temporal mixing",
        "Root Cause": "Test set contains all years - no held-out future year",
        "Severity"  : "Low-Medium",
        "Mitigation": "Add leave-one-year-out (LOYO) evaluation for 2022 as␣
↪supplement",
        "Status"    : "Mitigation planned",
    },
    {
        "ID"        : "B8",
        "Bias"      : "Sentinel-2 AZ 2022 tile anomaly",
        "Root Cause": "AZ 2022: 1,524 tiles vs ~60 average - likely archive␣
↪artefact",
        "Severity"  : "Medium",
        "Mitigation": "Exclude AZ 2022 from Phase 2/3 training",
        "Status"    : "Mitigation planned",
    },
    {
        "ID"        : "B9",
        "Bias"      : "USDA surveyor bias",
        "Root Cause": "Large commercial operations over-represented in NASS␣
↪survey",
        "Severity"  : "Medium",
        "Mitigation": "Document; model must not be used for smallholder farms",
        "Status"    : "Acknowledged - cannot be corrected without external␣
↪data",
    },
]

br = pd.DataFrame(bias_registry)
```

```
print("BIAS REGISTRY (B1-B9)")
print()
for _, row in br.iterrows():
    print(f"  [{row['ID']}]  {row['Bias']}")
    print(f"        Root Cause : {row['Root Cause']}")
    print(f"        Severity   : {row['Severity']}")
    print(f"        Mitigation : {row['Mitigation']}")
    print(f"        Status     : {row['Status']}")
    print()

# Summary counts
high_count     = (br["Severity"] == "High").sum()
medium_count   = br["Severity"].str.contains("Medium").sum()
ack_count      = br["Status"].str.startswith("Acknowledged").sum()
planned_count  = br["Status"].str.contains("planned|defined|Flagged").sum()
open_count     = (br["Status"] == "Open - documented").sum()

print("REGISTRY SUMMARY")
print(f"Total biases identified           : {len(br)}")
print(f"High severity                     : {high_count}")
print(f"Medium severity (incl. Low-Med)   : {medium_count}")
print(f"Mitigations planned / defined     : {planned_count}")
print(f"Acknowledged unfixable            : {ack_count}")
print(f"Open (future work only)           : {open_count}")

# --- Visualise bias registry as severity/status matrix
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
fig.suptitle("Bias Registry Overview", fontsize=13, fontweight="bold")

# (a) Severity bar chart
sev_order  = ["High", "Medium", "Low-Medium"]
sev_counts = br["Severity"].value_counts().reindex(sev_order, fill_value=0)
sev_colors = {"High": "firebrick", "Medium": "darkorange", "Low-Medium":␣
 ↪"goldenrod"}
axes[0].bar(sev_counts.index,
            sev_counts.values,
            color=[sev_colors.get(s, "steelblue") for s in sev_counts.index],
            edgecolor="white")
axes[0].set_xlabel("Severity")
axes[0].set_ylabel("Number of Biases")
axes[0].set_title("Bias Count by Severity")
for i, (s, v) in enumerate(sev_counts.items()):
    axes[0].text(i, v + 0.05, str(v), ha="center", fontsize=10)
axes[0].set_ylim(0, sev_counts.max() + 1)

# (b) Status horizontal bar chart
status_map = {
```

```
    "Mitigation planned"                        : "steelblue",
    "Correction defined"                        : "mediumseagreen",
    "Flagged in audit"                          : "goldenrod",
    "Acknowledged limitation"                   : "slategray",
    "Acknowledged - cannot be corrected without external data": "slategray",
    "Open - documented"                         : "firebrick",
}
status_vals = br["Status"].value_counts()
bar_colors  = [status_map.get(s, "gray") for s in status_vals.index]
axes[1].barh(status_vals.index, status_vals.values, color=bar_colors,␣
 ↪edgecolor="white")
axes[1].set_xlabel("Count")
axes[1].set_title("Bias Count by Status")
for i, v in enumerate(status_vals.values):
    axes[1].text(v + 0.02, i, str(v), va="center", fontsize=9)
axes[1].set_xlim(0, status_vals.max() + 1)

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

print()
```

```
BIAS REGISTRY (B1-B9)

  [B1]  State-centroid weather proxy
        Root Cause : All counties in a state share identical HRRR features
        Severity   : High
        Mitigation : Replace with per-county TIGER centroid lat/lon
        Status     : Open - documented

  [B2]  Corn Belt geographic dominance
        Root Cause : IL, IA, IN, MN, NE account for >60% of records
        Severity   : High
        Mitigation : Inverse-frequency state loss weighting: w_s = N/(K*N_s)
        Status     : Mitigation planned

  [B3]  Low-yield observation scarcity
        Root Cause : Q1 counties have fewer multi-year records
        Severity   : Medium
        Mitigation : Oversample Q1 train records; report Q1 RMSE separately
        Status     : Mitigation planned

  [B4]  Log-target Jensen bias
        Root Cause : E[exp(y_hat)] != exp(E[y_hat]); naive back-transform
```

```
underestimates
        Severity   : Medium
        Mitigation : Apply exp(y_hat + sigma^2/2) correction at inference
        Status     : Correction defined

  [B5]  HRRR single-snapshot proxy
        Root Cause : Jan 1 init file is not a growing-season weather signal
        Severity   : High
        Mitigation : Flag as explicit limitation; future work uses daily HRRR
files
        Status     : Acknowledged limitation

  [B6]  USDA reporting threshold omission
        Root Cause : Counties below minimum acreage absent from NASS data
        Severity   : Medium
        Mitigation : Flag FIPS with < 3 years; exclude from policy claims
        Status     : Flagged in audit

  [B7]  Random 80/20 temporal mixing
        Root Cause : Test set contains all years - no held-out future year
        Severity   : Low-Medium
        Mitigation : Add leave-one-year-out (LOYO) evaluation for 2022 as
supplement
        Status     : Mitigation planned

  [B8]  Sentinel-2 AZ 2022 tile anomaly
        Root Cause : AZ 2022: 1,524 tiles vs ~60 average - likely archive
artefact
        Severity   : Medium
        Mitigation : Exclude AZ 2022 from Phase 2/3 training
        Status     : Mitigation planned

  [B9]  USDA surveyor bias
        Root Cause : Large commercial operations over-represented in NASS survey
        Severity   : Medium
        Mitigation : Document; model must not be used for smallholder farms
        Status     : Acknowledged - cannot be corrected without external data

REGISTRY SUMMARY
Total biases identified           : 9
High severity                     : 3
Medium severity (incl. Low-Med)   : 6
Mitigations planned / defined     : 6
Acknowledged unfixable            : 2
Open (future work only)           : 1
```



3.1 Ethical Use Constraints

The following constraints govern any deployment or reporting of model predictions in terms of ethics.

Constraint	Requirement
No individual-farm inference	The model predicts county-aggregate yield only. Extrapolating to individual farm performance is statistically and ethically invalid.
Food security embargo	Predictions must be reviewed before the corresponding official USDA NASS report to prevent commodity market speculation.
Climate nonstationarity	The model trained on 2016–2022 may not be assumed valid beyond 2025 without recalibration. Yield distributions shift under progressive climate change.
Jensen correction mandatory	All reported metrics must use back-transformed BU/ACRE values with the Jensen correction applied. `JENSEN_SIGMA2` must be updated later for residual variance on the train split.

3.1.1 Key Tradeoffs

Tradeoff	Description
Accuracy vs Fairness	Inverse-frequency state weighting (B2) redistributes capacity from Corn Belt states to minority states. Expected: +0.5–2.0 BU/ACRE RMSE increase on IL/IA; −3–5 BU/ACRE reduction on AL/DE. Both weighted and unweighted metrics must be reported.
Model Complexity vs Interpretability	The GMU is the highest-accuracy architecture but least interpretable. The Ridge Regression baseline is the recommended option for any policy or advisory context.
Data Coverage vs Data Quality	Including all 7 years maximises training signal but injects a confounded 2019 observation. Train on all years; evaluate 2019 and 2022 in isolation.
Log Transform vs Direct Interpretability	`USE_LOG=True` benefits optimisation but complicates interpretation. Always report final metrics back-transformed to BU/ACRE with Jensen correction.

The ethics and bias audit framework is established. All fairness metric functions (`sliced_rmse`, `bias_per_group`, `cv_error_per_state`, `fairness_report`) and the Jensen correction constant (`JENSEN_SIGMA2`) are ready for use later.